

Statistical Methods for Respiratory Research

Foundations through High-Dimensional, Longitudinal, and Causal Analysis

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Welcome

A **working handbook** for respiratory research statistics: question first, method second, reproducible R code.

New here? Read [HANDBOOK_GUIDE.md](#) · **Choosing a method?** Open [QUICK_REFERENCE.md](#)

Handbook navigation (start here)

I need to...	Open
Choose a test or model	QUICK_REFERENCE.md → method decision tree
See all methods by chapter	METHOD_MAP.md
Find a figure	FIGURE_INDEX.md
Look up a term	GLOSSARY.md
Cite a guideline or textbook	REFERENCES.md · references.qmd (Quarto appendix)
Follow CASTOR examples	RECURRING_COHORT.md
Practice	exercises/ · solutions/

Reader paths

Path	Chapters	Best for
Core path	1–12	Trials, regression, prediction, CASTOR capstone (Cases A–C)
Advanced discovery	13–17	Omics DE/FDR, batch, flow, antibody screens, integrated CASTOR-HD
Longitudinal & causal	18–21	Repeated FEV1, time-to-event, missing data, IPW/DAGs

Role	Path
Clinician / trialist	QUICK_REFERENCE → relevant chapter technique card → Reporting template
Analyst	Ch 2 → Ch 3 → METHOD_MAP
Course instructor	Preface → Core 1–12 + exercises ; optional 13–21 blocks

Role	Path
Writing a paper	QUICK_REFERENCE → Ch 8 reporting → REFERENCES
Omics / discovery	Ch 13 → Ch 14 batch → Ch 17 pipeline

Eight parts

Part	Chapters	Content
I Foundations	0–2	Preface, statistical thinking, data types
II Describe & compare	3–4	Summaries, <i>t</i> -tests, ANOVA, proportions
III Regression	5–7	Linear models, GLMs, model building
IV Validate & predict	8–9	Reporting, bootstrap, ML metrics
V Discovery (core)	10–12	Dimension reduction, clustering, CASTOR cases
VI High-dimensional biology	13–16	DE/FDR, batch, flow, antibody screens
VII Integrated capstone	17	End-to-end CASTOR-HD + elastic net
VIII Longitudinal & causal	18–21	Mixed models, survival, missing data, causality

Chapters

Ch	Title	Key navigation
0	Preface	How to use this handbook
1	Statistical thinking	Estimands, PICO, inference vs prediction
2	Respiratory data	Outcome types → method routing
3	Descriptive analysis	Table 1, plots
4	Comparing groups	<i>t</i> -test vs Wilcoxon vs chi-square
5	Linear models	Gaussian / <code>lm()</code>
6	GLMs	Logistic, Poisson, NB
7	Model building	Selection, LASSO
8	Validation & reporting	CONSORT, STROBE, TRIPOD
9	Prediction vs inference	AUC, calibration
10	Dimensionality reduction	PCA + alternatives menu
11	Clustering	Phenotype discovery
12	Case studies	End-to-end CASTOR (Cases A–D)
13	Differential analysis + FDR	Omics DE, volcano, BH

Ch	Title	Key navigation
14	Batch effects	PCA, overlap, sensitivity
15	Flow cytometry	Proportions, pseudo-replication
16	Antibody discovery	Screens, PPV, stability tiers
17	Integrated CASTOR-HD	Full pipeline + elastic net
18	Longitudinal mixed models	Repeated FEV1
19	Survival analysis	Time to exacerbation
20	Missing data	MAR, imputation sensitivity
21	Causal inference	Confounding, toy IPW

CASTOR core datasets (Ch 3–12)

File	Use
data/spirometry.csv	FEV1, trials, ANOVA
data/exacerbation.csv	Logistic regression
data/exacerbation_counts.csv	Poisson / NB
data/bronchodilator_paired.csv	Paired <i>t</i> -test
data/spirometry_trial.csv	ANCOVA
data/marker_panel.csv	PCA, clustering

CASTOR extensions (Ch 18–19, Case E)

File	Use
data/longitudinal_spirometry.csv	Mixed models, trajectories
data/time_to_exacerbation.csv	Kaplan–Meier, Cox

CASTOR-HD datasets (Ch 13–17)

File	Use
data/proteomics_olink_like.csv	DE, batch, elastic net
data/rnaseq_counts.csv	NB differential expression
data/flowcytometry_summary.csv	Participant-level proportions
data/antibody_screen.csv	Hit calling, PPV
data/antibody_confirmation.csv	Confirmation assays

R setup

```
setwd("/path/to/respiratory-research-methods")
source("R/00_setup.R")
source("R/generate_data.R")
source("R/examples/generate_figures.R") # all handbook figures
source("R/run_all_examples.R")          # full chapter scripts
```

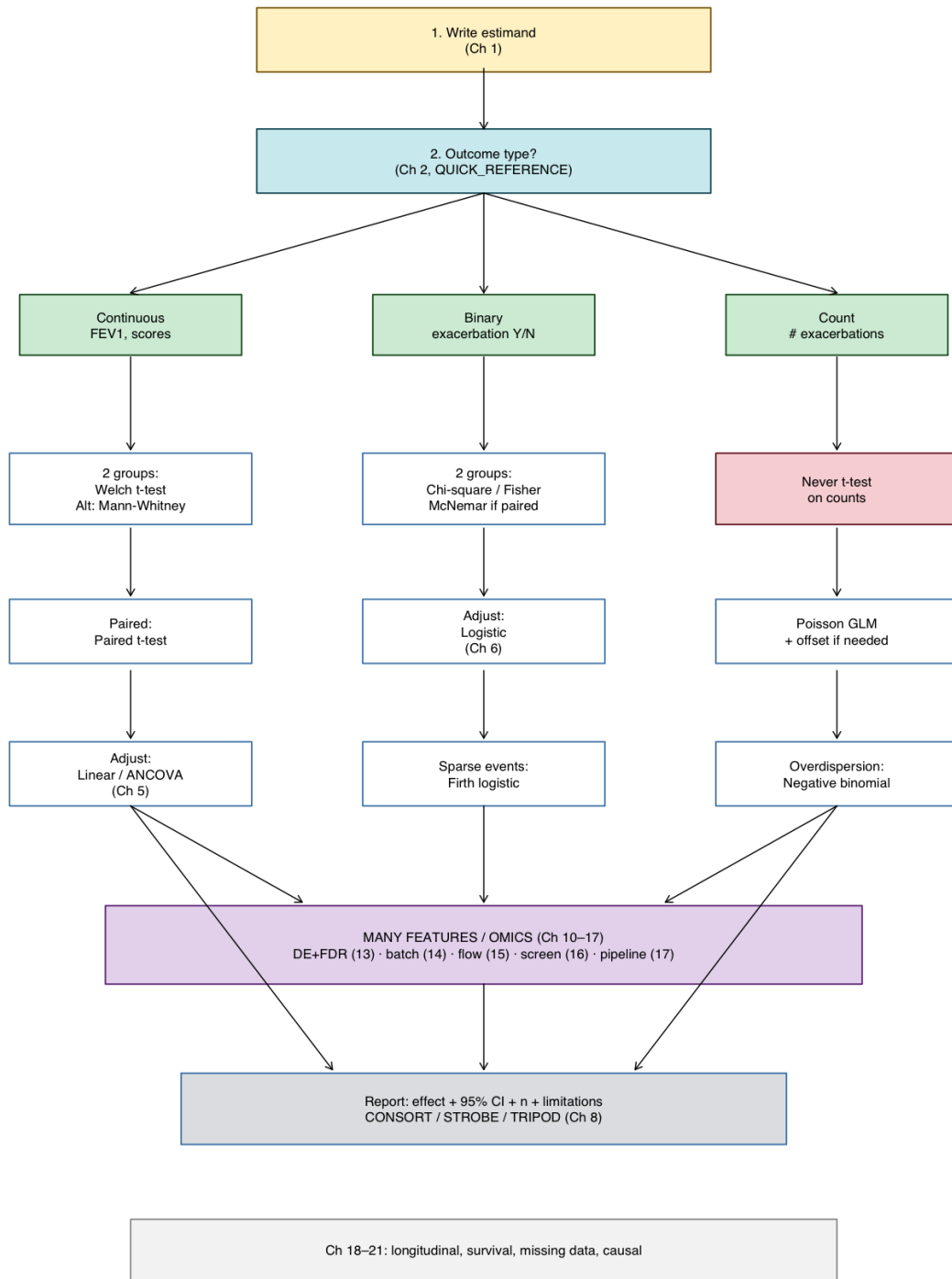
Datasets

See [RECURRING_COHORT.md](#) for full CASTOR / CASTOR-HD map.

Figures

Decision tree and chapter plots: [FIGURE_INDEX.md](#). Regenerate anytime with `source("R/examples/generate_figures.R")`.

Handbook — Method decision tree



See QUICK_REFERENCE.md and METHOD_MAP.md

Figure 1: Method decision tree

Edition: Single-volume handbook (Ch 0–21) with unified navigation, reproducible figures, and CASTOR + CASTOR-HD datasets. See [CHAPTER_TEMPLATE.md](#).

PDF build: `quarto render --to pdf → _book/Statistical-Methods-for-Respiratory-Research.pdf`

Part I: Foundations

This part makes the rest of the handbook *cohere*.

You will learn the shared workflow that every later chapter reuses:

1. **Write the estimand** (what number answers the question?)
2. **Classify the outcome and design** (continuous / binary / count; independent / paired / clustered)
3. **Choose a method** using [QUICK_REFERENCE](#) and [METHOD_MAP](#)
4. **Describe first**, then infer
5. **Report estimates + uncertainty + limitations** (not p-values alone)

If you are using this volume as a handbook, you can skim Part I once and then jump directly by outcome type.

Start here: [QUICK_REFERENCE](#)

Preface

Respiratory research generates an unusual breadth of data: spirometry and lung function, symptom scores, exacerbation counts, imaging features, biomarker panels, trial endpoints, and intensive-care time series. The statistical methods available are equally broad. The gap is not a shortage of techniques. It is a shortage of **clear connections** between clinical questions, valid analysis, and reproducible implementation.

This book was written to fill that gap.

Why question-first?

In practice, analysts often learn methods in reverse: they know how to run a t-test or fit a logistic model before they have articulated what is being estimated, for whom, and under what assumptions. That order produces technically correct code and scientifically weak conclusions.

We reverse it.

Before any formula appears, we ask:

- What would change in clinical or scientific practice if we knew the answer?
- What quantity are we estimating or comparing?
- What data structure do we actually have?
- What could bias the result?

This mirrors good applied biostatistics. We make it explicit and repeatable.

Accessibility and validity

These are not opposites. A method explained well is not a method explained loosely. Throughout the book, we use layered exposition:

- a plain-language statement of the idea
- a precise statistical statement nearby
- assumptions stated openly
- respiratory examples where things go wrong in real studies

We avoid jargon without definition. We also avoid replacing assumptions with metaphors that mislead.

R as a teaching tool

R is used here not as a software manual but as a **proof of understanding**. If you can simulate data, fit a model, check assumptions, and report an estimand with code, you understand the method at a level that reading alone cannot provide.

We use the tidyverse for readability. Where base R is clearer, we use base R.

What this book is not

- Not a catalogue of every test
- Not a machine-learning hype volume
- Not a substitute for protocol design, ethics approval, or clinical judgment
- Not a copy of existing textbooks with lung disease examples pasted in

Acknowledgements

Replace this section with your own acknowledgements before publication. Review by a biostatistician and a respiratory clinician is strongly recommended.

How to use this handbook

Resource	Purpose
HANDBOOK_GUIDE.md	Reader paths and workflow
QUICK_REFERENCE.md	Which test / which model (open first)
METHOD_MAP.md	Full technique inventory
FIGURE_INDEX.md	All plots and decision figures
REFERENCES.md	Citations and reporting guidelines

You do not need to read cover-to-cover. Most analysts open **QUICK_REFERENCE**, jump to one chapter, run the R script, and copy the reporting template.

“Alternatives & extensions” boxes (how to use them)

Each chapter includes an **Alternatives & extensions** section. This is where you’ll find:

- method variants that apply when assumptions break (non-normality, sparse events, overdispersion)
- modern alternatives used in respiratory papers (penalization, bootstrap validation, supervised reduction)
- pointers to [Ch 18–21](#) when the data structure requires them (longitudinal, survival, missing data, causal)

These sections are not an invitation to “try everything until something is significant.” They are a **menu** for defensible method choice: pick one primary analysis and prespecify a small number of sensitivity analyses.

About the data

All examples use simulated data with realistic structure. This keeps the book fully reproducible. When you apply the methods, replace synthetic datasets with your study data and revisit every assumption.

Chapter 1: Statistical Thinking in Respiratory Research

Part I: Foundations

At a glance

Focus	How to think before analysing
Key ideas	Estimands, design, bias, inference vs prediction
Tools	PICO, method map, reporting frameworks
Navigation	HANDBOOK_GUIDE · QUICK_REFERENCE · METHOD_MAP · Decision tree · REFERENCES
Exercises	ch01 · Solutions

Learning objectives

1. Separate clinical, statistical, and data layers of an analysis.
2. Define an **estimand** in plain language and formal terms.
3. Classify respiratory study designs and what each can support.
4. Distinguish **inference** from **prediction** ([Shmueli 2010](#)).
5. Navigate the handbook method map and quick reference.

Prerequisites

None - start here.

Opening question

A pulmonologist asks: “*Is FEV1 lower in smokers?*”

Before opening R, you need three answers:

1. **Clinical:** What decision changes if we know the answer?
2. **Statistical:** What exact quantity are we estimating (estimand)?
3. **Data:** Cross-sectional snapshot or longitudinal trial? Which confounders?

This chapter teaches that order. Methods come in [Chapter 4](#) onward; **thinking** comes first ([Harrell 2015](#)).

Handbook rule: Open [QUICK_REFERENCE.md](#) only **after** you can write the estimand in one sentence.

Three layers of every analysis

Technique card

Answers	Are clinical question, statistical estimand, and data structure aligned?
Clinical layer	What would change practice or knowledge?
Statistical layer	What parameter or contrast is estimated?
Data layer	What was measured, in whom, when, with what missingness?
When to use	At the start of every project - before software
When NOT to use	Never skip - even for “simple” t-tests
Does NOT prove	That a method is correct - only that the question is clear

Dual interpretation

Plain language: make sure you are answering the question the clinician actually asked.

Precise language: the estimand must be well-defined relative to the target population, intervention/exposure, and outcome measure ([Celli et al. 2015](#)).

Clinician read: if the statistician analyses post-bronchodilator FEV1 but the trial protocol specifies pre-bronchodilator, the answer may not apply to your decision.

Layer	Question	Example error
Clinical	What decision or knowledge?	Using the wrong endpoint (symptom score when regulatory approval needs exacerbations)
Statistical	What estimand/hypothesis?	Testing mean FEV1 when the protocol specifies risk difference
Data	What was measured, in whom?	Ignoring missing spirometry or mixed BD protocols

Wrong analysis

Mistake	Open software first; pick “Compare Means” from a menu
Why it fails	Method follows question, not the reverse
Do instead	Write estimand → check QUICK_REFERENCE → then code

PICO for respiratory studies

Technique card

Element	Question	COPD example
P Population	Who?	Moderate-severe COPD, GOLD II-III, adults 40
I Intervention/Exposure	What treatment or factor?	Triple therapy vs dual bronchodilator
C Comparator	Compared to what?	Standard care, placebo, active control
O Outcome	What endpoint?	FEV1 at 12 weeks; 1 moderate/severe exacerbation in 12 months

Write PICO **before** choosing software. PICO does not specify the test - it specifies the **target** (Celli et al. 2015).

Reporting template

Methods (study question):

We asked whether [intervention] improves [outcome] compared with [comparator] in [population] over [timeframe]. The primary estimand was [one sentence].

Estimands: the target of inference

Technique card

Definition	The precise numerical summary the analysis should estimate for a defined population
Examples	Mean FEV1 difference at 12 weeks (RCT); adjusted odds of exacerbation (cohort); rate ratio per person-year
Report with	Point estimate, 95% CI, population, timeframe
Does NOT mean	The p-value; the test statistic; “significance”

Worked examples (respiratory)

Study	Estimand
COPD RCT	Mean difference in FEV ₁ (L) at 12 weeks: intervention – control, ITT population
Observational cohort	Adjusted odds ratio for 1 exacerbation comparing current smokers to never-smokers, conditional on age and FEV ₁ % predicted
Bronchodilator test	Mean change in FEV ₁ (post – pre) on same visit
Prediction model (Ch 9)	12-month exacerbation risk for a patient with specified covariates - not the same as an OR estimand

Dual interpretation

Plain language: the estimand is the number you would put in the abstract if you could know the truth.

Precise language: under the ICH E9(R1) framework, estimands link treatment, population, variable, and intercurrent events - this handbook uses the plain-language version; trials should follow protocol definitions ([Schulz et al. 2010](#)).

Clinician read: ask your analyst “what one number answers my question?” If they cannot say, pause.

Wrong analysis

Mistake	Report $p = 0.06$ with no effect size
Do instead	Report estimand + CI; see Ch 8

Study designs: what each can support

Design	Strength	Limit for causal claims	Vol I methods
RCT	Randomisation balances confounders	ITT vs per-protocol; non-adherence	Ch 4-8
Cohort	Temporal order; incidence	Residual confounding	Ch 5-6 (associations)
Case-control	Efficient for rare outcomes	OR not RR; selection bias	Ch 6 logistic
Cross-sectional	Prevalence, description	No temporal order for causation	Ch 3-5
Clustered	Real-world multi-centre	Wrong SEs if ignored	Vol II

Design **limits language**. An adjusted logistic OR from an observational cohort is an **association**, not proof that stopping smoking **causes** fewer exacerbations ([Elm et al. 2007](#)).

Full data-structure detail: [Chapter 2](#).

Type I error, Type II error, and power

Term	Plain language	Typical setting
Type I error ()	False alarm - declare effect when none	Often $= 0.05$ for confirmatory tests
Type II error ()	Miss a real effect	Depends on sample size and effect size
Power	$1 -$; chance to detect effect if true	Target 80-90% at design stage

Term	Plain language	Typical setting
------	----------------	-----------------

Underpowered exacerbation studies produce **wide CIs** compatible with both null and clinically important effects ([Harrell 2015](#)). A non-significant p-value does **not** prove “no effect” - see [Ch 8 §8.7](#).

Clinician “so what?”

If a trial has 60 patients and a soft endpoint, a “negative” result may mean **inconclusive**, not **futile**.

Bias: threats to valid inference

Bias	Respiratory example	Mitigation
Confounding	Smoking distorts therapy-FEV association	Adjust measured confounders; design (RCT)
Selection	Sicker patients drop out of follow-up spirometry	Report attrition; sensitivity analyses
Information	Misclassified exacerbations (patient recall)	Standardised definitions (Hurst et al. 2010)
Lead-time	Screening detects mild disease earlier	Survival artefacts - Vol II
Measurement	Mixed pre/post bronchodilator FEV	Protocol standardisation (Graham et al. 2019)

Wrong analysis

Claim causation from observational adjustment alone → use associational language; cite design limits ([Elm et al. 2007](#)).

Inference vs prediction

	Inference / explanation	Prediction
Goal	Estimate adjusted associations; test hypotheses	Rank or classify new patients
CAS-TOR	Smoking OR for exacerbation (Ch 6)	Exacerbation risk score (Ch 9)
ex-ample		
Eval-uation	CI, LRT, prespecified estimand	Calibration, AUC, external validation (Moons et al. 2015)

	Inference / explanation	Prediction
Variable selection	Prespecified confounders (Ch 7)	CV, LASSO - different rules

Do not evaluate an explanatory model **only** by AUC ([Shmueli 2010](#)). Do not treat a high-AUC predictor as proof of causation.

Full treatment: [Chapter 9](#).

Reporting frameworks (overview)

Guideline	Design	Key reference
CONSORT	RCT	(Schulz et al. 2010)
STROBE	Observational cohorts, case-control	(Elm et al. 2007)
TRIPOD	Prediction model development/validation	(Moons et al. 2015)
RECORD	Routinely collected health data	(Benchimol et al. 2015)

Checklists improve **transparency**; they do not replace correct analysis ([Harrell 2015](#)). Details: [Chapter 8](#).

Navigating the method map

This handbook provides three linked tools:

Tool	Use when
QUICK_REFERENCE.md	You know outcome type; need test/model now
METHOD_MAP.md	Full inventory and decision tree text
figures/method_decision_tree.png	Visual flowchart for teaching or slides

Workflow: estimand → outcome type (Ch 2) → QUICK_REFERENCE → chapter technique card → R script.

Catalog of wrong analyses (thinking chapter)

Wrong	Right
Menu-driven statistics	Estimand-first workflow
p-value without CI	Estimate + CI + n (Harrell 2015)

Wrong	Right
“Trend” without prespecified trend test	Pre-specify or label exploratory
Causal language from cohort OR	Associational language + limitations
Evaluate explanatory model by AUC only	Match metrics to goal (Shmueli 2010)
Skip protocol definitions for exacerbation	Cite definition (Hurst et al. 2010)

Alternatives & extensions (how to strengthen the thinking layer)

These are not “more statistics.” They are ways to make the same methods more defensible.

Need	Add	Where
Formal causal language	DAGs + identification assumptions	Ch 21
Trial estimand precision	ICH E9(R1) estimand framework	Trial protocol work
Clinical decision framing	Decision thresholds / net benefit	Ch 9 extensions
Reporting discipline	Protocol + checklist alignment	Ch 8

Rule: choose the smallest extension that solves the real problem (confounding, clustering, time-to-event), rather than adding complexity for its own sake.

R lab: first look at CASTOR

```
source("R/00_setup.R")
source("R/generate_data.R")
library(tidyverse)

spirometry <- read_csv(file.path(paths$data, "spirometry.csv"), show_col_types = FALSE)

# Describe before infer
spirometry %>%
  group_by(group) %>%
  summarise(n = n(), mean_fev1 = mean(fev1), sd_fev1 = sd(fev1), .groups = "drop")

# Estimand: mean FEV1 difference by group - method justified in Ch 4
t.test(fev1 ~ group, data = spirometry, var.equal = FALSE)
```

Always **describe** ([Ch 3](#)) before **compare** ([Ch 4](#)).

Chapter summary

- Align clinical, statistical, and data layers before any test.
- Write the **estimand** in one sentence; use PICO to frame the question.

- Design limits causal language; reporting guidelines limit hidden flexibility.
- Use [QUICK_REFERENCE](#) after the estimand is clear - not before.

Further reading

- Harrell, *Regression Modeling Strategies* ([Harrell 2015](#))
- Shmueli, “To explain or to predict?” ([Shmueli 2010](#))
- ATS/ERS COPD research statement ([Celli et al. 2015](#))

Exercises · [Solutions](#)

Next: [Chapter 2 - Data in Respiratory Research](#)

Chapter 2: Data in Respiratory Research

Part I: Foundations

At a glance

Focus	Classify variables, outcomes, and study structures
Key idea	Outcome type → method (see QUICK_REFERENCE)
Recurring cohort	CASTOR - data/*.csv
Navigation	HANDBOOK_GUIDE · QUICK_REFERENCE · METHOD_MAP · GLOSSARY · Decision tree
Exercises	ch02 · Solutions

Learning objectives

1. Distinguish outcome, exposure, and covariate roles.
2. Classify respiratory outcomes (continuous, binary, count, ordinal, time-to-event, high-dimensional).
3. Recognise data structures: cross-sectional, longitudinal, clustered, case-control.
4. Apply respiratory-specific quality checks (spirometry, exacerbations, omics).
5. Map outcome type to handbook chapter using the quick reference.

Prerequisites

[Chapter 1](#) - estimands and design.

Opening question

You receive a spreadsheet with patient ID, age, smoking status, therapy, and FEV1. Is that enough to know which methods apply?

No. You need the **outcome type**, **study design**, **unit of analysis**, and **research question**. This chapter classifies respiratory data so [QUICK_REFERENCE.md](#) and Chapters 3-11 can match methods correctly.

Variables: outcome, exposure, covariate

Technique card

Role	Definition	Respiratory examples
Outcome	What you measure as the result	FEV1, exacerbation, death, hospitalisation
Exposure/predictor	Factor hypothesised to influence outcome	Treatment, smoking, biomarker, air pollution
Covariate/confounder	Other factors to account for	Age, sex, height, severity, centre

The same variable **changes role by question**. *Therapy* is an **exposure** in a treatment-effect analysis and a **covariate** when studying biomarkers on a background of standard care.

Dual interpretation

Plain language: know which column is the “answer” and which columns might distort the link.

Precise language: causal diagrams ([Ch 21](#)) formalise which variables must be adjusted; earlier chapters use subject-matter knowledge and protocol ([Harrell 2015](#)).

Clinician read: if you adjust for a variable **caused by** the exposure (a mediator), you may hide a real effect - discuss with your analyst.

Wrong analysis

Mistake	Treat every column as a predictor in a fishing expedition
Do instead	Prespecify outcome, exposure, confounders (Ch 7)

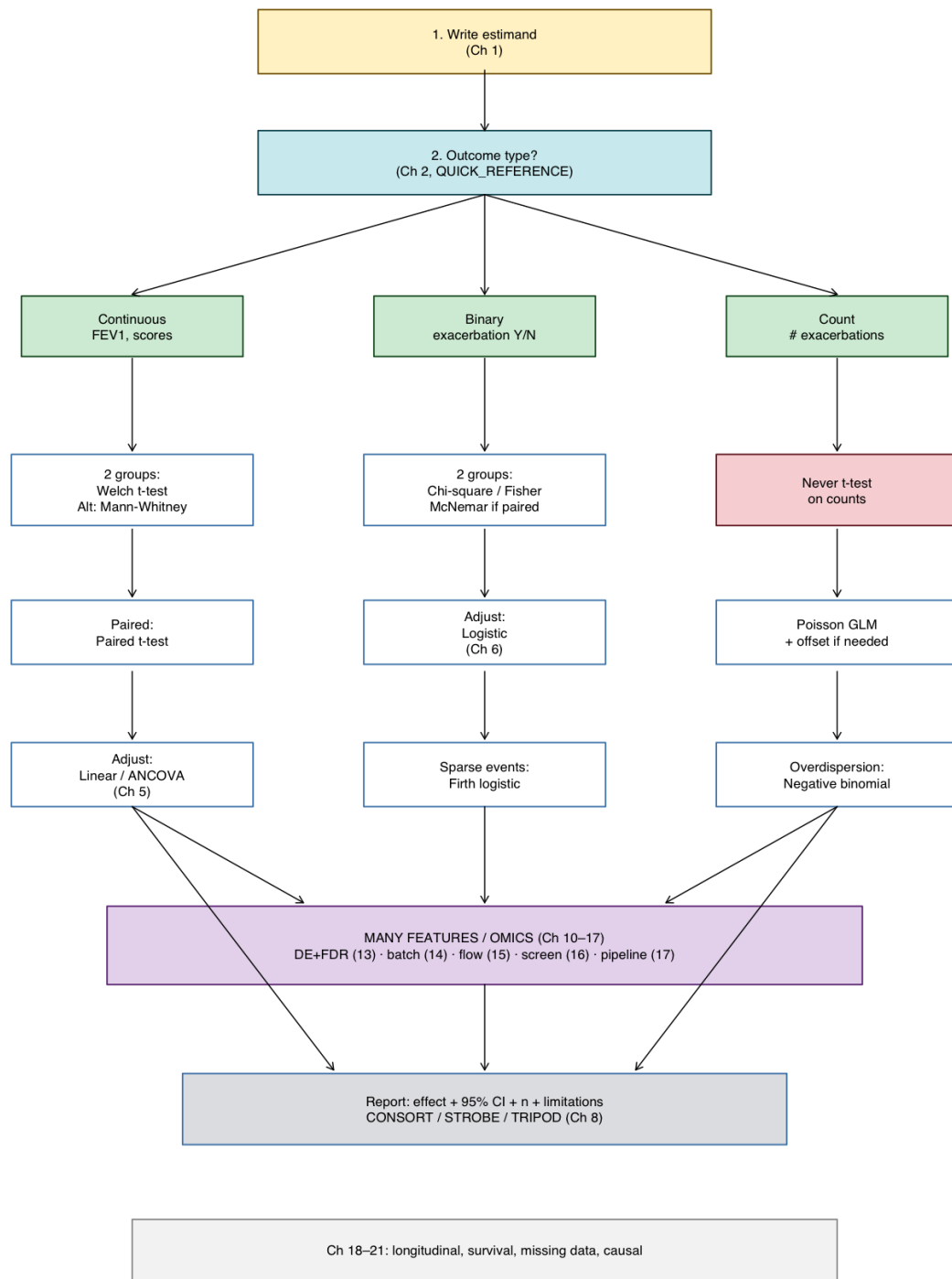
Outcome types: the master routing table

Choosing the wrong outcome type - treating a **count** as **binary**, or a **binary** endpoint as **continuous** - is among the most common errors in applied work ([Stoltzfus 2019](#)).

Type	Properties	Examples	Handbook methods	Chapter
Continuous	Numeric, many values	FEV1 (L), FVC, 6MWD, scores	Mean, <i>t</i> -test, linear regression	Ch 3-5
Binary	Two categories	1 exacerbation Y/N; died Y/N	Proportions, Fisher, logistic	Ch 4, 6

Type	Properties	Examples	Handbook methods	Chapter
Count	Non-negative integers	Exacerbations per year	Poisson, negative binomial	Ch 6
Proportion	Rate in large samples	% with response	Often logistic or linear (large n)	Ch 4
Ordinal	Ordered categories	mMRC dyspnoea 0-4	Ordinal logistic	Ch 6 (extensions)
Time-to-event	Time until event/censoring	Time to first exacerbation	Kaplan-Meier, Cox	Ch 19
High-dimensional	Many correlated features	Proteomics, transcriptomics	PCA, clustering, penalized ML	Ch 10-11

Handbook — Method decision tree



See QUICK_REFERENCE.md and METHOD_MAP.md

Figure 1: Method decision tree - start from outcome type

Handbook link: full routing tables in [QUICK_REFERENCE.md](#).

Clinician “so what?”

If the protocol defines exacerbations as a **count** but the analyst runs a t -test on raw counts, the p-value may be wrong and the effect size uninterpretable → insist on Poisson/NB models ([Ch 6](#)).

Data structures

Structure	Definition	Unit of analysis	Vol I note
Cross-sectional	One row per person, one time point	Patient	Independence assumed in Ch 4-6
Longitudinal	Repeated measures per person	Patient-visit	Mixed models
Clustered	Units nested (patients in hospitals)	Patient (with cluster adjustment)	Cluster-robust SE / mixed models
Case-control	Sample enriched for cases	Case or control	OR natural; incidence not estimable (Agresti 2018)
Paired	Two measurements same subject	Patient	Paired t , McNemar - Ch 4

Dual interpretation

Plain language: repeated FEV1 on the same patients is **not** the same as two independent groups of people.

Precise language: longitudinal and clustered data violate independence assumptions of standard t -tests and GLMs unless extended ([Harrell 2015](#)).

Respiratory-specific data domains

Domain	Common variables	Pitfalls	Reference
Spirometry	FEV1, FVC, FEV1/FVC, % predicted	Pre- vs post-bronchodilator; quality grades	(Graham et al. 2019)
Symptoms	CAT, SGRQ, ACT	Patient-reported missingness; ordinal nature	
Exacerbations	Binary, count, severity	Definition varies by trial	(Hurst et al. 2010)

Domain	Common variables	Pitfalls	Reference
Imaging	Emphysema fraction, airway wall area	High dimensionality; segmentation error	(McShane et al. 2011)
Omics	Thousands of markers	$p \gg n$; batch effects	
ICU / ventilation	Waveforms, ventilator settings	Missing not at random; high frequency	

Always document **definitions** in Methods. An exacerbation in one COPD trial is not necessarily the same in another (Hurst et al. 2010).

Technique card: spirometry essentials

Must specify	Pre- or post-bronchodilator; reference equation for % predicted; quality flags
Common error	Mix pre-BD in one arm and post-BD in another
When to use % predicted	Comparing across age/sex/height
When to use litres	Within-trial mean differences with narrow eligibility
Standard	ATS/ERS spirometry standardisation (Graham et al. 2019)

Data quality checks (pre-analysis)

Before any model:

Check	Action	Example failure
Range	Plausible min/max	FEV1 < 0; FEV1/FVC > 1
Consistency	Age, sex, height vs lung function	Adult with FEV1 6 L and height 150 cm
IDs	One row per estimand unit	Duplicate patient_id
Missingness	Pattern and amount	40% missing FEV1 in one arm only
Units	Litres vs mL; pack-years definition	Mixed units in one column
Protocol	BD timing, visit windows	Visit 2 spirometry outside window

Reporting template

Methods (data):

Spirometry was performed according to ATS/ERS standards ([Graham et al. 2019](#)). Pre-bronchodilator FEV1 (litres) was the primary lung function outcome. Exacerbations were defined as [prespecified definition] ([Hurst et al. 2010](#)). Data quality checks included range validation and duplicate ID review.

From question to data checklist

Complete **before** opening [QUICK_REFERENCE](#):

#	Question	If unclear →
1	What is the outcome variable and its type ?	Re-read protocol
2	What is the unit of analysis (patient, visit, cluster)?	Ch 2 §2.3
3	Are observations independent , paired , or clustered ?	Ch 4 design row
4	Was randomisation used?	Limits causal language
5	What confounders are available and justified?	Ch 7
6	How much missing data and what pattern?	Ch 7 / Vol II MI
7	What is the estimand in one sentence?	Ch 1

CASTOR datasets: outcome type map

File	Key variables	Outcome type	Primary chapter
spirometry.csv	fev1, group, smoking	Continuous	Ch 3-5
spirometry_trial.csv	baseline and follow-up FEV1	Continuous (longitudinal; ANCOVA)	Ch 4, 5
bronchodilator_paired.csv	fev1_pre, fev1_post	Continuous, paired	Ch 4
exacerbation.csv	exacerbation_12m	Binary	Ch 6
exacerbation_counts.csv	exacerbations_12m, person_years	Count	Ch 6
marker_panel_mdsm30, processing_batch		High-dimensional	Ch 10-11

See [RECURRING_COHORT.md](#).

CASTOR-HD datasets (high-dimensional biology preview)

These files are designed for later “advanced discovery” chapters in a single-volume book. They illustrate recurring problems: (p n), batch/plate/run effects, censored/missing values, multiplicity (FDR), and honest validation.

File	What it represents	Key teaching points
proteomics_olink_proteins.csv	Olink-like protein panel (~1000 proteins)	LOD missingness, plate/batch effects, differential abundance, FDR
rnaseq_counts.csv	RNA-seq gene counts (~1200 genes)	library size, NB models, DE, batch runs, FDR
flowcytometry_summary.csv	per-subject flow summaries	proportions, marker medians, drift, group comparisons
flowcytometry_cells_toy.csv	small per-cell flow toy dataset	gating vs clustering, visualization vs inference
antibody_screen.csv	screening signals (replicates, batch)	hit calling, replicate agreement, ranking stability
antibody_confirmation.csv	confirmation assay (KD + positives)	screen PPV, confirmation discipline

Reporting templates for these analyses are in [HIGH_DIM_REPORTING_TEMPLATES.md](#).

Catalog of wrong analyses (data chapter)

Wrong	Why it fails	Right
<i>t</i> -test on binary exacerbation Y/N	Wrong outcome type	Fisher / logistic [Ch 4, 6]
<i>t</i> -test on exacerbation counts	Count, skewed, bounded	Poisson / NB [Ch 6]
lm() on 0/1 outcome	Predictions outside [0,1]	Logistic [Ch 6]
Ignore pairing in pre/post BD	Inflated false positives	Paired <i>t</i> [Ch 4]
Analyse repeated FEV1 as independent	Wrong SEs	Mixed models Vol II
Cluster on markers + FEV1, then “predict” FEV1	Circular	Cluster on markers only [Ch 11]
Omics heatmap without batch check	Technical artefacts	Check <code>processing_batch</code> [Ch 11]

Alternatives & extensions (data structures that change the method)

If any of these apply, the “default” methods in Vol I need an extension.

Data feature	What breaks	What to do
Repeated FEV1 over time	independence	Ch 18 : mixed models / GEE
Time to exacerbation	censoring	Ch 19 : survival analysis
Multi-centre clustering	SEs too small	Ch 18 : cluster-robust / mixed
High-dimensional omics p»n	unstable estimates	Vol I: penalization (Ch 7), PCA toolkit (Ch 10), careful validation
Routine EHR data	selection/measurement bias	RECORD-style reporting (Benchimol et al. 2015) + sensitivity

R lab: inspect CASTOR data

```
source("R/00_setup.R")
source("R/generate_data.R")
library(tidyverse)

spirometry <- read_csv(file.path(paths$data, "spirometry.csv"), show_col_types = FALSE)
exacerbation <- read_csv(file.path(paths$data, "exacerbation.csv"), show_col_types = FALSE)
counts <- read_csv(file.path(paths$data, "exacerbation_counts.csv"), show_col_types = FALSE)

glimpse(spirometry)
summary(spirometry$fev1)

# Quality summary
spirometry %>%
  summarise(
    n = n(),
    pct_missing_fev1 = mean(is.na(fev1)),
    min_fev1 = min(fev1),
    max_fev1 = max(fev1)
  )

# Outcome type routing
message("fev1: continuous → Ch 4–5")
message("exacerbation_12m: binary → Ch 6 logistic")
message("exacerbations_12m: count → Ch 6 Poisson/NB")
```

Chapter summary

- Classify **outcome type** and **data structure** before choosing any test.
- Respiratory data require domain definitions (spirometry, exacerbations, omics).

- Use the pre-analysis checklist (§2.6) every time.
- Route to methods via [QUICK_REFERENCE.md](#).

Further reading

- ATS/ERS spirometry standardisation ([Graham et al. 2019](#))
- COPD exacerbation impact ([Hurst et al. 2010](#))
- Stoltzfus, *Biostatistics for Health and Biological Science Users of R* ([Stoltzfus 2019](#))

Exercises · [Solutions](#)

Next: [Chapter 3 - Descriptive Analysis and Visualization](#)

Part II: Describe, then Compare

Respiratory papers often fail for predictable reasons:

- the sample is not described (missingness, units, protocol differences)
- the wrong outcome type is analysed (counts treated as continuous)
- comparisons ignore pairing, clustering, or multiplicity

This part is the antidote:

- **Chapter 3** makes description explicit (Table 1 + plots + missingness)
- **Chapter 4** is the “which test?” chapter for common outcomes, with alternatives and sensitivity options

If you only read two chapters before running an analysis, read **Ch 3–4**.

Chapter 3: Descriptive Analysis and Visualization

Part II: Description Before Inference

At a glance

Recurring cohort	CASTOR - data/spirometry.csv
Format	Technique cards + Caveats + Wrong analysis + Reporting (template)
R	R/examples/ch03_descriptive.R
Figures	FIGURE_INDEX - ch03_*.png
Navigation	QUICK_REFERENCE · Describe before infer: Ch 4-6
Exercises	ch03

Learning objectives

1. Choose summaries matched to variable type and distribution.
2. Build Table 1 before any inferential analysis.
3. Create diagnostic plots that inform method choice (Ch 4-6).
4. Report missingness and n transparently.
5. Write descriptive Methods/Results text for a respiratory paper.

Prerequisites

Chapters 1-2.

Opening question (CASTOR)

Before comparing FEV1 between trial arms - who is in the CASTOR cohort, and are groups similar at baseline?

Description is not optional preamble. It is how you catch errors and justify the methods that follow ([Harrell 2015](#); [Stoltzfus 2019](#)).

The descriptive workflow

1. **Inventory** - n, variables, types, missingness
2. **Summarise** - Table 1 by exposure/arm
3. **Visualise** - distributions and relationships
4. **Decide** - symmetric → mean/t-test path; skewed → median/nonparametric
5. **Report** - before primary analysis

Technique: Table 1 (baseline characteristics)

Technique card

Answers	What does the study sample look like? Are groups balanced?
Outcome	None - purely descriptive
Design	Any; essential for RCT and cohort papers
Data required	Demographics, key clinical vars, stratification factor
R	<code>gtsummary::tbl_summary(by = group)</code>
When to use	Always before primary inference (Schulz et al. 2010 ; Elm et al. 2007)
When NOT to use	As primary evidence of treatment effect
Does NOT prove	Causation; that adjustment is unnecessary

Dual interpretation

Plain language: Table 1 shows age, sex, smoking, and lung function in each group.

Precise language: descriptive frequencies and summaries; between-group p-values are optional and not substitute for prespecified primary analysis.

Clinician read: are groups similar enough that unadjusted comparison is plausible? Large imbalances → need adjustment (Ch 5-6).

Caveats box

Caveat	Why it matters
Table 1 p-values	Not primary endpoint in RCT; can mislead (Schulz et al. 2010)
Missing data	Different n across rows hides attrition
Collapsed categories	Hides severity spectrum (GOLD, mMRC)
Single time point	Baseline only - not longitudinal status
Synthetic vs real	CASTOR is simulated - replace with your cohort

Wrong analysis

Mistake	Skip Table 1; jump to p-value on outcome
Do instead	Describe first; state n and missingness

Mistake	“Baseline $p < 0.05$ proves groups differ on outcome”
Why wrong	Table 1 tests are not the trial primary analysis
Do instead	Prespecified outcome model (Ch 4-6)

Reporting template

Methods: Baseline characteristics were summarised by treatment arm. Continuous variables are mean (SD); categorical variables n (%). Analyses used R (gtsummary).

Results: Table 1 shows 400 participants (198 intervention, 202 standard). Mean age 58.7 years; 33% smokers; mean FEV1 3.80 L. Groups were similar on observed baseline variables.

R lab

```
source("R/examples/ch03_descriptive.R")
```

Technique: Summary statistics (mean, median, SD, IQR)

Technique card

Answers	What is typical and how variable is this measure?
Continuous symmetric	Mean (SD)
Continuous skewed	Median [IQR]
Binary	n (%)
Does NOT prove	Normality (use QQ plot)

Caveats: FEV1 specifically

Caveat	Detail
Mean hides bimodality	Mixed asthma+COPD cohorts
SD CI	SD describes patients; SE/CI describes mean uncertainty
Units	Litres vs % predicted - label clearly (Graham et al. 2019)

Wrong analysis

Report mean (SD) for ICU length-of-stay with extreme outliers → use median [IQR].

Technique: Histogram and density plot

Technique card

Answers	What is the shape of the distribution?
Use	Continuous outcomes before t-test
R	<code>ggplot(aes(x=fev1)) + geom_histogram + geom_density</code>
CASTOR figure	<code>volume-01/figures/ch03_fev1_histogram.png</code>

Dual interpretation

Plain language: where do most patients' FEV1 values fall?

Clinician read: bimodal or floor effects (severe obstruction cluster) suggest mean alone may mislead.

Caveats

Overlapping histograms by group need clear colours; bin width affects appearance.

Wrong analysis

Use bar chart with only mean $\pm$ SE for $n < 30$ without showing raw data.

Technique: Boxplot and violin plot

Technique card

Answers	Compare distribution between groups
Boxplot	Median, IQR, outliers
Violin	Full density shape
R	<code>geom_boxplot() + geom_jitter(); geom_violin()</code>
Figure	<code>ch03_fev1_violin.png</code>

Caveats

Outliers may be real severe patients - investigate before deleting. Jitter reveals sample size visually.

Technique: QQ plot (normality check)

Technique card

Answers	Are data roughly normal for mean-based methods?
Use before	t-test, ANOVA on small/m moderate n
R	<code>stat_qq() + stat_qq_line()</code>
Figure	<code>ch03_fev1_qq.png</code>

Dual interpretation

Plain language: do points follow the diagonal line?

Precise language: visual check of agreement with normal distribution; Shapiro test often overpowers large n ([Harrell 2015](#)).

Caveats

Mild tail deviation may be acceptable with large n (CLT) ([Harrell 2015](#)). QQ on **residuals** after regression more relevant than on raw FEV1 alone.

Wrong analysis

Reject t-test solely because Shapiro $p < 0.05$ with $n = 500$ - consider Welch t and effect size.

Technique: Scatterplot and correlation

Technique card

Answers	Is FEV1 associated with age (linearly)?
Pearson r	Linear association
Spearman	Monotonic, robust
R	<code>ggplot + geom_point + geom_smooth; cor()</code>
Figure	<code>ch03_fev1_scatter.png</code>

Caveats

Correlation $\neq$ causation. Influential points drive r. Smoking colour on scatter can reveal confounding structure.

Wrong analysis

Correlate binary smoking (0/1) with FEV1 and call it “effect” - use regression (Ch 5) for adjusted statement.

Missing data in descriptives

Report n for each variable. Note if complete-case n drops. Missing FEV1 in spirometry trials often informative (sicker patients skip test) - see [Ch 20](#).

CASTOR worked example

Step 1: Table 1 by group.

Step 2: Histogram and violin of FEV1.

Step 3: QQ plot → roughly symmetric → Welch t reasonable (Ch 4) ([Welch 1947](#)).

Step 4: Scatter FEV1 vs age, coloured by smoking → motivates adjusted regression (Ch 5).

Catalog of wrong analyses (descriptive chapter)

Wrong	Right
Hide missing n	Report n per variable
Table 1 p as primary result	Prespecified outcome analysis
Mean for heavy skew	Median [IQR]
No units on axes	FEV1 (L), age (years)
Skip plots	Plot before test

R lab

```
source("R/examples/ch03_descriptive.R")
```

Chapter summary

- Describe before you infer ([Harrell 2015](#)).
- Table 1 + distribution plots drive method choice.
- Every summary needs n, units, and missingness note.

Alternatives & extensions (choose by data)

Descriptives are where you decide what later methods are plausible. Use these alternatives when the data demand it.

Technique: Standardized mean differences (SMD) instead of Table 1 p-values

Use when	You want to describe baseline balance without hypothesis testing
Why	SMD is scale-free; p-values depend heavily on n
Common R	<code>cobalt</code> package; or compute SMD directly

Technique: Robust summaries for skew/outliers

Data pattern	Prefer
Heavy right tail (LOS, ICU days)	Median [IQR], trimmed mean
Outliers likely	Median absolute deviation (MAD), winsorized summaries

Technique: Missingness visualization

Use when	Many variables with missing data; patterns may be informative
Common R	<code>naniar::vis_miss()</code> , <code>mice::md.pattern()</code> (Vol II expands MI)

Technique: Distributional comparison plots

Plot	Use when
ECDF	Compare full distributions across groups
Ridgeline density	Many groups; need compact distribution comparison

Technique: Transformations (for later modelling)

Use when	Positive skewed outcomes; multiplicative variability
Examples	log-transform biomarkers; $\log(1+y)$ for counts <i>only descriptively</i>
Caution	Transformation changes estimand; report scale clearly

Further reading

- Harrell, *Regression Modeling Strategies* - descriptive summaries before modelling ([Harrell 2015](#))
- Stoltzfus, *Biostatistics for Health and Biological Science Users of R* ([Stoltzfus 2019](#))
- Wickham, *ggplot2* ([Wickham 2016](#))
- CONSORT / STROBE baseline reporting ([Schulz et al. 2010](#); [Elm et al. 2007](#))

Exercises · [Solutions](#)

Next: [Chapter 4](#)

Chapter 4: Comparing Groups

Part II: Inference for Common Respiratory Outcomes

At a glance

Question type	Are groups different? By how much?
Typical outcomes	FEV1, FVC, exacerbation Y/N, exacerbation counts
Recurring cohort	CASTOR
Key methods	t-tests, ANOVA, nonparametric, chi-square, Fisher, ANCOVA, permutation, power
Format	Technique cards + Caveats + Wrong analysis + Reporting (template)
R scripts	R/examples/ch04_comparing_groups.R
Figures	ch04_fev1_by_group · paired BD · comparison panel
Navigation	QUICK_REFERENCE § Step 2-3 · Master table
Exercises	Chapter 4 exercises

Learning objectives

After this chapter you should be able to:

1. State the estimand before choosing any test.
2. Select a comparison method based on outcome type, number of groups, and pairing.
3. Explain assumptions and know nonparametric alternatives.
4. Report effect sizes and confidence intervals, not only p-values.
5. Implement comparisons in R and interpret results for clinical audiences.
6. Recognise respiratory-specific pitfalls (spirometry timing, clustering, multiplicity).

Prerequisites

- [Chapter 1](#) - estimands and study design
- [Chapter 2](#) - outcome types
- [Chapter 3](#) - summaries and plots

Opening question

A COPD trial asks: *Does mean FEV1 differ between intervention and standard care at 12 weeks?*

A second study asks: *Does the proportion of patients with 1 exacerbation differ by therapy line?*

These are both **comparison** questions, but they require **different methods** (Stoltzfus 2019). This chapter is the reference for that choice.

We use the **CASTOR cohort** throughout - same patients you described in Chapters 3 and will model in Chapters 5-9.

The comparison workflow

Every group comparison follows five steps:

1. **Estimand** - mean difference? median difference? risk difference?
2. **Design** - independent, paired, or clustered?
3. **Outcome type** - continuous, binary, count?
4. **Method** - matched to 1-3 with stated assumptions
5. **Report** - effect estimate, 95% CI, n, test/ model, limitations

Continuous outcomes: two independent groups

Technique: Welch two-sample t-test

Answers	Is the mean of a continuous outcome different between two independent groups?
Outcome	Continuous (FEV1, FVC, 6MWD)
Design	Two independent groups, one measurement per patient
Assumptions	Independence; approximate normality within groups or large n; Welch does not require equal variances
Effect measure	Mean difference (group A – group B)
R	<code>t.test(y ~ group, var.equal = FALSE)</code>
Report	Mean difference, 95% CI, n per group, Welch t
Avoid when	Outcome binary/count; paired measurements; strong skew with small n; clustered data

Plain language: compare average lung function between two groups.

Precise language: test $H_0: \mu_A = \mu_B$ with CI for $\mu_A - \mu_B$; robust to unequal variances (Welch-Satterthwaite df) (Welch 1947).

Clinical relevance: ask whether the mean difference exceeds the **MCID** (minimum clinically important difference) for that outcome - literature-based, disease-specific (Cazzola et al. 2008).

```
t.test(fev1 ~ group, data = spirometry, var.equal = FALSE)
```

Dual interpretation (CASTOR trial arms)

Plain language: is average FEV1 higher or lower in the intervention group?

Precise language: estimate of $\mu_{\text{intervention}} - \mu_{\text{standard}}$ with 95% CI; Welch test does not assume equal variances.

Clinician read: is the mean difference (e.g. 0.09 L) large enough to matter given MCID (~0.10 L in many COPD trials)? The CI answers that as well as p-values (Cazzola et al. 2008).

Caveats box: Welch t-test

Caveat	Why it matters in respiratory research
Compares means, not medians	Skewed FEV1 in severe COPD → consider Mann-Whitney or report median
Independence assumed	ICU patients in same ward, or repeated FEV1 → wrong test
One time point	Does not model FEV1 decline over time (Vol II)
Not causal in observational data	Group differences may reflect confounding
MCID	Non-significant p with CI including MCID is inconclusive, not “no effect”
p-value	
Spirometry protocol	Pre- vs post-bronchodilator must match between groups (Graham et al. 2019)

Wrong analysis

Mistake	Welch t-test on exacerbation counts (0, 1, 2, 3...)
Why wrong	Count data, skewed, bounded at zero; mean difference hard to interpret
Do instead	Poisson / negative binomial (Ch 6)

Mistake	Compare post-BD FEV1 in one arm vs pre-BD in another
Why wrong	Confounds treatment with bronchodilator response
Do instead	Same BD protocol in both arms; prespecify pre or post

Mistake	“No significant difference → treatments equivalent”
Why wrong	Underpowered studies cannot prove equivalence
Do instead	Report CI; equivalence needs prespecified margin (TOST)

Reporting template

Methods: FEV1 (L) at 12 weeks was compared between intervention and standard care using Welch’s t-test (two-sided $\alpha = 0.05$). The primary estimand was the mean difference (intervention – standard).

Results: Mean FEV1 was 3.85 L (SD 0.64) in intervention ($n = 198$) and 3.76 L (SD 0.64) in standard care ($n = 202$). Mean difference 0.09 L (95% CI -0.04 to 0.21 ; $p = 0.20$).

Do not say: “No effect of intervention”; “trend toward benefit” without prespecified trend test.

Technique: Pooled two-sample t-test

Assumes **equal variances** in both groups. Rarely needed; Welch is the default. Use pooled version only with strong domain reason and Levene test supporting equality.

```
t.test(fev1 ~ group, data = spirometry, var.equal = TRUE)
```

Technique: Mann–Whitney U (Wilcoxon rank-sum)

Answers	Do two independent groups differ in location/distribution?
Use when	Continuous outcome, small n, clear skew, normality doubtful
Effect measure	Difference in medians (descriptive); rank-based test statistic
R	<code>wilcox.test(y ~ group, exact = FALSE)</code>
Caution	Tests a different null than t-test; does not compare means

```
wilcox.test(fev1 ~ group, data = spirometry)
```

Dual interpretation

Plain language: do smokers and non-smokers differ in typical FEV1 (rank-based)?

Precise language: tests whether distributions differ; null differs from equal means unless distributions are symmetric shifts ([Mann and Whitney 1947](#)).

Clinician read: if t-test and Mann-Whitney disagree, report both estimates (mean diff and median diff).

Caveats box

Caveat	Detail
Not a mean test	Median interpretation; tie handling affects results
Loses power	If data actually normal, t-test more powerful
Ties common	FEV1 tied values - use exact=FALSE carefully

Wrong analysis

Use Mann-Whitney then report mean difference only → report median [IQR] or both.

Reporting template

Results: Mann-Whitney $p = 0.30$; median FEV1 3.82 L (IQR ...) vs 3.75 L (IQR ...).

Continuous outcomes: one sample

Technique: One-sample t-test

Answers	Is mean outcome different from a reference value ?
CASTOR use	Compare cohort mean FEV1 to protocol benchmark (with caution)
R	<code>t.test(fev1, mu = 2.5)</code>
Does NOT prove	That reference is clinically appropriate for this population

```
t.test(spirometry$fev1, mu = 3.0)
```

Caveats

Reference equations for FEV1 % predicted are population-specific. One-sample tests on subgroups need clear estimand.

Wrong analysis

Compare clinic mean to textbook “normal” FEV1 without age/sex/height standardization → use % predicted or regression.

Continuous outcomes: paired measurements

Technique: Paired t-test

Answers	Is mean change (or paired difference) 0?
Design	Same patient, two measurements
CASTOR example	Pre/post bronchodilator FEV1 (bronchodilator_paired.csv)
R	<code>t.test(pre, post, paired = TRUE)</code>
Avoid when	Order effects in crossover; independent groups

```
bronchodilator <- read_csv("data/bronchodilator_paired.csv", show_col_types = FALSE)
t.test(bronchodilator$fev1_pre, bronchodilator$fev1_post, paired = TRUE)
```

Dual interpretation

Plain language: did bronchodilator increase FEV1 on average?

Precise language: tests mean of pairwise differences; assumes approximate normality of differences.

Clinician read: mean change ~0.25 L - compare to MCID for bronchodilator response in relevant disease ([Cazzola et al. 2008](#)).

Caveats box

Caveat	Detail
BD protocol	Must be standardised (200-400 mcg salbutamol etc.) (Graham et al. 2019)
Carryover	Not for crossover without washout
One visit	Learning/spirometry effort may affect 2nd blow

Wrong analysis

Mistake	Welch t on pre vs post as independent groups
Do instead	Paired t or Wilcoxon signed-rank

Reporting template

Results: Mean FEV1 increased 0.25 L post-bronchodilator (95% CI 0.24 to 0.27; paired t, $p < 0.001$; $n = 80$).

Technique: Wilcoxon signed-rank (paired)

For skewed paired differences: `wilcox.test(pre, post, paired = TRUE)`.

Continuous outcomes: three or more groups

Technique: One-way ANOVA

Answers	Do any group means differ?
CASTOR example	FEV1 by diagnosis (no/moderate/severe obstruction)
H	All group means equal
R	<code>aov(fev1 ~ diagnosis) + TukeyHSD</code>
Follow-up	Prespecified contrasts only

```
fit_aov <- aov(fev1 ~ diagnosis, data = spirometry)
summary(fit_aov)
TukeyHSD(fit_aov)
```

Dual interpretation

Plain language: does lung function differ across severity categories?

Precise language: omnibus F-test; does not say which pairs differ without contrasts.

Caveats box

Caveat	Detail
Omnibus only	Significant F → need post-hoc/contrasts
Multiplicity	Many pairwise comparisons inflate error
Ordinal diagnosis	Categories ordered - trend tests alternative
Imbalanced n	ANOVA robust with moderate imbalance

Wrong analysis

ANOVA significant → test all pairs without adjustment → Tukey or prespecified contrasts.

Reporting template

Results: FEV1 differed by diagnosis (ANOVA $F = 49.6$, $p < 0.001$). Tukey: no obstruction vs moderate mean diff 1.52 L (95% CI ...).

Technique: Kruskal-Wallis

Answers	Do 3 groups differ in distribution (nonparametric)?
R	<code>kruskal.test(fev1 ~ diagnosis, data = spirometry)</code>
Use when	Skew, outliers, ordinal severity

Caveats

Does not locate which groups differ - use Dunn post-hoc if needed.

Adjusting for baseline: ANCOVA preview

When groups differ at baseline (observational) or you want greater precision (RCT with baseline covariate):

Question: Is FEV1 at follow-up different between groups **after adjusting for baseline FEV1 and prespecified covariates?**

```
# If follow-up FEV1 available:  
# lm(fev1_followup ~ group + fev1_baseline + age + sex, data = dat)
```

Full treatment in [Chapter 5](#). ANCOVA is **not** a substitute for randomisation in causal claims.

Binary outcomes: comparing proportions

Technique: Chi-square test of independence

Answers	Are row and column categories independent?
CASTOR example	Therapy class $\times$ exacerbation
R	<code>chisq.test(table(therapy, exacerbation_12m))</code>
Requires	Expected counts ~5 per cell (rule of thumb) (Agresti 2018)

```
tab <- table(exacerbation$therapy, exacerbation$exacerbation_12m)  
chisq.test(tab)
```

Caveats box

Caveat	Detail
Sparse tables	Chi-square p invalid $\rightarrow$ Fisher
Unadjusted	Confounding by severity
Large samples	Tiny effects become “significant”

Wrong analysis

Chi-square only without RD/RR/OR $\rightarrow$ always add effect measure with CI.

Technique: Fisher's exact test

For small samples or sparse tables:

```
fisher.test(tab, simulate.p.value = TRUE, B = 10000) # large tables
```

Dual interpretation (CASTOR smoking × exacerbation)

Plain language: is the exacerbation rate different in smokers?

Precise language: tests independence in 2×2 table; Fisher exact valid for small expected counts (Agresti 2018).

Clinician read: report **risk difference** (absolute %) alongside p-value - “2.9% more events” is clearer than p alone.

Caveats box: Fisher / chi-square for binary outcomes

Caveat	Why it matters
Unadjusted	Table ignores age, FEV1, prior history → confounding
OR RR when common	High exacerbation rate → OR overstates RR
Sparse cells	Chi-square invalid; use Fisher or exact methods
Multiple therapies	3+ level tables need careful contrast specification
Not paired	Same patient before/after needs McNemar

Wrong analysis

Mistake	Chi-square only, no effect size
Do instead	RD, RR, or OR with 95% CI; logistic for adjustment (Ch 6)

Mistake	Report OR from unadjusted 2×2 as “adjusted effect”
Why wrong	No covariates in table
Do instead	Logistic with prespecified confounders

Reporting template

Results: Exacerbation occurred in 13/171 smokers (7.6%) and 5/179 non-smokers (2.8%). Fisher exact $p = 0.05$; unadjusted odds ratio 2.86 (95% CI 0.93 to 10.5). Confounders not adjusted.

Effect measures for binary outcomes

Measure	Meaning	R (concept)
Risk difference	Absolute difference in proportions	<code>prop.test</code> , <code>riskdifference</code> packages
Risk ratio	Relative proportion	log-binomial or <code>epiR</code>
Odds ratio	Ratio of odds	<code>glm(..., family = binomial)</code>

Plain language: risk difference tells you how many more patients per 100 experience the event.

Precise language: $RD = p - p$; $RR = p / p$; $OR = [p / (1-p)] / [p / (1-p)]$.

```
prop.test(
  x = c(sum(exacerbation$exacerbation_12m[exacerbation$smoking]),
        sum(!exacerbation$exacerbation_12m[exacerbation$smoking])),
  n = c(sum(exacerbation$smoking), sum(!exacerbation$smoking))
)
```

Technique: McNemar test (paired binary)

Answers	Did paired binary outcome change in same patients?
Design	Before/after on same patient
Example	Sputum culture positive before vs after therapy
R	<code>mcnemar.test(matrix(c(a,b,c,d), nrow=2))</code>
Does NOT	Compare independent groups

```
# Example structure: rows=before, cols=after
# mcnemar.test(table(before, after))
```

Dual interpretation

Plain language: did the proportion with positive culture change after treatment?

Precise language: tests marginal homogeneity in 2×2 paired table; uses discordant pairs only.

Caveats

Requires true pairing; missing one time point drops patient.

Wrong analysis

Chi-square on stacked before/after as independent groups $\rightarrow$ McNemar.

Reporting template

Results: McNemar test $p = \dots$; 12 patients improved, 3 worsened among discordant pairs ($n = \dots$).

Count outcomes: comparing rates

For **exacerbation counts** between groups, Poisson or negative binomial regression is preferred over t-tests on raw counts. See [Chapter 6](#).

Quick two-group comparison (equal follow-up):

```
counts <- read_csv("data/exacerbation_counts.csv", show_col_types = FALSE)
wilcox.test(exacerbations_12m ~ factor(smoking), data = counts) # descriptive
# Inferential: Poisson GLM (Ch 6)
```

Effect sizes

Cohen's d (two groups)

$$d = \frac{\bar{x}_1 - \bar{x}_2}{s_p}$$

where s_p is pooled SD.

```
cohen_d <- function(x, g) {
  stats <- tapply(x, g, function(v) c(mean = mean(v), sd = sd(v), n = length(v)))
  m1 <- stats[[1]]["mean"]; m2 <- stats[[2]]["mean"]
  s1 <- stats[[1]]["sd"]; s2 <- stats[[2]]["sd"]
  n1 <- stats[[1]]["n"]; n2 <- stats[[2]]["n"]
  sp <- sqrt(((n1 - 1) * s1^2 + (n2 - 1) * s2^2) / (n1 + n2 - 2))
  (m1 - m2) / sp
}
cohen_d(spirometry$fev1, spirometry$group)
```

Rule-of-thumb (Cohen): $|d|$ 0.2 small, 0.5 medium, 0.8 large - context matters more in clinical research.

Do not: run many tests and report only significant ones.

ANCOVA: comparing groups with baseline adjustment

Technique: ANCOVA (analysis of covariance)

Answers	Is follow-up FEV1 different between groups after adjusting baseline FEV1 and prespecified covariates?
Outcome	Continuous follow-up (e.g. FEV1 at 12 weeks)
Key covariate	Baseline measure of same outcome
R	<code>lm(fev1_followup ~ group + fev1_baseline + age + sex)</code>
Report	Adjusted mean difference for group, 95% CI

Caution	Not a substitute for randomisation; regression to the mean; define estimand (change vs follow-up)
----------------	---

Plain language: compare lung function at end of study while accounting for where patients started.

Precise language: test group effect in linear model with baseline and covariates; estimand should be prespecified (analysis of follow-up vs analysis of change scores - they differ unless balanced RCT).

```
trial <- read_csv("data/spirometry_trial.csv", show_col_types = FALSE)
lm(fev1_followup ~ group + fev1_baseline + age + sex, data = trial)
```

Dual interpretation

Plain language: after accounting for starting FEV1, is follow-up lung function different by arm?

Clinician read: often increases precision in RCT; does not replace intention-to-treat primary unless prespecified.

Caveats box

Caveat	Detail
Estimand choice	Follow-up adjusted vs change score
Regression to mean	Extreme baselines regress
Same BD protocol	Mandatory
Not causal fix	In observational data

Wrong analysis

Use change scores without checking correlation baseline-follow-up → ANCOVA often more efficient in RCT.

Reporting template

Methods: Follow-up FEV1 analysed with ANCOVA adjusting for baseline FEV1, age, and sex.

Results: Adjusted mean difference intervention vs standard = ... L (95% CI ...).

Respiratory note: common in COPD/asthma trials. See also [Chapter 5](#).

Permutation (randomisation) tests

Technique card

Answers	Is observed mean difference extreme under label exchangeability?
Use	Small n; skew; robustness check for Welch t
R	Resample group labels (see script)
Does NOT	Replace prespecification or MCID reasoning

When distributional assumptions are doubtful or as a **robustness check**, permutation tests simulate the null by shuffling group labels.

Idea: if groups were exchangeable under H_0 , relabelling should not produce larger differences than observed.

```
set.seed(101)
obs_diff <- diff(tapply(spirometry$fev1, spirometry$group, mean))
perm_diff <- replicate(5000, {
  g <- sample(spirometry$group)
  diff(tapply(spirometry$fev1, g, mean))
})
mean(abs(perm_diff) >= abs(obs_diff)) # two-sided permutation p-value
```

When to use: small n, skewed outcomes, or to complement parametric tests.

Caveats

Computationally intensive; ties and small n need sufficient permutations.

Wrong analysis

Permutation $p < 0.05$ but CI includes MCID → still report CI and clinical context.

Sample size and power (design stage)

Technique card

Answers	How many patients needed to detect effect size with power 1– ?
Inputs	MCID or Cohen's d, , power, design
R	<code>pwr::pwr.t.test(...)</code>
When	Before data collection - not post hoc unless labeled exploratory

Power analysis belongs **before** data collection.

Inputs: expected effect size (e.g. MCID for FEV1), α , desired power (often 80%), design (two-sample, paired).

Cohen's d links mean difference to standardized effect for power formulas.

```
# install.packages("pwr")
pwr::pwr.t.test(d = 0.25, power = 0.8, sig.level = 0.05, type = "two.sample")
```

Plain language: how many patients per arm to have a fair chance of detecting a clinically meaningful difference?

Precise language: under specified alternative d , probability of rejecting H_0 at level α .

Respiratory trials: anchor d or mean difference on published MCID, not only statistical tradition ([Cazzola et al. 2008](#); [Harrell 2015](#)).

Multiplicity

Testing FEV1, FVC, symptoms, and exacerbations without adjustment inflates false positives ([Benjamini and Hochberg 1995](#)).

Strategies:

- One **prespecified primary endpoint** ([Schulz et al. 2010](#))
- Holm or Bonferroni for secondary endpoints
- Gatekeeping procedures in trials

Do not: run many tests and report only significant ones.

Master decision table

Outcome	Groups	Pairing	Primary method	Alternative
Continuous	1 vs reference	-	One-sample t	Wilcoxon signed-rank vs median
Continuous	2	Independent	Welch t	Mann-Whitney
Continuous	2	Paired	Paired t	Wilcoxon signed-rank
Continuous	3+	Independent	ANOVA + contrasts	Kruskal-Wallis
Binary	2	Independent	Chi-square / Fisher	Logistic regression
Binary	2	Paired	McNemar	-
Count	2+	Independent	Poisson/NB GLM	-

Full map: [METHOD_MAP.md](#) · Visual: [method_decision_tree.png](#)

FEV1 by trial arm (CASTOR)

Diamond = mean; use Welch t-test for independent groups (Ch 4)

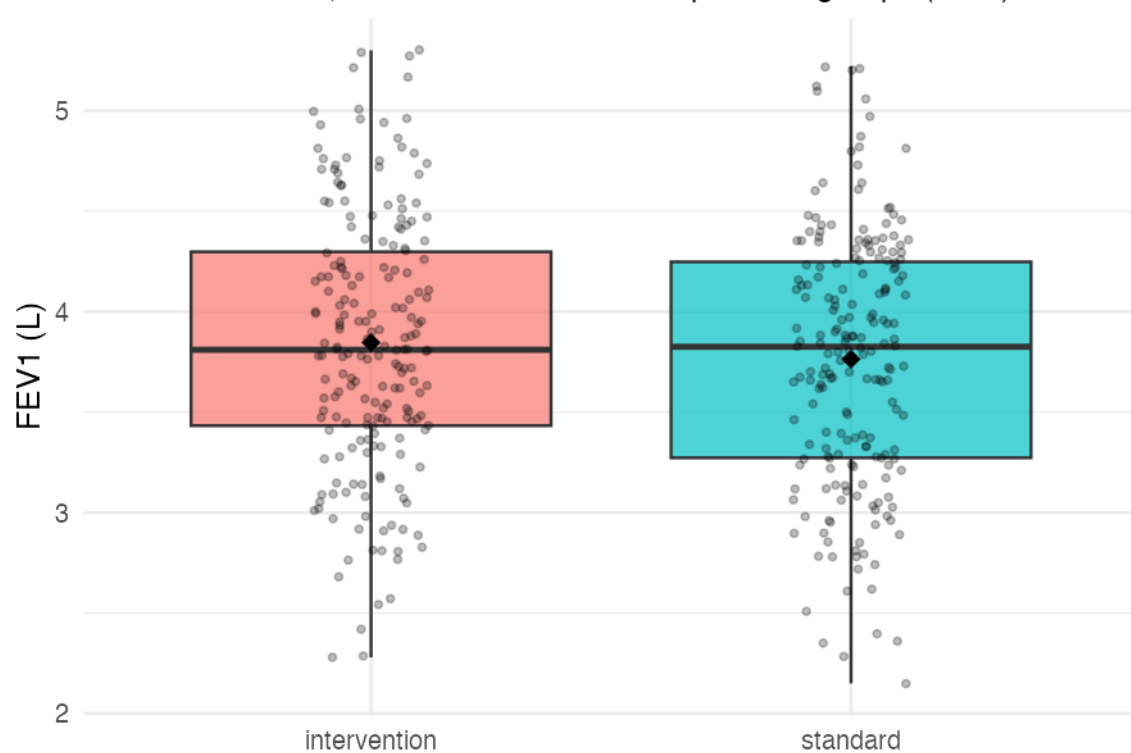


Figure 1: FEV1 by group

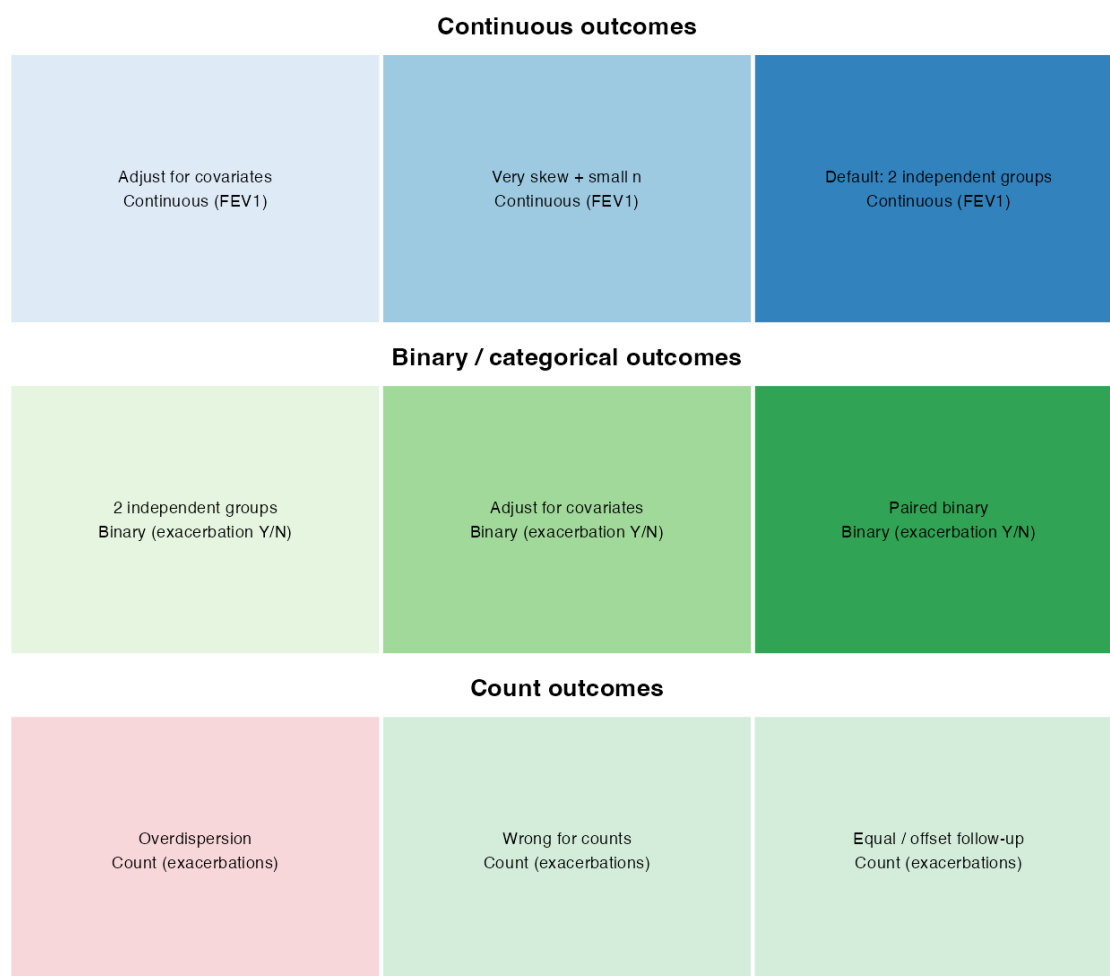


Figure 2: Method comparison panel

Worked example: COPD trial FEV1

Design: Parallel RCT, n = 200 per arm.

Estimand: Difference in mean FEV1 (litres) at 12 weeks (intervention — standard).

Analysis: Welch t-test (prespecified); linear model with baseline covariates as sensitivity.

Results template:

Mean FEV1 was 3.85 L (SD 0.64) in the intervention arm and 3.76 L (SD 0.64) in standard care (n = 198 and 202). The mean difference was 0.09 L (95% CI −0.04 to 0.21; Welch t, p = 0.20). The difference is compatible with no effect and with clinically small benefits; this trial was not powered for a prespecified MCID of 0.10 L.

Clinician read: not statistically significant; CI includes values that may or may not matter clinically.

Statistician read: non-significant p does not prove no effect - interval estimation preferred ([Harrell 2015](#)).

R lab

Run the full script:

```
source("R/examples/ch04_comparing_groups.R")
```

Includes: Welch t, Wilcoxon, ANOVA, Tukey, ANCOVA, permutation test, power calculation, chi-square, Fisher, Cohen's d, paired bronchodilator test.

Common pitfalls in respiratory research

1. **Mixed spirometry standards** - pre- vs post-bronchodilator compared across groups.
2. **Ignoring clustering** - patients within hospitals analysed as independent.
3. **Multiple endpoints** - fishing without multiplicity control.
4. **Mean vs median** - skewed ICU length-of-stay analysed with t-test on small n.
5. **Baseline imbalance** - unadjusted observational comparison of FEV1 by exposure.

Alternatives & extensions (choose by data and design)

Chapter 4 covers the default comparisons. Use these alternatives when the **assumptions, design, or estimand** differ.

Continuous outcomes: beyond “t vs Wilcoxon”

Situation	Alternative	Why / note
Heavy tails/outliers	Trimmed-mean tests (e.g. Yuen)	Robust mean-like estimand; report trimming
Small n; unclear distribution	Permutation test (already in Ch 4)	Minimal assumptions; prespecify statistic
Want CI without formulas	Bootstrap CI	See Ch 8 (Efron and Tibshirani 1993)
Want distributional effect size	Cliff's delta / rank-biserial	Nonparametric effect size

Binary outcomes: beyond chi-square/Fisher

Situation	Alternative	Why / note
Stratified 2×2 tables	Mantel-Haenszel OR/RR	Adjust for one stratifier without full regression
Common outcome; want RR	RR model (log-binomial / modified Poisson)	See Ch 6
Small samples	Exact CIs for OR/RR	Fisher test is exact; CI choice still matters

Count outcomes: compare *rates*, not raw counts

Situation	Alternative	Why / note
Different follow-up time	Poisson/NB + offset(log person-time)	Ch 6
Many zeros	Zero-inflated / hurdle models	Ch 6 (ZIP/ZINB)

Design extensions (Ch 18–19)

Design feature	Why Ch 4 methods fail	Handbook chapter
Repeated measures	observations not independent	Ch 18 mixed models / GEE
Multi-centre clustering	SEs too small if ignored	Ch 18 cluster-robust / mixed
Time-to-event endpoints	censoring	Ch 19 survival analysis

Equivalence / non-inferiority (when “no difference” is the goal)

If the scientific goal is **equivalence** (or non-inferiority), prespecify a margin and use an appropriate framework (e.g. TOST). A non-significant p-value from a superiority test is not evidence of equivalence ([Harrell 2015](#)).

Catalog of wrong analyses (comparison chapter)

#	Wrong	Right
1	t-test on binary exacerbation Y/N	Logistic / compare proportions
2	t-test on count of exacerbations	Poisson / NB
3	Ignore pairing in pre/post BD	Paired t or Wilcoxon signed-rank
4	ANOVA then all pairwise without plan	Prespecified contrasts or Tukey with multiplicity awareness
5	Claim equivalence from $p > 0.05$	Equivalence trial with margin
6	Pool sites without clustering check	Mixed model or cluster-robust SE (Vol II)

Chapter summary

- Define estimand and outcome type first.
- Welch t-test is the default for two independent continuous groups.
- ANCOVA adjusts follow-up comparisons for baseline lung function.
- Permutation tests and power analysis support robust and planned inference.
- Binary outcomes need RD/RR/OR with CIs, not only chi-square p-values.
- Always report effect size and CI.

Exercises

[Chapter 4 exercises](#) · [Solutions](#)

Further reading

- Agresti, *An Introduction to Categorical Data Analysis* ([Agresti 2018](#))
- Harrell, *Regression Modeling Strategies* - comparison and MCID context ([Harrell 2015](#))
- ATS/ERS spirometry standardisation ([Graham et al. 2019](#))
- Welch (1947); Mann & Whitney (1947) - original test papers ([Welch 1947](#); [Mann and Whitney 1947](#))

Next: [Chapter 5 - Linear Models](#)

Part III: Regression (continuous, binary, count)

This part unifies model choice under one idea:

- **Gaussian / linear models** for continuous outcomes (Ch 5)
- **Generalized linear models** for binary and count outcomes (Ch 6)
- **Model building discipline** (Ch 7) so we do not fool ourselves

The goal is not “run a model,” but:

1. pick the right outcome family
2. state the estimand
3. fit the model defensibly
4. report the estimate, uncertainty, and limitations

Chapter 5: Linear Models for Continuous Respiratory Outcomes

Part III: Regression for Continuous Outcomes

At a glance

Recurring cohort	CASTOR - data/spirometry.csv
Question type	How is FEV1 associated with predictors, adjusted for others?
Key methods	SLR, MLR, interactions, ANCOVA, diagnostics, VIF
R scripts	R/examples/ch05_linear_models.R
Figures	residual diagnostics · adjusted means
Navigation	QUICK_REFERENCE § Step 5 · When <i>t</i> -test vs regression: Ch 4 master table
Exercises	ch05

Learning objectives

1. Specify a linear model that matches the scientific question and estimand.
2. Interpret coefficients as adjusted mean differences or slopes.
3. Use dummy coding and interactions correctly.
4. Diagnose violations with residual plots, VIF, and influence.
5. Distinguish association from causation in observational spirometry data.
6. Write Methods and Results sentences for a regression paper.

Prerequisites

Chapters 3-4.

Opening question (CASTOR cohort)

Among adults in the CASTOR cohort, is FEV1 lower in smokers after adjusting for age, sex, and height?

A *t*-test compares smokers vs non-smokers on FEV1 alone. Regression estimates the **smoking effect while holding other factors fixed** - the estimand is an adjusted mean difference ([Harrell 2015](#)).

When linear regression is the right tool

You have	You want	Use
Continuous outcome (FEV1, FVC, 6MWD)	Adjusted association	Multiple linear regression
Continuous outcome + baseline	Trial follow-up comparison	ANCOVA
Continuous outcome, 2 groups, no covariates	Unadjusted comparison	t-test (Ch 4) - regression equivalent
Binary/count outcome	-	GLM (Ch 6) - not linear regression on 0/1

Technique: Multiple linear regression (MLR)

Technique card

Answers	What is the adjusted association between predictors and mean continuous outcome?
Outcome	Continuous (FEV1 litres, change in FEV1, symptom score)
Design	Cross-sectional, trial follow-up (with care), cohort
Data required	Outcome + predictors; complete cases or planned imputation
Assumptions	Linear in parameters; independent errors; homoscedasticity; approximate normality of residuals (small n) (Harrell 2015)
Effect measure	= mean difference or slope; 95% CI
R	<code>lm(fev1 ~ smoking + age + sex + height_cm, data = spirometry)</code>
When to use	Multiple confounders; adjusted comparison; continuous outcome
When NOT to use	Binary/count outcome; repeated measures without extension; causal claims from observational data without design
Does NOT prove	Causal effect of smoking; prediction accuracy (use Ch 9); mechanism

Dual interpretation

Plain language: after accounting for age, sex, and height, smokers have lower average FEV1 by about 0.39 L.

Precise language: the coefficient for smoking is the expected difference in mean FEV1 between smokers and non-smokers with other covariates held fixed; inference assumes linear model and independent homoscedastic errors.

Clinician read: is ~0.4 L a meaningful gap given MCID (~0.1 L in many COPD contexts)? Possibly - but this is observational; unmeasured confounding may remain ([Cazzola et al. 2008](#)).

Caveats box: MLR for FEV1

Caveat	Why it matters
Association causation	Smokers differ in many unmeasured ways
Linearity	FEV1 vs age is not always linear - consider splines
Collinearity	FEV1 and FVC together inflate SEs
Extrapolation	Do not predict outside observed age/height range
One time point	Cross-sectional model longitudinal decline
% predicted vs litres	Mixing units confounds interpretation

Wrong analysis

Mistake	lm(exacerbation_12m ~ smoking) on 0/1 outcome
Why wrong	Predictions outside [0,1]; wrong variance model
Do instead	Logistic regression (Ch 6)

Mistake	Drop non-significant covariates after fitting (stepwise)
Why wrong	Inflates type I error; biased coefficients for confirmatory inference
Do instead	Prespecify confounders (Ch 7)

Mistake	Say “smoking causes lower FEV1” from cross-sectional data
Why wrong	Design does not support causal language
Do instead	“Smoking was associated with lower FEV1 after adjustment...”

Reporting template

Methods:

FEV1 (litres) was modelled with multiple linear regression adjusting for age (years), sex, and height (cm). Smoking was coded yes/no. We report regression coefficients with 95% confidence intervals. Residual diagnostics were examined. Analysis used R 4.x (lm).

Results:

In 400 participants, smoking was associated with 0.39 L lower mean FEV1 (95% CI −0.47 to −0.32; $p < 0.001$) after adjustment for age, sex, and height. Residuals showed no major departures from linear model assumptions.

Do not say: “Smoking reduced FEV1” (causal); “ $R^2 = 0.68$ proves good model” (R^2 is not primary evidence).

R lab

```
source("R/00_setup.R")
library(tidyverse)
library(broom)

spirometry <- read_csv("data/spirometry.csv", show_col_types = FALSE)
fit <- lm(fev1 ~ smoking + age + sex + height_cm, data = spirometry)
tidy(fit, conf.int = TRUE)
par(mfrow = c(2, 2)); plot(fit); par(mfrow = c(1, 1))
```

Sensitivity: log(FEV1) if distribution heavily skewed; spline on age:

```
lm(fev1 ~ smoking + splines::ns(age, df = 3) + sex + height_cm, data = spirometry)
```

Technique: Simple linear regression (SLR)

One continuous predictor: `lm(fev1 ~ age, data = spirometry)`.

__age: mean change in FEV1 per additional year.

Use SLR for bivariate exploration; prefer MLR when confounders matter (almost always in respiratory epidemiology).

Dummy coding and reference categories

R default treatment contrasts: first level alphabetically is reference.

```
spirometry$sex <- factor(spirometry$sex) # reference: female
# sexmale coefficient = mean difference male vs female
```

Always state reference category in tables and text.

Interactions

`fev1 ~ smoking * age` - smoking effect **varies by age**.

Plain language: the FEV1 gap between smokers and non-smokers may be larger in older patients.

Report: stratified coefficients or marginal effects - not only the interaction p-value.

```
fit_int <- lm(fev1 ~ smoking * age + sex + height_cm, data = spirometry)
anova(lm(fev1 ~ smoking + age + sex + height_cm, data = spirometry), fit_int)
```

ANCOVA: follow-up FEV1 adjusting baseline

See [Chapter 4](#) for trial context.

```
trial <- read_csv("data/spirometry_trial.csv", show_col_types = FALSE)
lm(fev1_followup ~ group + fev1_baseline + age + sex, data = trial)
```

Estimand choice: follow-up adjusted for baseline vs change score - prespecify in protocol.

Caveats: ANCOVA

Caveat	Detail
Regression to the mean	Extreme baseline values pull follow-up toward mean
Balanced RCT	ANCOVA adds precision; unadjusted comparison also valid if prespecified
Same spirometry protocol	Pre/post BD must match

Model diagnostics (required)

Plot / test	Checks
Residuals vs fitted	Linearity, homoscedasticity
Q-Q plot	Normality of residuals
Scale-location	Heteroscedasticity
Cook's distance	Influential points

```
plot(cooks.distance(fit), type = "h", main = "Cook's distance")
```

Investigate data errors before removing points. One patient with FEV1 = 0.1 L may be entry error.

Multicollinearity (VIF)

When predictors correlate strongly (FEV1 model with FVC and FEV1/FVC):

```
car::vif(fit) # values > 5-10 warrant attention
```

Fix: drop redundant predictors, combine, or use ridge (prediction context, Ch 7).

CASTOR worked example: full narrative

Question: Is smoking associated with lower FEV1 after adjustment?

Steps:

1. Table 1 by smoking (Ch 3)
2. Fit `lm(fev1 ~ smoking + age + sex + height_cm)`
3. Check residuals
4. Report `_smoking` with CI
5. Contextualize with MCID

Three-reader summary:

- **Statistician:** adjusted mean difference -0.39 L (95% CI -0.47 to -0.32); model assumptions reasonable.
- **Clinician:** ~ 400 mL lower FEV1 in smokers - clinically substantial if causal; observational design limits causal claim.
- **General reader:** smokers in this dataset have notably lower lung function even after accounting for age, sex, and height.

What linear regression does NOT do

- Model binary exacerbations (Ch 6)
- Handle repeated FEV1 visits ([Ch 18](#) mixed models)
- Prove smoking **caused** lower FEV1 in observational data
- Automatically select important predictors (Ch 7 - prespecification)

R lab

```
source("R/examples/ch05_linear_models.R")
```

Alternatives & extensions (choose by data)

Linear regression (Gaussian errors) is the default for continuous outcomes, but many respiratory endpoints push you to variants.

Continuous outcome but strong skew (e.g., biomarker concentration)

Option	When to use	Note
Transform outcome (log)	Multiplicative variability; positive skew	Changes estimand; report on original scale if needed
Gamma regression (GLM)	Positive continuous with mean-variance link	Often better than forcing normality
Quantile regression	Median (or other quantiles) is the estimand	Robust to outliers; different interpretation

Outliers or heavy tails

Option	When to use	Note
Robust regression	Outliers expected; want slope resistant to leverage	Sensitivity, not a magic fix
Winsorization / trimming	Clear artefacts; documented rule	Must be prespecified for confirmatory work

Clear nonlinearity

Option	When to use	Note
Splines / GAM	Curvature in age-FEV1 or exposure-response	Prespecify df/smoothness; avoid p-hacking

Design requires Ch 18–19

Feature	Why <code>lm()</code> fails	What to use
Repeated FEV1 visits	correlated residuals	Ch 18 mixed models / GEE
Multi-centre clustering	SEs too small if ignored	Ch 18 cluster-robust / mixed

Chapter summary

Chapter summary

- Linear regression answers **adjusted** questions about continuous outcomes ([Harrell 2015](#)).
- Every coefficient needs a reference category or unit definition.
- Diagnostics and caveats are part of the analysis, not optional extras.
- Report coefficients + CI; avoid causal language without design support.

Further reading

- Harrell, *Regression Modeling Strategies* ([Harrell 2015](#))
- Venables & Ripley, *Modern Applied Statistics with S* ([Venables and Ripley 2002](#))

Exercises · [Solutions](#)

Next: [Chapter 6 - GLMs](#)

Chapter 6: Generalized Linear Models

Part III: Regression for Non-Continuous Outcomes

At a glance

Question type	What factors are associated with binary/count outcomes?
Recurring cohort	CASTOR
Format	Technique cards + Caveats + Wrong analysis + Reporting (template)
Key methods	Logistic, Firth, log-binomial, Poisson, NB, zero-inflated, offset
R scripts	R/examples/ch06_glm.R
Figures	logistic forest · Poisson RR
Navigation	QUICK_REFERENCE § Step 3-5 · Decision table
Exercises	Chapter 6 exercises

Learning objectives

1. Explain the three components of a GLM (distribution, link, linear predictor).
2. Fit and interpret logistic regression for binary respiratory endpoints.
3. Distinguish odds ratios from risk ratios and when each matters.
4. Model exacerbation counts with Poisson and negative binomial regression.
5. Use offsets for varying follow-up time.
6. Compute adjusted predictions and recognize separation/overdispersion issues.

Prerequisites

- [Chapter 4](#) - comparing proportions
- [Chapter 5](#) - linear predictor, confounding

Opening question

Among COPD patients, is smoking associated with at least one exacerbation in 12 months after adjusting for age, lung function, and prior history?

The outcome is **binary**. Fitting `lm(exacerbation ~ ...)` predicts values outside [0,1] and wrong variance structure. We need a **GLM** ([Hosmer et al. 2013](#)).

We continue with **CASTOR** - `data/exacerbation.csv`.

GLM framework

A GLM assumes:

$$g(\mu) = \eta = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p$$

where $\mu = E(Y)$ and g is the **link function**.

Component	Role
Random component	Distribution of Y (binomial, Poisson, ...)
Systematic component	Linear predictor
Link	Connects to

Estimation: **maximum likelihood** (iteratively reweighted least squares for many GLMs) (Venables and Ripley 2002).

Binary outcomes: logistic regression

Technique: Logistic regression

Answers	How is binary outcome associated with predictors on log-odds scale?
Outcome	Binary (0/1, TRUE/FALSE)
Link	logit: $\log(p/(1-p))$
Effect measure	Odds ratio = $\exp(\)$
R	<code>glm(y ~ x, family = binomial)</code>
Report	OR, 95% CI, n, number of events
Avoid when	Outcome is count; data clustered; interpreting OR as RR when outcome common

Plain language: smoking multiplies the odds of exacerbation by $\exp(\)$, holding other variables fixed.

Precise language: under the model, a one-unit increase in X adds to the log-odds of Y=1 (Hosmer et al. 2013).

```
logit_fit <- glm(
  exacerbation_12m ~ smoking + age + fev1_percent_predicted + prior_exacerbations,
  data = exac,
  family = binomial
)
broom::tidy(logit_fit, conf.int = TRUE, exponentiate = TRUE)
```

Dual interpretation (CASTOR)

Plain language: after adjusting for age, lung function, and past flare-ups, prior exacerbations are linked to higher odds of another event in 12 months.

Precise language: adjusted odds ratio for prior exacerbations = $\exp(\cdot)$; conditions on other linear predictors; assumes logit link and independent observations.

Clinician read: history of exacerbations is among the strongest predictors - consistent with clinical practice. OR is not the same as “X% more likely” unless events are rare.

Caveats box: logistic regression

Caveat	Why it matters
OR RR when events common	Overstates relative risk
Low events (CASTOR ~18/350)	Unstable estimates; wide CIs; consider Firth (Firth 1993)
EPV	~18/4 4.5 events per variable - below ideal 10-15
Observational	Smoking-exacerbation not causal without design
No clustering	Patients at same centre may correlate
Linear on logit scale	May miss nonlinear age effects - splines

Wrong analysis

Mistake	<code>lm(exacerbation_12m ~ smoking)</code>
Why wrong	Linear model for binary Y
Do instead	<code>glm(..., family = binomial)</code>

Mistake	Report OR as “smokers have 30% higher risk”
Why wrong	OR risk difference; misstates scale
Do instead	Marginal predicted risks via <code>emmeans</code> ; or log-binomial RR

Mistake	Stepwise selection of 20 predictors with 18 events
Why wrong	Massive overfitting; invalid p-values
Do instead	Prespecify confounders; penalized methods for prediction

Reporting template

Methods: Logistic regression modelled 12-month exacerbation (yes/no) adjusting for smoking, age, FEV1 % predicted, and prior exacerbation count. We report odds ratios with 95% CIs (Wald). Model fit assessed by event count and residual deviance.

Results: Among 350 patients (18 events), prior exacerbations were associated with higher odds of a new event (OR 1.70, 95% CI 1.12 to 2.59). FEV1 % predicted OR 0.95 per 1% (95% CI 0.91 to 0.99). Smoking OR imprecise (95% CI included 1).

Do not say: “Smoking causes exacerbation” (observational); “logistic proved prediction model” (use Ch 9 metrics).

Odds vs risk

	Odds ratio	Risk ratio
Scale	Odds	Probability
Common in	Case-control, logistic output	Cohort, trials
Approximation	OR RR when outcome rare (<10%)	Exact for cohort

When events are common, OR exaggerates RR. Consider log-binomial or reporting **marginal** adjusted risks (`marginalEffects`, `emmeans`).

```
# Adjusted predicted probabilities by smoking (conceptual)
if (requireNamespace("emmeans", quietly = TRUE)) {
  emmeans::emmeans(logit_fit, ~ smoking, type = "response")
}
```

Separation and sparse data: Firth penalized logistic

Technique: Firth penalized logistic

Answers	Stable OR estimates when MLE diverges (separation, sparse events)
CASTOR trigger	~18 events / 350 - wide CIs; separation possible in subgroups
R	<code>logistf::logistf(...)</code>
Effect measure	OR (penalized)
When to use	Complete/quasi separation; small event counts
Does NOT prove	Stronger evidence - estimates are biased low

When a predictor perfectly predicts outcome in a category (**complete separation**), MLE coefficients diverge.

```
logistf::logistf(
  exacerbation_12m ~ smoking + age + fev1_percent_predicted + prior_exacerbations,
  data = exac
)
```

Dual interpretation

Plain language: Firth gives finite ORs when ordinary logistic “blows up.”

Precise language: Jeffreys-prior penalized likelihood; reduces bias in small samples at cost of slight shrinkage.

Caveats box

Caveat	Detail
Penalization	State in Methods
Still low EPV	Does not create information from 18 events
Not for prediction	Use calibration/AUC separately (Ch 9)

Wrong analysis

Report Firth OR with narrow causal language and no event count.

Reporting template

Methods: Firth penalized logistic regression was used due to sparse events (n events = ...).

Log-binomial regression: direct risk ratios

Technique: Log-binomial GLM

Answers	Adjusted risk ratio (not odds ratio)
Link	log
R	<code>glm(y ~ x, family = binomial(link = "log"))</code>
When to use	Cohort/trial; outcome common (>10%)
Problem	Convergence failures → modified Poisson + robust SE

```
glm(exacerbation_12m ~ smoking + age, data = exac, family = binomial(link = "log"))
```

Dual interpretation

Plain language: smokers have $RR \times \exp(\)$ times the risk of exacerbation, adjusted.

Clinician read: RR often closer to “percent increase in risk” than OR when events common.

Caveat	Detail
--------	--------

Caveats box

Caveat	Detail
Convergence	May need Poisson trick with robust variance
Bounded	Predicted risks must stay ≤ 1
vs logistic	OR easier to fit; RR more interpretable

Wrong analysis

Report logistic OR as RR when 30% event rate $\rightarrow$ log-binomial or marginal RD.

Reporting template

Results: Adjusted RR for smoking = ... (95% CI ...) from log-binomial model.

Model	Effect measure	When
Logistic	OR	Rare outcome; case-control
Log-binomial	RR	Cohort/trial, common outcome
Marginal predictions	RD	Absolute risk for clinicians

Technique: Probit regression

Answers	Same as logistic on latent scale
Link	probit
R	<code>family = binomial(link = "probit")</code>
Use when	Field standard; latent variable interpretation

Probit and logistic usually rank predictors similarly; coefficients not directly comparable across links.

Caveats

Interpretation less intuitive than OR for clinicians.

Count outcomes: Poisson regression

Technique: Poisson GLM

Answers	How do rates of count outcomes change with predictors?
Outcome	Non-negative integer (exacerbations, ED visits)
Link	log
Effect measure	Rate ratio = $\exp()$
Assumption	Mean = variance (equidispersion)
R	<code>glm(y ~ x, family = poisson)</code>

```
pois_fit <- glm(exacerbations_12m ~ smoking + ics_adherence,
               data = counts, family = poisson)
broom::tidy(pois_fit, conf.int = TRUE, exponentiate = TRUE)
```

Dual interpretation

Plain language: each unit increase in ICS adherence is associated with lower expected exacerbation counts (if rate ratio < 1).

Precise language: $\exp()$ multiplies expected count on multiplicative scale; assumes Poisson variance = mean ([Cameron and Trivedi 2013](#)).

Caveats box: Poisson regression

Caveat	Why it matters
Overdispersion	Exacerbation counts often more variable than Poisson
Equal follow-up	Without offset, unequal person-time biases rates
Zeros	Many zero counts → consider ZIP/ZINB
Rate vs count	Model is for counts; interpret as rates only with offset
Exposure window	Exacerbation definition must match follow-up

Wrong analysis

Mistake	t-test on exacerbation counts
Do instead	Poisson / NB GLM

Mistake	Poisson without checking dispersion
Do instead	Check Pearson χ^2/df ; use NB if > 1

Reporting template

Results: In Poisson regression, ICS adherence was associated with lower exacerbation rate (rate ratio 0.28 per unit adherence scale, 95% CI ...). Pearson dispersion 1.09 suggested mild overdispersion; negative binomial sensitivity gave similar inference.

Technique: Poisson offset (person-time)

Answers	Rate per person-year when follow-up varies
R	<code>glm(... + offset(log(person_years)), family = poisson)</code>
CASTOR data	<code>exacerbation_counts.csv</code> includes <code>person_years</code>
Does NOT	Fix overdispersion alone

When follow-up varies:

$$\log(\mu_i) = \log(t_i) + \beta_0 + \beta_1 X_i$$

```
glm(exacerbations_12m ~ smoking + ics_adherence + offset(log(person_years)),
    data = counts, family = poisson)
```

Dual interpretation

Plain language: models exacerbation **rate** accounting for different follow-up length.

Caveats

Offset must be `log(person-time)`; zero follow-up invalid.

Wrong analysis

Compare total counts without standardizing for 6 vs 12 month follow-up.

Overdispersion and alternatives

Technique: Quasi-Poisson

Answers	Scale SEs when variance > mean
R	<code>family = quasipoisson</code>
Use	Quick fix; not full generative model

Technique: Negative binomial (NB)

Answers	Count regression with extra dispersion parameter
R	<code>MASS::glm.nb(...)</code>
When to use	Overdispersed exacerbation counts (default over Poisson)
Effect measure	Rate ratio = $\exp()$

```
MASS::glm.nb(exacerbations_12m ~ smoking + ics_adherence, data = counts)
```

Dual interpretation

Plain language: allows exacerbation counts to vary more than Poisson expects ([Hilbe 2011](#)).

Precise language: gamma mixing distribution adds overdispersion parameter .

Caveats box

Caveat	Detail
Still need offset	If follow-up varies
Zeros	NB may still miss excess zeros → ZI models
Interpretation	Rate ratios similar to Poisson

Wrong analysis

Poisson p-values when dispersion = 1.09+ without sensitivity NB → anti-conservative inference.

Reporting template

Results: NB rate ratio for smoking = ... (95% CI ...). Poisson sensitivity similar; Pearson dispersion 1.09.

Technique: Zero-inflated Poisson (ZIP) / ZINB

Answers	Separate “structural zeros” from count process
R	<code>pscl::zeroinfl(count ~ x x)</code>
Data	<code>exacerbation_zero_inflated.csv</code>
Use	Excess zeros beyond Poisson/NB
Does NOT prove	“Never exacerbator” phenotype without validation

```
pscl::zeroinfl(exacerbations_12m ~ smoking | smoking, data = counts_zi)
```

Dual interpretation

Plain language: some patients never exacerbate (zero-inflation), others follow count model.

Caveats

Identifiability problems; exploratory; validate clinically.

Wrong analysis

Label ZIP clusters as validated COPD phenotypes from one dataset.

Model comparison for GLMs

Nested models: **likelihood ratio test**

```
anova(reduced_model, full_model, test = "Chisq")
```

Non-nested: **AIC/BIC** - predictive/in-sample comparison, not formal test.

Goodness of fit

Context	Tool
Inference	Residual deviance, Pearson χ^2 , careful interpretation
Prediction	Calibration plot, Brier score (Ch 9)
Count	Check overdispersion; compare Poisson vs NB

Hosmer-Lemeshow for logistic calibration - useful but sensitive to group choice; prefer calibration plots for prediction.

Decision table: which GLM?

Outcome	First choice	If problem
Binary	Logistic	Separation → Firth; common outcome → consider RR model
Count, equal follow-up	Poisson	Overdispersion → NB or quasi
Count, varying follow-up	Poisson + offset	Overdispersion → NB + offset
Ordinal (mMRC 0-4)	Ordinal logistic (Vol II)	-

See [QUICK_REFERENCE.md](#) for the full outcome → model table.

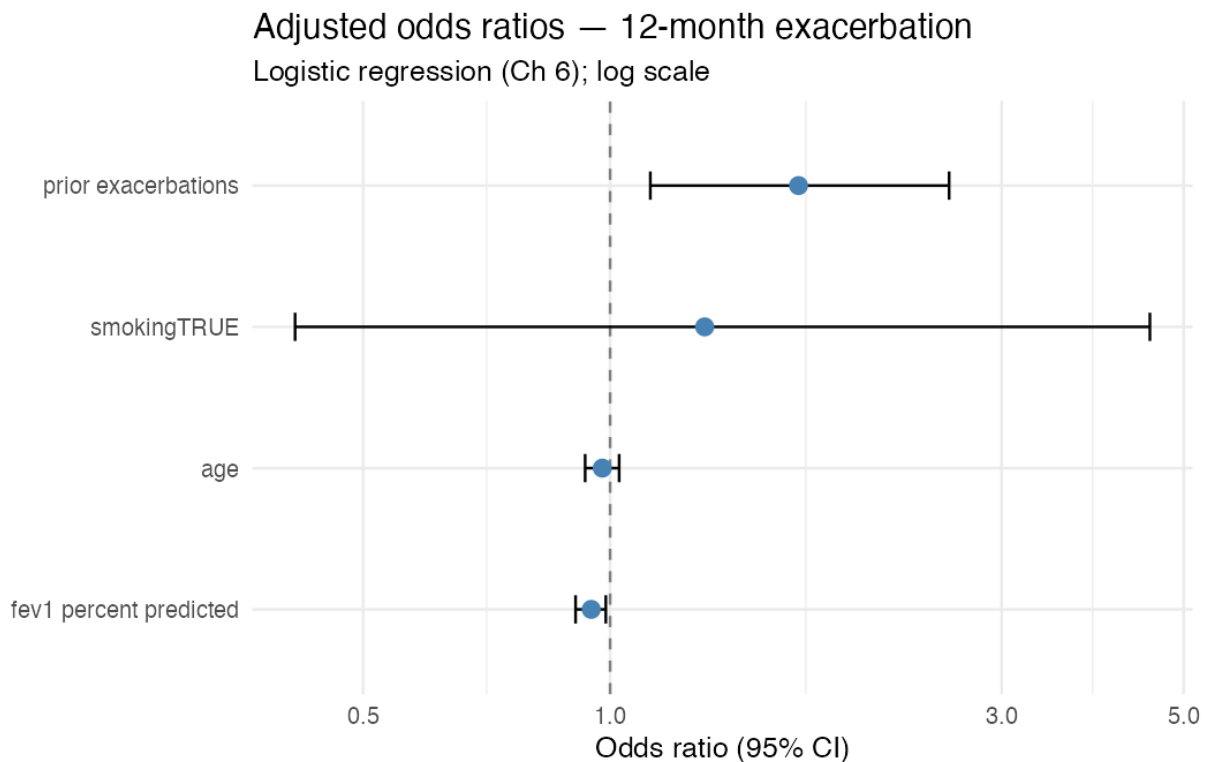


Figure 1: Logistic regression forest plot

Worked example: exacerbation logistic model

Estimand: Adjusted odds ratio for smoking and 12-month exacerbation.

Model: `exacerbation_12m ~ smoking + age + fev1_percent_predicted + prior_exacerbations`

Interpretation template:

After adjustment, prior exacerbations were associated with higher odds of a new event (OR 1.70, 95% CI 1.12 to 2.59). A 1-unit increase in prior count is not necessarily one extra event - verify coding. FEV1 % predicted showed association (OR 0.95 per 1% increase). Smoking OR was imprecise in this sample.

Always state **event count and n**.

Sensitivity analyses to report:

1. Firth logistic if separation / sparse events
2. Log-binomial if OR would mislead (common outcome)
3. NB instead of Poisson if overdispersed
4. Marginal risks for clinicians (`emmeans`)

Catalog of wrong analyses (GLM chapter)

#	Wrong	Right
1	Linear regression on 0/1 outcome	Logistic
2	OR as RR in common outcomes	Log-binomial or marginal RD
3	Poisson ignoring person-time	Offset log(person-years)
4	Ignore overdispersion	Quasi-Poisson / NB
5	Causal claim from adjusted logistic	Associative language + design limits
6	50 predictors, 20 events	Penalized / reduce predictors

R lab

```
source("R/examples/ch06_glm.R")
```

Covers: logistic, probit, Firth (`logistf`), log-binomial, Poisson with offset, quasi-Poisson, negative binomial, zero-inflated (`pscl`), LRT, emmeans.

Pitfalls

1. OR interpreted as RR when events are common.
2. Poisson without offset when follow-up varies.
3. Ignoring overdispersion → false precision.
4. Causal language from observational logistic models.
5. Reporting model without number of events.

Alternatives & extensions (choose by outcome nuance)

Binary outcomes: link and estimand alternatives

Goal / nuance	Alternative	Why / note
Want risk ratio (not OR)	Log-binomial or modified Poisson	OR can mislead when outcome common
Need absolute risks	Marginal risks / risk differences	Often more clinician-friendly
Complementary hazard-type interpretation	complementary log-log link	Ch 19
Clustered binary outcomes	GEE / mixed logistic	Ch 18

Count outcomes: beyond Poisson/NB

Data pattern	Alternative	Why / note
Only patients with 1 event included	Zero-truncated Poisson/NB	Selection changes likelihood

Data pattern	Alternative	Why / note
Two processes (any event vs number of events)	Hurdle model	Separates “ever” from “how many”
High exposure heterogeneity	Offset + interaction / stratification	Person-time is essential

Model family beyond GLM (Ch 18–19)

Outcome	Why GLM may be wrong	Handbook chapter
Ordinal (mMRC 0-4)	ordered categories	ordinal logistic (Ch 6 extensions)
Time-to-event	censoring	Ch 19 survival analysis
Repeated counts over time	correlation + time trends	Ch 18 mixed models / GEE

Chapter summary

- GLMs extend regression to binary and count outcomes via link functions.
- Logistic → odds ratios; Poisson/NB → rate ratios.
- Check overdispersion; use offsets for person-time.
- Separation and sparse events need special handling.

Exercises

[Chapter 6 exercises](#) · [Solutions](#)

Further reading

- Agresti, *An Introduction to Categorical Data Analysis* ([Agresti 2018](#))
- Hosmer, Lemeshow & Sturdivant, *Applied Logistic Regression* ([Hosmer et al. 2013](#))
- Hilbe, *Modeling Count Data* ([Hilbe 2014](#))
- Cameron & Trivedi, *Regression Analysis of Count Data* ([Cameron and Trivedi 2013](#))
- TRIPOD statement for prediction models using binary outcomes.

Next: [Chapter 7 - Model Building](#)

Chapter 7: Model Building and Selection

Part IV: Building Defensible Models

At a glance

Recurring cohort	CASTOR
Format	Technique cards + Caveats + Wrong analysis + Reporting (template)
Key methods	Prespecification, LRT, AIC/BIC, LASSO, splines, missing-data overview
R scripts	<code>R/examples/ch07_model_building.R</code>
Navigation	QUICK_REFERENCE § Step 6 · Inference vs prediction: Ch 1
Exercises	ch07 exercises

Learning objectives

1. Match model-building strategy to inference vs prediction goal.
2. Select confounders by subject-matter knowledge, not p-hacking.
3. Compare nested models correctly (LRT / F-test).
4. Use penalization and CV for prediction - not stepwise for confirmatory p-values.
5. Handle nonlinearity and missing data at introductory level.

Prerequisites

Chapters 5-6.

Opening question (CASTOR)

Which variables should enter the CASTOR exacerbation logistic model - and how do we choose without fooling ourselves?

Model building is where good studies become invalid: data dredging, stepwise selection, and tuning on test data ([Harrell 2015](#)).

Three model-building modes (choose first)

Mode	Variable selection	Evaluation	CASTOR example
Confirmatory inference	Prespecified confounders	CI, LRT for prespecified extras	Smoking + age + FEV1% + prior exac
Exploratory Prediction	Flexible CV / LASSO / RF	Generate hypotheses only AUC, calibration on held-out data	Subgroup scans - label exploratory Ch 9 shootout (James et al. 2023)

Never mix modes without labelling which is which ([Shmueli 2010](#)).

Technique: Prespecified confounder adjustment

Technique card

Answers	Which covariates adjust the exposure-outcome association?
Basis	Causal DAG, clinical knowledge, protocol - not $p < 0.20$ screening
When to use	Observational COPD studies; secondary adjusted RCT analyses
When NOT to use	To “shop” covariates until exposure is significant
Does NOT prove	Causal effect - only reduces confounding if confounders measured

Dual interpretation

Plain language: we adjust for factors that could distort the smoking-exacerbation link.

Precise language: conditional association given measured covariates; unmeasured confounding may remain.

Caveats box

Caveat	Detail
Colliders	Adjusting for mediators or colliders biases
Overadjustment	Adjusting for variables on causal path
Table 2 fallacy	Changing coefficients by adding variables without theory
MRC / GOLD	Severity measures may be mediators depending on question

Wrong analysis

Mistake	Univariate screen: keep vars with $p < 0.2$
Why wrong	Data-driven; inflates false positives
Do instead	Prespecify minimal sufficient adjustment set

Reporting template

Methods: The primary model adjusted for age, FEV1 % predicted, and prior exacerbation count, prespecified in the analysis plan. Smoking was the exposure of interest.

Technique: Nested model comparison (LRT / F-test)

Technique card

Answers	Does adding predictors improve fit significantly?
Linear	<code>anova(m_small, m_big)</code> - F-test
GLM	<code>anova(m_small, m_big, test = "Chisq")</code> - LRT
Requires	Nested models (big contains small)
When to use	Prespecified nested hypotheses (e.g. interaction term)
Does NOT prove	That added variables are causal

CASTOR example

```
reduced <- glm(exacerbation_12m ~ smoking + age, data = exac, family = binomial)
full <- glm(exacerbation_12m ~ smoking + age + fev1_percent_predicted + prior_exacerbations,
            data = exac, family = binomial)
anova(reduced, full, test = "Chisq")
```

Caveats

Multiple LRTs without multiplicity control inflate error. LRT after stepwise invalid.

Wrong analysis

Compare non-nested models with LRT → use AIC or CV instead.

Technique: AIC / BIC

Technique card

Answers	Which model balances fit and complexity in-sample?
Lower	Preferred (with caution)
Use for	Exploratory comparison; prediction prototyping
Avoid for	Confirmatory p-values; causal inference

Caveats

AIC favours prediction; BIC penalizes complexity more. Neither replaces prespecification in trials.

Technique: LASSO (penalized regression)

Technique card

Answers	Which predictors predict outcome when p is large relative to n?
Method	L1 penalty; some coefficients $\rightarrow 0$
R	<code>glmnet::cv.glmnet(..., alpha = 1)</code>
When to use	Prediction; many weak predictors (James et al. 2023)
When NOT to use	Confirmatory inference on prespecified OR
Does NOT prove	Causal importance of selected variables

Dual interpretation

Plain language: LASSO picks a sparse set of predictors that predict exacerbation in cross-validation.

Precise language: penalized logistic regression with λ chosen by CV; coefficients biased but prediction may improve.

Caveats box

Caveat	CASTOR note
Low events	~18 events - LASSO unstable; mostly teaching here
Tuning leakage	Tune only on training folds
Selected vars change	Bootstrap stability recommended

Wrong analysis

Mistake	LASSO-selected model → report unpenalized p-values
Why wrong	Selection ignored in inference
Do instead	Report prediction metrics (Ch 9) or use debiased methods

R lab

```
source("R/examples/ch07_model_building.R")
```

Technique: Splines for nonlinearity

Technique card

Answers	Is the age-FEV1 relationship linear?
R	<code>lm(fev1 ~ smoking + splines::ns(age, df = 3) + sex, data = spirometry)</code>
When to use	Clear curvature; large n
Caution	Overfitting with small n; df prespecify

Wrong analysis

Add spline, pick df that minimizes p-value without prespecification.

Technique: Why NOT stepwise selection

Technique card

Methods	forward, backward, stepAIC
Problem	Inflated type I error; biased coefficients; overfit (Harrell 2015)
Use	Avoid for confirmatory clinical trials and primary publications
Alternative	Prespecification; LASSO for prediction (James et al. 2023)

Wrong analysis (signature example)

Mistake: stepwise logistic on 30 variables with 18 events → “significant” OR for exposure.

Do instead: prespecify 4 confounders; report OR with CI; label exploratory scans separately.

Missing data (introductory)

Complete-case analysis drops rows with any missing covariate - may bias if missing not random.

Report: n analysed vs n enrolled. **Missing data:** [Ch 20](#) (multiple imputation).

Caveats

Missing FEV1 often sicker patients - MNAR. Do not silently complete-case without note.

CASTOR worked example: model building path

Inference path (prespecified):

`exacerbation_12m ~ smoking + age + fev1_percent_predicted + prior_exacerbations`

Sensitivity: add therapy class if not on causal path; Firth if separation.

Prediction path (Ch 9): same predictors → train/test → LASSO by CV - evaluate AUC, not stepwise p.

Catalog of wrong analyses

Wrong	Right
Stepwise for primary endpoint	Prespecified model
Tune on test set	CV on training only
30 predictors, 18 events	Reduce predictors / penalize
Adjust for collider	DAG-informed adjustment
Report AIC-min model as confirmatory	Label exploratory

Reporting template

Methods: Confounders (age, FEV1 % predicted, prior exacerbations) were prespecified. Nested models compared with likelihood ratio tests where stated. No stepwise selection was used for primary inference.

R lab

```
source("R/examples/ch07_model_building.R")
```

Alternatives & extensions (model building menus)

Penalization family (prediction-focused)

Method	When to use	Note
Ridge	many correlated predictors; keep all	stabilizes coefficients; less sparse
Elastic net	correlated blocks; want some sparsity	between ridge and LASSO
Stability selection	want reproducible feature set	still needs external validation

Uncertainty about model form

Option	When to use	Note
Bootstrap model averaging	avoid single “best” exploratory model	report as exploratory
Pre-registered analysis plan	confirmatory inference	reduces researcher degrees of freedom

Causal selection ([Ch 21](#))

Concept	Why it matters	Chapter
DAG-informed adjustment	avoid colliders/mediators	Ch 21

Chapter summary

- Goal determines method: inference → prespecify; prediction → CV/penalize ([Shmueli 2010](#)).
- Stepwise is not a substitute for thinking ([Harrell 2015](#)).
- LRT for nested prespecified comparisons; LASSO for prediction with low EPV caution ([James et al. 2023](#)).

Further reading

- Harrell, *Regression Modeling Strategies* ([Harrell 2015](#))
- James et al., *An Introduction to Statistical Learning* ([James et al. 2023](#))
- Shmueli, “To explain or to predict?” ([Shmueli 2010](#))

Exercises · [Solutions](#)

Next: [Chapter 8](#)

Part IV: Validation, Reporting, Prediction

Two analyses can use the *same* model (e.g., logistic regression) but require different standards:

- **Inference:** CI validity, prespecification, sensitivity analyses
- **Prediction:** held-out evaluation, calibration, transportability

This part provides those standards:

- **Ch 8:** uncertainty, bootstrap, multiplicity, reporting checklists
- **Ch 9:** prediction and ML foundations (AUC + calibration; leakage)

Chapter 8: Uncertainty, Validation, and Reporting

Part IV: Evidence Quality and Transparency

At a glance

Recurring cohort	CASTOR
Format	Technique cards + Caveats + Wrong analysis + Reporting (template)
Methods	CI, bootstrap, multiplicity, CONSORT/STROBE/TRIPOD, sensitivity, reproducibility
Navigation	QUICK_REFERENCE · REFERENCES · Reporting checklists below
Exercises	ch08

Learning objectives

1. Report estimates with uncertainty, not p-values alone.
2. Use bootstrap when asymptotic CIs are doubtful.
3. Apply the correct reporting guideline for your design.
4. Prespecify sensitivity analyses.
5. Document reproducible analysis workflows in R.

Prerequisites

Chapters 4-7.

Opening question (CASTOR)

You found $p = 0.20$ for FEV1 difference between arms. Has the trial “failed,” or is that the wrong question?

This chapter is about **how to report evidence** - intervals, validation, transparency - so CASTOR analyses (and yours) can be trusted and interpreted correctly ([Harrell 2015](#)).

Technique: Confidence interval reporting

Technique card

Answers	What range of effect sizes is compatible with the data?
Report with	Point estimate + 95% CI + n (+ events for binary)
When to use	Always, for every primary and key secondary analysis
When NOT to use	CI alone without clinical context (MCID)
Does NOT prove	No effect if CI includes null - may be underpowered

Dual interpretation

Plain language: the true mean FEV1 difference might plausibly be anywhere from -0.04 to $+0.21$ L.

Precise language: 95% CI procedure; under repeated sampling, ~95% of such intervals cover the true estimand.

Clinician read: is the entire CI below MCID? above MCID? straddling it? That drives interpretation more than $p = 0.20$ ([Cazzola et al. 2008](#)).

Caveats box

Caveat	Detail
95% is convention	Not 95% probability for this one interval (frequentist)
Multiple CIs	Do not cherry-pick the narrowest
Model-based CI	Assumes model correct
Small n	Wald CI for OR can be poor - profile/likelihood or bootstrap (Efron and Tibshirani 1993)

Wrong analysis

Mistake	" $p > 0.05$ therefore no effect"
Do instead	Report CI; discuss power and MCID

Mistake	Report only significant secondary endpoints
Do instead	Pre-specify or label exploratory

Reporting template

Results: Mean difference 0.09 L (95% CI -0.04 to 0.21 ; $n = 400$). The interval includes no effect and clinically small benefits.

Technique: Bootstrap confidence intervals

Technique card

Answers	CI for statistic without formula (complex estimands)
Method	Resample rows with replacement B times; percentile CI
R	<code>replicate</code> + <code>quantile</code> or <code>boot</code> package
When to use	Non-normal statistic; median difference; small n robustness
Does NOT fix	Bad design or wrong estimand

```
B <- 2000
set.seed(1)
boot_diff <- replicate(B, {
  d <- spirometry[sample(nrow(spirometry), replace = TRUE), ]
  diff(tapply(d$fev1, d$group, mean))
})
quantile(boot_diff, c(0.025, 0.975))
```

Dual interpretation

Plain language: if we repeated the study many times, the mean difference would fall in this range about 95% of the time (percentile bootstrap interpretation) ([Efron and Tibshirani 1993](#)).

Caveats

Clustered data need cluster bootstrap (Vol II). Percentile bootstrap can fail for skewed ORs - consider BCa.

Wrong analysis

Bootstrap on 15 patients and treat as definitive → report wide CI honestly.

Technique: Multiplicity control

Technique card

Answers	How to control false positives across many tests?
Methods	Prespecify primary endpoint; Holm; Bonferroni; gatekeeping (Benjamini and Hochberg 1995)
CASTOR context	FEV1, FVC, symptoms, exacerbations - pick one primary
Does NOT mean	Never run secondary analyses

Caveats

Bonferroni conservative; Holm less conservative. Exploratory analyses must be labelled.

Wrong analysis

Test 20 biomarkers at $\alpha = 0.05$; publish the three “significant” ones without correction.

Reporting template

Methods: The primary endpoint was 12-week FEV1. Secondary endpoints were pre-specified with Holm adjustment. Exploratory biomarker analyses were unadjusted and hypothesis-generating.

Technique: Reporting guidelines (CONSORT / STROBE / TRIPOD)

Technique card

Guideline	Design	Key items	Reference
CONSORT	RCT	Flow diagram, randomisation, primary outcome	(Schulz et al. 2010)
STROBE	Observational	Eligibility, confounding, missing data	(Elm et al. 2007)
TRIPOD	Prediction	Validation, calibration, model specification	(Moons et al. 2015)

Dual interpretation

Plain language: checklists help reviewers and readers see what you did and what you might have hidden.

Statistician read: not a substitute for correct analysis - you can follow CONSORT and still misuse ANCOVA.

Caveats

Checklists evolve - cite version/year. Supplementary materials for full protocol.

Wrong analysis

Report CONSORT flow diagram but change primary endpoint post hoc without disclosure.

Reporting template

Methods: We followed CONSORT 2010 guidelines ([Schulz et al. 2010](#)). The analysis plan was prespecified before unblinding (supplementary protocol).

Technique: Reproducible reporting

Technique card

Minimum	R version, key packages, seed, analysis script path
Better	Quarto/R Markdown compiling Tables and Figures
CASTOR	<code>source("R/run_all_examples.R")</code> reproduces book outputs

```
sessionInfo()
```

Caveats

`sessionInfo()` alone insufficient without shared data and script. Proprietary GUI-only steps not reproducible.

Wrong analysis

“Data available on request” with no code → not reproducible science.

Technique: Sensitivity analysis

Technique card

Answers	Are conclusions robust to reasonable analysis choices?
Examples	Complete-case vs MI; Welch vs Mann-Whitney; logistic vs Firth
Rule	Prespecify in protocol when possible

Does NOT mean	Run analyses until one is significant
---------------	---------------------------------------

CASTOR examples

Primary	Sensitivity
Welch t on FEV1	Mann-Whitney; ANCOVA with baseline
Logistic OR smoking	Firth; log-binomial RR
Poisson counts	Negative binomial

Wrong analysis

Run 12 models; report only the one with $p < 0.05$.

Reporting template

Results: Primary analysis ... Sensitivity analyses using [method] yielded similar conclusions (Table S2).

CASTOR worked example: interpreting “non-significant” trial

Primary: Welch t, mean FEV1 difference 0.09 L (95% CI -0.04 to 0.21), $p = 0.20$.

Interpretation:

- **Statistician:** compatible with null and with effects up to ~ 0.21 L.
- **Clinician:** if MCID = 0.10 L, CI includes values above MCID - inconclusive, not negative ([Cazzola et al. 2008](#)).
- **Wrong:** “trial failed” / “no benefit.”

Sensitivity: permutation p; bootstrap CI; ANCOVA on `spirometry_trial.csv`.

Catalog of wrong analyses (reporting chapter)

Wrong	Right
p-value only	Estimate + CI + n
Post hoc primary endpoint	Prespecified + protocol
No flow diagram (RCT)	CONSORT
Hide missing n	Report attrition
No code/data	Reproducible script

R lab

```
source("R/00_setup.R")
library(tidyverse)
s <- read_csv("data/spirometry.csv", show_col_types = FALSE)
t.test(fev1 ~ group, data = s)$conf.int
# Bootstrap - see bootstrap section
sessionInfo()
```

Alternatives & extensions (evidence quality toolkit)

Uncertainty intervals beyond the default

Option	When to use	Note
BCa bootstrap CI	skewed statistics; boundary effects	more stable than percentile in some settings
Profile likelihood CI	logistic ORs; small samples	often better than Wald CI
Bayesian credible intervals	prior information; small samples	requires explicit prior; optional extensions

Multiplicity beyond Bonferroni/Holm

Option	When to use	Note
FDR control	many biomarkers/omics	report method and q-values (Benjamini and Hochberg 1995)
Gatekeeping	trials with endpoint hierarchy	prespecify family structure

Validation options

Context	Option	Why / note
Prediction models	optimism correction / bootstrap validation	complements single split
Clustering / PCA	stability + external replication	Ch 10-11

Missing data ([Ch 20](#))

Multiple imputation (MICE) is the default modern sensitivity tool when missingness is plausibly MAR. [Ch 20](#) introduces patterns and sensitivity; production analyses should use `mice` with pooled estimates.

Chapter summary

- Intervals over dichotomous significance ([Harrell 2015](#)).
- Bootstrap and sensitivity analyses support robust conclusions ([Efron and Tibshirani 1993](#)).
- Match reporting guideline to design ([Schulz et al. 2010](#); [Elm et al. 2007](#); [Moons et al. 2015](#)); document reproducibility.

Further reading

- Harrell, *Regression Modeling Strategies* ([Harrell 2015](#))
- Efron & Tibshirani, *An Introduction to the Bootstrap* ([Efron and Tibshirani 1993](#))
- Steyerberg, *Clinical Prediction Models* ([Steyerberg 2019](#))

Exercises · [Solutions](#)

Next: [Chapter 9](#)

Chapter 9: From Explanation to Prediction

Part V: Prediction and Machine Learning (Foundations)

At a glance

Recurring cohort	CASTOR - data/exacerbation.csv
Question type	What is this patient's 12-month exacerbation risk?
Methods	Logistic, LASSO, rpart, RF, AUC, Brier, calibration
R	R/examples/ch09_prediction.R
Figures	ch09_calibration_logistic
Navigation	QUICK_REFERENCE § Step 6 · REFERENCES · Inference vs prediction: Ch 1
Exercises	ch09

Learning objectives

1. State whether the goal is inference or prediction before modelling.
2. Compare models on the **same** held-out data.
3. Report discrimination (AUC) and calibration together.
4. Explain sensitivity, specificity, PPV, NPV at a chosen threshold.
5. Identify data leakage in respiratory prediction studies.
6. Write TRIPOD-aligned prediction methods and results text.

Prerequisites

Chapters 6-8.

Opening question (CASTOR cohort)

Can we predict which CASTOR participants will have 1 exacerbation in 12 months using baseline smoking, age, FEV1 % predicted, and prior exacerbation history?

That is a **prediction** question. The answer is a **risk score**, not an odds ratio paragraph ([Shmueli 2010](#); [Steyerberg 2019](#)).

Inference vs prediction: choose your lane

	Inference (Ch 6)	Prediction (this chapter)
Question	Is X associated with Y?	What is Y for a new patient?
Output	OR, CI, p-value	Probability or class
Success metric	Valid CI, prespecified model	Calibration + discrimination on new data
Complexity	Parsimony often preferred	Flexible models OK if validated
CAS-TOR example	“Smoking OR for exacerbation”	“Predicted 12-month risk 8%”

You can use logistic regression for both - but evaluation differs ([Hosmer et al. 2013](#); [Shmueli 2010](#)).

Technique: Binary prediction model (general)

Technique card

Answers	What is P(exacerbation in 12 months baseline predictors)?
Outcome	Binary within prespecified horizon
Predictors	Measured before outcome window
Design	Cohort with follow-up; case-control only with care
Evaluation	Train/test or CV; AUC + calibration on held-out data (Steyerberg 2019)
R	<code>glm(..., family=binomial) + predict(..., type="response")</code>
When to use	Risk stratification, screening tools, ML comparison
When NOT to use	Causal treatment effects; tiny event count without regularization
Does NOT prove	Causal impact of predictors; clinical utility without decision analysis

Caveats box: prediction in respiratory research

Caveat	Why it matters
Small event count	CASTOR has ~18 events / 350 - models overfit easily
EPV rule	Aim 10-15 events per predictor; be skeptical with low EPV (Harrell 2015)
Leakage	Post-exacerbation labs must not enter baseline model
AUC alone	High AUC with poor calibration misleads clinicians (Steyerberg 2019)
Prevalence	PPV/NPV change when you apply model to new populations
Transportability	Model trained in one clinic may fail in another

Wrong analysis

Mistake	Report training AUC as model performance
Why wrong	Overoptimistic; model memorized training noise
Do instead	Test-set or cross-validated AUC

Mistake	Select predictors using full dataset, then split train/test
Why wrong	Information from test set leaked into feature selection
Do instead	Feature selection inside training folds only

Mistake	“Random forest discovered exacerbation drivers”
Why wrong	Prediction causal inference; black-box importance mechanism
Do instead	“RF achieved AUC X for 12-month exacerbation prediction”

Mistake	Use OR from logistic model as if it were individual risk without calibration check
Why wrong	Rankings may be OK while absolute risks are wrong
Do instead	Calibration plot on held-out data

TRIPOD-aligned workflow (CASTOR)

Follow TRIPOD for transparent prediction reporting ([Moons et al. 2015](#)):

1. **Population:** CASTOR synthetic COPD-style cohort
2. **Outcome:** exacerbation_12m within 12 months

3. **Predictors:** smoking, age, FEV1 % predicted, prior exacerbations - all baseline
4. **Split:** 70% train / 30% test (prespecified seed)
5. **Metrics:** AUC, Brier, calibration deciles
6. **Report:** n, events, EPV, software

Model shootout: four models, one split

Model	Role	CASTOR script output
Logistic	Interpretable baseline	AUC, Brier
LASSO	Penalized selection	AUC, Brier
rpart tree	Nonlinear rules	AUC, Brier
Random forest	Ensemble	AUC, Brier (Breiman 2001)

```
source("R/examples/ch09_prediction.R")
```

Rule: same `train/test` indices for all models. Differences in AUC are often modest with small event counts - report uncertainty (bootstrap CV in extensions).

Technique: Discrimination (AUC / ROC)

Technique card

Answers	How well does the model rank cases above non-cases?
Metric	$AUC = P(\text{score_case} > \text{score_noncase})$
Interpretation	0.5 random; 1.0 perfect ranking
R	<code>pROC::roc(y, pred); pROC::auc(roc)</code>
Does NOT tell you	Absolute risk accuracy (need calibration)

Dual interpretation

Plain language: if you pick one patient who exacerbated and one who did not, the model assigns higher risk to the right patient 93% of the time (example AUC).

Precise language: AUC is rank correlation between scores and outcomes; threshold-independent ([Steyerberg 2019](#)).

Clinician read: good for triage ordering; not enough to trust the exact percentage shown to patients.

Technique: Calibration

Technique card

Answers	Do predicted probabilities match observed event rates?
Method	Group by risk decile; plot mean predicted vs observed
Perfect	Points on 45° line
Figure	volume-01/figures/ch09_calibration_logistic.png

Brier score: $\text{mean}((y - p)^2)$. Lower is better. Penalizes both discrimination and calibration ([Steyerberg 2019](#)).

```
mean((test$exacerbation_12m - test$pred_log)^2)
```

Classification metrics at a threshold

Default 0.5 is often **not** clinical optimum. Choose threshold by cost of false positives vs false negatives.

Metric	Formula	Respiratory note
Sensitivity	$TP/(TP+FN)$	Catch exacerbations
Specificity	$TN/(TN+FP)$	Avoid false alarms
PPV	$TP/(TP+FP)$	Low when events rare
NPV	$TN/(TN+FN)$	High when events rare

```
# See ch09_prediction.R - class_metrics() helper
```

Cross-validation

Single split is noisy. **k-fold CV** or **bootstrap** for stable performance estimates.

For LASSO: `cv.glmnet` selects using CV - **never** use test set for tuning ([James et al. 2023](#)).

CASTOR worked example: full narrative

Goal: Predict 12-month exacerbation.

Train logistic model on 70% of CASTOR. Evaluate on 30%.

Results template:

Among 105 test-set patients (11 events), logistic regression achieved AUC 0.93 and Brier score 0.04. Calibration by decile showed reasonable agreement except in the highest-risk bin where event count was sparse. Prior exacerbations and FEV1 % predicted contributed most in the penalized and tree models. Given low event numbers, external validation would be required before clinical use.

What this does NOT prove: that lowering FEV1 will prevent exacerbations (causal); that RF “found” biological pathways.

Reporting template (TRIPOD-style)

Methods:

We developed prediction models for 12-month exacerbation (yes/no) using baseline smoking, age, FEV1 % predicted, and prior exacerbation count. The cohort was split 70/30 into training and test sets (seed 42). We fitted logistic regression, LASSO (10-fold CV for), classification trees, and random forests (500 trees). Performance was assessed on the held-out test set using AUC, Brier score, and calibration by predicted risk decile ([Moons et al. 2015](#)).

Results:

The test set included $n = \dots$ patients with $\dots$ events. Logistic regression AUC = ... (95% CI ...); Brier = Calibration plot showed At threshold 0.5, sensitivity = ..., specificity = ..., PPV = ..., NPV =

Do not say: “AI model proves exacerbation risk factors”; “validated biomarker panel” without external cohort.

When ML adds value: and when it does not

Adds value: many weak predictors; unknown nonlinearities; explicit risk tool after validation.

Does not add value: 18 events and 4 predictors (CASTOR) - logistic often sufficient ([Harrell 2015](#)); causal questions; deployment without calibration check.

R lab

```
source("R/examples/ch09_prediction.R")
```

Alternatives & extensions (prediction reporting and validation)

These are common in modern respiratory prediction papers; use them when the goal is deployment-quality prediction rather than a classroom example.

Option	When to use	Note
--------	-------------	------

Calibration beyond one plot

Option	When to use	Note
Calibration intercept/slope	detect systematic under/overprediction	complements decile plot
Recalibration	applying model to new clinic/population	must report transportability limits

Clinical usefulness

Option	When to use	Note
Decision curve analysis	threshold-based clinical decisions	separates AUC from utility
Net benefit	screening/triage policies	requires explicit harm/benefit

Validation designs

Validation	When to use	Note
Temporal validation	later cohort in same site	mimics future deployment
Geographic validation	different centres	key for respiratory multi-centre generalization
External validation	independent cohort	required for clinical claims (Moons et al. 2015)

Model families (when logistic may be insufficient)

Situation	Alternative	Note
Many weak predictors	penalized GLM / gradient boosting	must validate; avoid leakage
Nonlinear interactions expected	tree ensembles	interpretability trade-off

Chapter summary

- Prediction needs held-out evaluation, calibration, and clear predictor timing ([Moons et al. 2015](#); [Steyerberg 2019](#)).
- Compare models fairly; report AUC **and** Brier/calibration.
- Avoid leakage and causal overinterpretation of ML output ([Shmueli 2010](#)).

Further reading

- Moons et al., TRIPOD statement ([Moons et al. 2015](#))
- Steyerberg, *Clinical Prediction Models* ([Steyerberg 2019](#))
- Shmueli, “To explain or to predict?” ([Shmueli 2010](#))
- James et al., *An Introduction to Statistical Learning* ([James et al. 2023](#))
- Breiman, random forests ([Breiman 2001](#))

Exercises · [Solutions](#)

Next: [Chapter 10 - PCA](#)

Part V: Structure Discovery (many markers)

Unsupervised methods are where overclaiming is most common in respiratory research.

This part teaches a defensible discovery workflow:

1. **Reduce dimensions** (Ch 10) to visualize and stabilize noisy marker panels
2. **Cluster defensibly** (Ch 11): stability, batch checks, endotype claim ladder
3. **Integrate end-to-end** (Ch 12): write publication-style narratives and say what was *not* proven

Ch 10 and Ch 11 include “alternatives menus” so you can choose methods based on marker type (continuous, binary, mixed), dimensionality ($p \gg n$), and goal (exploration vs supervised reduction).

Chapter 10: Dimensionality Reduction (Toolkit)

Part VI: Structure Discovery

At a glance

Recurring cohort	CASTOR - data/marker_panel.csv
Format	Technique cards + Caveats + Wrong analysis + Reporting (template)
Methods	PCA + alternatives menu (PLS, sparse/robust PCA, MCA/FAMD, logistic PCA), scree, loadings, biplots
R	R/examples/ch10_pca.R
Figures	ch10_scree · ch10_pca_biplot
Navigation	QUICK_REFERENCE · Omics reporting: (McShane et al. 2011)
Exercises	ch10

Learning objectives

1. Use PCA for exploratory summarisation of many correlated markers.
2. Scale appropriately and choose number of components defensively.
3. Interpret loadings and scores without overclaiming biology.
4. Recognise exploratory limits and need for replication.
5. Report PCA methods transparently.

Prerequisites

Chapters 3, 5.

Opening question (CASTOR marker panel)

Thirty blood markers were measured in 120 CASTOR participants. Can we visualise patient similarity without thirty separate plots?

PCA is **exploratory**. It does not test hypotheses by itself ([Jolliffe and Cadima 2016](#)).

Start here: choose a dimension-reduction method

PCA is the default for **many continuous markers**, but respiratory datasets often violate PCA's assumptions (mixed data types, $p \gg n$, batch effects, binary markers).

Use this menu first, then read the relevant technique section.

Your data / goal	Recommended method	Why	Notes
Many continuous correlated markers (p moderate)	PCA	Simple, transparent	This chapter §10.2
$p \gg n$ (many markers, few patients)	Sparse PCA / regularized PCA	Stabilize loadings	“Advanced options” §10.9
Strong outliers / artefacts	Robust PCA	Reduce outlier leverage	§10.9
Mixed continuous + categorical variables	FAMD	Handles mixed types	§10.9
Mostly categorical (binary/ordinal)	MCA	PCA analogue for categorical	§10.9
Binary data but want “PCA-like” latent structure	Logistic PCA	Uses Bernoulli likelihood	§10.9
Want supervised reduction to predict Y	PLS / PLS-DA	Uses outcome information	§10.10 (with warnings)

Wrong analysis : pick the method that gives the “best-looking separation” on the same cohort, then name an endotype. Discovery needs stability and external validation (Ch 11) ([McShane et al. 2011](#); [Wenzel 2012](#)).

When PCA is appropriate

Use	Avoid
Visualisation, noise reduction	Confirmatory “biomarker axis” without replication
Preprocessing before clustering (train only)	Replacing clinical diagnosis
Hypothesis generation	p-values on components without plan

Technique: Principal component analysis (PCA)

Technique card

Answers	What orthogonal directions capture most joint variation?
Input	$n \times p$ matrix of continuous features (markers)

Output	Scores (patients), loadings (variables), eigenvalues
R	<code>prcomp(X, scale. = TRUE)</code>
When to use	Many correlated continuous features
When NOT to use	Confirmatory endpoint; mixed binary/continuous without encoding
Does NOT prove	Biological mechanism; validated endotype

Dual interpretation

Plain language: PC1 is the weighted combination of markers that separates patients most.

Precise language: eigenvector of covariance/correlation matrix; scores are projections; components uncorrelated ([Jolliffe and Cadima 2016](#)).

Clinician read: a statistical summary axis - name it only after independent validation.

Caveats box

Caveat	Detail
Scaling	Unscaled PCA dominated by largest units
Outliers	Single patient can pull PC1
$p \gg n$	Unstable; regularized PCA needed
Batch effects	Can drive top PCs in omics (McShane et al. 2011)
Circular analysis	Fit PCA on full data then “predict” outcome on same data

Wrong analysis

Mistake	“PC1 is the asthma endotype axis” from one cohort
Do instead	Label exploratory; replicate externally

Mistake	PCA on training+test before split
Do instead	Fit PCA on training; apply rotation to test

Reporting template

Methods: Markers were standardised (z-scores). PCA used the correlation matrix. Scree plot and cumulative variance guided component retention (exploratory) ([Jolliffe and Cadima 2016](#)).

Results: PC1 explained 27% of variance. Loadings were highest on M1-M5 (Table S3). No confirmatory inference was drawn.

R lab

```
source("R/examples/ch10_pca.R")
omics <- read_csv("data/marker_panel.csv", show_col_types = FALSE)
X <- scale(omics %>% select(starts_with("M")))
pca <- prcomp(X)
summary(pca)
```

Technique: Choosing number of components

Technique card

Method	Rule
Scree plot	Elbow in eigenvalues
Cumulative variance	e.g. 80% (arbitrary)
Kaiser	Eigenvalue > 1 (correlation PCA)
Domain	Clinical interpretability - weak alone

Caveats

Many rules disagree; components are retained for **description**, not hypothesis tests.

Wrong analysis

Keep components until outcome regression is significant → circular.

Technique: Loadings and scores

Object	Meaning
Score	Patient position on component k
Loading	Variable weight on component k

High loading on M3 → M3 contributes strongly to that axis.

Caveats

Loadings sign arbitrary; rotation changes interpretation (varimax).

Technique: Varimax rotation

Technique card

Purpose	Simpler loading structure for interpretation
Use	Exploratory only
R	<code>psych::principal(..., rotate = "varimax")</code> or similar

Still not validation of biology.

Technique: Biplot

Joint plot of patients (scores) and variables (loadings). Useful for CASTOR `true_phenotype` visual check - teaching only; real studies lack truth labels. See [ch10_pca_biplot.png](#).

Technique: PC regression (preview)

Regress outcome on first k PCs instead of all markers.

Caveat: components lose direct marker interpretability; risk of overfitting if k chosen from same data.

Advanced options (short technique cards)

These are common in modern respiratory biomarker papers. This handbook includes the *decision logic* and minimal R pointers; do not treat them as confirmatory proof.

Technique: Sparse PCA

Answers	PCA-like components with many loadings forced to zero
When to use	$p \gg n$; interpretability focus
Common R	<code>elasticnet::spca</code> , <code>PMA::SPC</code>
Caveat	Tuning choices are flexible; report how chosen
Does NOT prove	Biological axis; validated marker subset

Technique: Robust PCA

Answers	Reduce influence of outliers/artefacts on components
When to use	Outliers/batch artefacts likely; QC imperfect
Common R	<code>rrcov::PcaHubert</code> , robust covariance approaches
Caveat	Different robust methods yield different PCs

Technique: MCA (multiple correspondence analysis): “PCA for categorical data”

Data	Categorical variables (including binary)
When to use	Many categorical features; survey/symptom patterns
Common R	FactoMineR::MCA, factoextra plotting
Caveat	Interpret with care; axes reflect category frequencies

Technique: FAMD: mixed continuous + categorical

Data	Mixed feature types
When to use	Combine symptoms (categorical) + labs (continuous)
Common R	FactoMineR::FAMD
Caveat	Scaling/weighting decisions matter; document them

Technique: Logistic PCA (binary matrix)

Data	Binary features (0/1)
When to use	Binary “marker present/absent” panels
Common R	logisticPCA-style packages / GLM-based latent factors
Caveat	More complex; convergence and identifiability issues

Technique: Kernel / nonlinear PCA (mention)

When to consider	Strong nonlinear manifolds suspected
Caveat	Harder to interpret; higher overfitting risk
Modern alternatives	UMAP/t-SNE for visualization only (do not call axes “biology”)

Supervised dimension reduction: Partial Least Squares (PLS)

PLS is **not** an “upgrade of PCA.” It changes the question: components are chosen to explain *covariation with an outcome*.

Technique card

Answers	Which linear combinations of markers best predict/associate with Y?
Use when	Many correlated predictors; prediction/exploration goal
Common variants	PLS regression (continuous Y), PLS-DA (classification)

Common R	<code>pls::plsr</code> ; <code>mixOmics</code> for supervised omics workflows
Does NOT prove	Causal mechanism; validated biomarker signature without external validation

Caveats box (PLS in respiratory biomarker papers)

Caveat	Why it matters
Leakage	PLS must be fit inside training folds (like LASSO)
Overfitting	$p \gg n$ with few events $\rightarrow$ inflated performance
Interpretation	“VIP” importance is not causality
Reporting	Use TRIPOD if goal is prediction; otherwise label exploratory (Moons et al. 2015)

Wrong analysis

Fit PLS on the full dataset, show perfect separation, and call it “validated endotypes” $\rightarrow$ external validation required (Ch 11) ([McShane et al. 2011](#); [Wenzel 2012](#)).

CASTOR worked example (PCA as baseline)

1. Scale 30 markers (`marker_panel.csv`).
2. PCA $\rightarrow$ PC1 27% variance.
3. Plot PC1 vs PC2 coloured by `true_phenotype` (synthetic ground truth for teaching).
4. **Conclusion:** separation visible in simulation; would require replication in real omics.

Sensitivity: correlation vs covariance PCA; compare scree.

Catalog of wrong analyses

Wrong	Right
PCA then test outcome without prespecification	Exploratory label
No scaling	<code>scale. = TRUE</code> when units differ
Name PCs “Th2 axis” immediately	Validate clinically
Use PCA scores as definitive subtypes	Clustering + external cohort

Chapter summary

- PCA summarises correlated markers for exploration ([Jolliffe and Cadima 2016](#)).
- Scale, scree, and loadings need careful reporting.
- Never skip external validation for clinical claims ([McShane et al. 2011](#)).

Further reading

- Jolliffe & Cadima, PCA review ([Jolliffe and Cadima 2016](#))
- McShane et al., REMARK biomarker reporting ([McShane et al. 2011](#))
- Wenzel, asthma phenotypes/endotypes context ([Wenzel 2012](#))
- ISL (penalization / CV mindset carries over to PLS/omics) ([James et al. 2023](#))

Exercises · [Solutions](#)

Next: [Chapter 11](#)

Chapter 11: Clustering and Phenotype Discovery

Part VI: Structure Discovery

At a glance

Recurring cohort	CASTOR - data/marker_panel.csv
Format	Technique cards + Caveats + Wrong analysis + Reporting (template)
Methods	k-means, hierarchical, PAM, silhouette, bootstrap stability, validation ladder
R	R/examples/ch11_clustering.R
Figures	FIGURE_INDEX - ch11_*.png
Navigation	QUICK_REFERENCE · Endotype ladder §11.6 · REFERENCES
Exercises	ch11

Learning objectives

1. Distinguish unsupervised clustering from supervised classification.
2. Apply k-means, hierarchical, and PAM methods with appropriate scaling.
3. Choose k with silhouette, domain knowledge, and bootstrap stability - not post hoc storytelling.
4. Navigate the **endotype claim ladder**: exploratory clusters → replication → clinical utility.
5. Recognise batch effects, circularity, and overclaiming in respiratory phenotype papers.

Prerequisites

Chapter 10 (PCA). Clustering on principal components is covered in §11.11.

Opening question (CASTOR)

Do CASTOR participants fall into distinct marker-based subgroups - and if so, what can we legitimately claim?

Clustering finds **structure in measured variables**. It does not, by itself, prove **disease subtypes**, **endotypes**, or **treatment-responsive groups**. Those require a validation path this chapter makes explicit.

For participants: “We looked for groups of patients with similar laboratory marker patterns. We do not yet know whether these groups need different treatments - that requires further studies.”

Clustering vs classification

	Clustering (unsupervised)	Classification (supervised)
Uses outcome label?	No	Yes
Goal	Discover groups of similar patients	Predict a known label
CASTOR example	k-means on M1-M30	Logistic model for 12-month exacerbation (Ch 9)
Typical claim	“Exploratory subgroups”	“Predictor of outcome”
Validation	Replication, stability, external cohort	Cross-validation, calibration, TRIPOD (Moons et al. 2015)

Clinician read: clustering is hypothesis-generating. It does not replace a diagnostic label or a validated risk score.

Technique: k-means clustering

Technique card

Answers	Partition n patients into k groups minimising within-cluster variation
Input	Continuous features; scale when units differ
Requires	k specified in advance; reasonable n per cluster
Assumptions	Roughly spherical, similar-sized clusters; Euclidean distance meaningful after scaling
R	<code>kmeans(X, centers = k, nstart = 25)</code>
When to use	Exploratory phenotyping; moderate n ; compact clusters
When NOT to use	Confirmatory proof of endotypes; elongated or irregular geometry
Does NOT prove	Validated endotype, causal mechanism, or differential treatment response

Dual interpretation

Plain language: split patients into k subgroups with the most similar marker profiles.

Precise language: iterative relocation to a local minimum of within-cluster sum of squares; random starts (`nstart`) reduce bad local optima.

Clinician read: useful for generating hypotheses about subgroups - not for changing care until replicated and linked to outcomes or treatment.

Precise math (optional)

Minimise within-cluster sum of squares:

$$\text{WCSS} = \sum_{c=1}^k \sum_{i \in C_c} \|\mathbf{x}_i - \mu_c\|^2$$

where $\mathbf{x}_i$ is the scaled marker vector for patient i , C_c is cluster c , and μ_c is the cluster centroid (mean vector).

Caveats box

Caveat	Why it matters in respiratory research
Choose k	Elbow, silhouette, and clinical priors disagree - document how k was chosen
Scale features	FEV1 (litres) and blood eosinophils (% or cells/ μL) must not dominate by unit
Outliers	Single severe patients can pull centroids; consider PAM (§11.4)
CASTOR teaching	true_phenotype is for learning only - real studies have no ground truth
Outcome circularity	Never cluster on markers and FEV1, then claim clusters “predict” lung function
Small n	Asthma/COPD omics with $n < 100 \rightarrow$ unstable clusters; avoid “precision medicine” language

Wrong analysis

Mistake	Try $k = 2 \dots 10$ until clusters “look clinical”
Why it fails	Multiple testing on cluster definitions; best-looking k is overfit
Do instead	Prespecify k , use bootstrap stability (§11.8), report all k explored

Mistake	Title: “Discovery of two asthma endotypes” from k-means on $n = 80$
Why it fails	No external replication or prospective treatment validation
Do instead	“Exploratory marker-based clusters; validation planned”

Reporting template

Methods: Thirty blood markers were z -scored. Exploratory k-means clustering ($k = 2, 25$ random starts) was applied. Mean silhouette width and bootstrap item stability (200 resamples) summarised cluster separation. External validation was not performed.

Results: Two clusters were identified (mean silhouette = 0.25; mean item stability = 0.XX). Cluster profiles differed on M1-M5 (Figure). Agreement with hierarchical clustering was high in CASTOR (adjusted Rand index = 1.0); agreement with processing batch was low (technical confounding not detected).

Do not say: “validated endotype”, “definitive subtype”, “precision medicine target” (from one cohort).

R lab

```
source("R/examples/ch11_clustering.R")
```

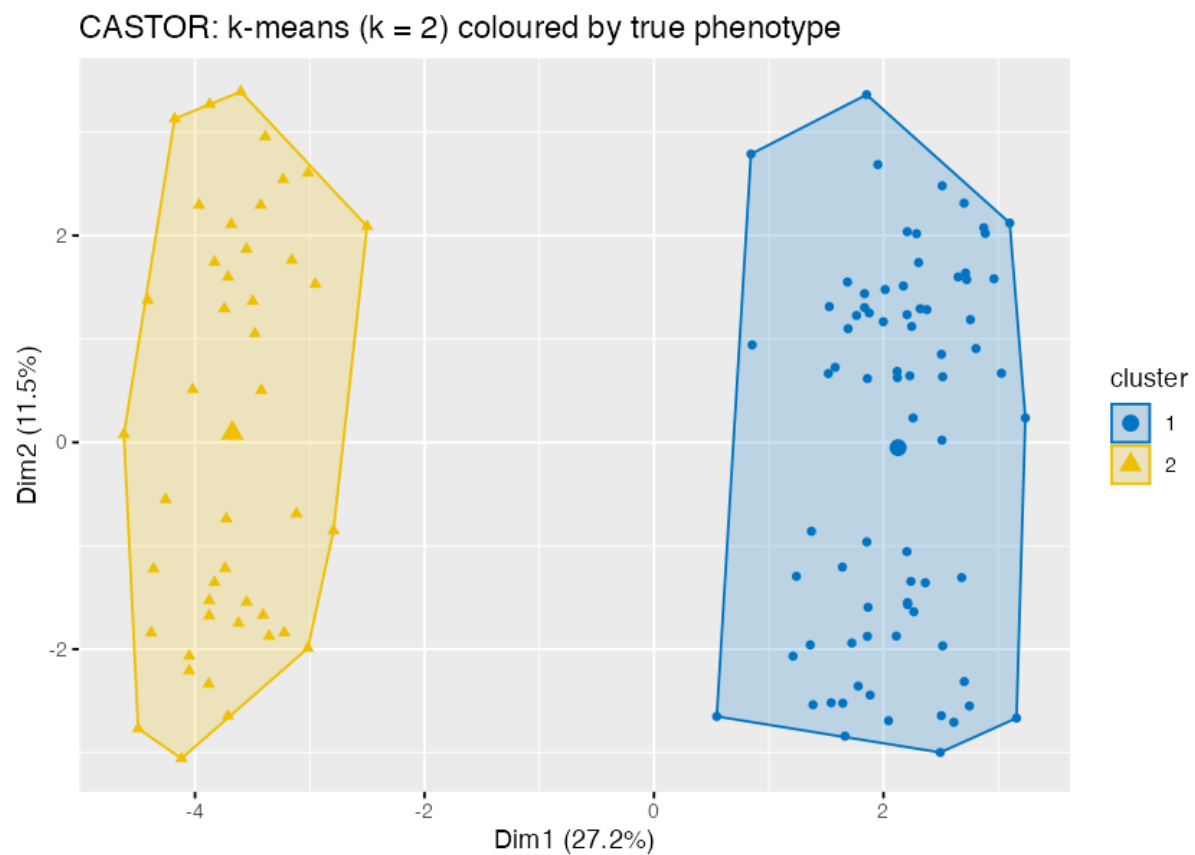


Figure 1: k-means clusters coloured by true phenotype (teaching only)

Clinician “so what?”

Would this change management today? Almost always **no** - until clusters replicate elsewhere and predict treatment response in a trial. Treat as a research finding for hypothesis generation.

Technique: Hierarchical clustering

Technique card

Answers	Nested hierarchy of patients via sequential merging (agglomerative)
Input	Scaled continuous features; distance matrix
Linkage	Ward.D2 (compact clusters), complete , average
R	<code>hclust(dist(X), method = "ward.D2");</code> <code>cutree(hc, k = k)</code>
Output	Dendrogram - cut at height to obtain k groups
When to use	Heatmaps; exploratory hierarchy; visualising nested structure
When NOT to use	Large n (memory $O(n^2)$); when you need soft/probabilistic membership
Does NOT prove	Optimal partition - early merges are irreversible

Dual interpretation

Plain language: build a family tree of patients from most similar to most different; cut the tree to get k groups.

Precise language: greedy agglomerative algorithm on pairwise distances; Ward minimises increase in within-cluster variance at each merge.

Clinician read: dendrograms are excellent for exploration and heatmaps; the cut height is subjective - agree k or validation criteria in advance.

Precise math (optional)

Ward linkage merges clusters A and B that minimise the increase in total within-cluster sum of squares when combined. Distances between clusters depend on linkage rule - **different rules** → **different trees**.

Caveats box

Caveat	Why it matters
Cut height	No unique "correct" cut - document k or height used
Linkage choice	complete vs Ward.D2 can yield different clinical stories
Irreversibility	An early wrong merge cannot be undone
CASTOR	With $n = 120$, hierarchical is feasible; omics studies with $n > 2000$ often cannot use full distance matrices
Batch	Dendrogram branches may reflect processing site, not biology

Wrong analysis

Mistake	Cut dendrogram until groups differ on outcome used to colour the plot
Why it fails	Circular - outcome informed the visual cut

Do instead

Prespecify k or cut height; validate on independent outcomes

Reporting template

Methods: Pairwise Euclidean distances on z -scored markers; agglomerative clustering with Ward.D2 linkage. Dendrogram cut at $k = 2$ for comparison with k-means.

Results: Hierarchical clustering yielded two groups (Figure: dendrogram). Adjusted Rand index vs k-means = 0.XX (partial agreement).

R lab

```
hc <- hclust(dist(X), method = "ward.D2")
cl_hc <- cutree(hc, k = 2)
plot(hc, labels = FALSE); rect.hclust(hc, k = 2, border = "red")
# See ch11_clustering.R - ch11_dendrogram.png
```

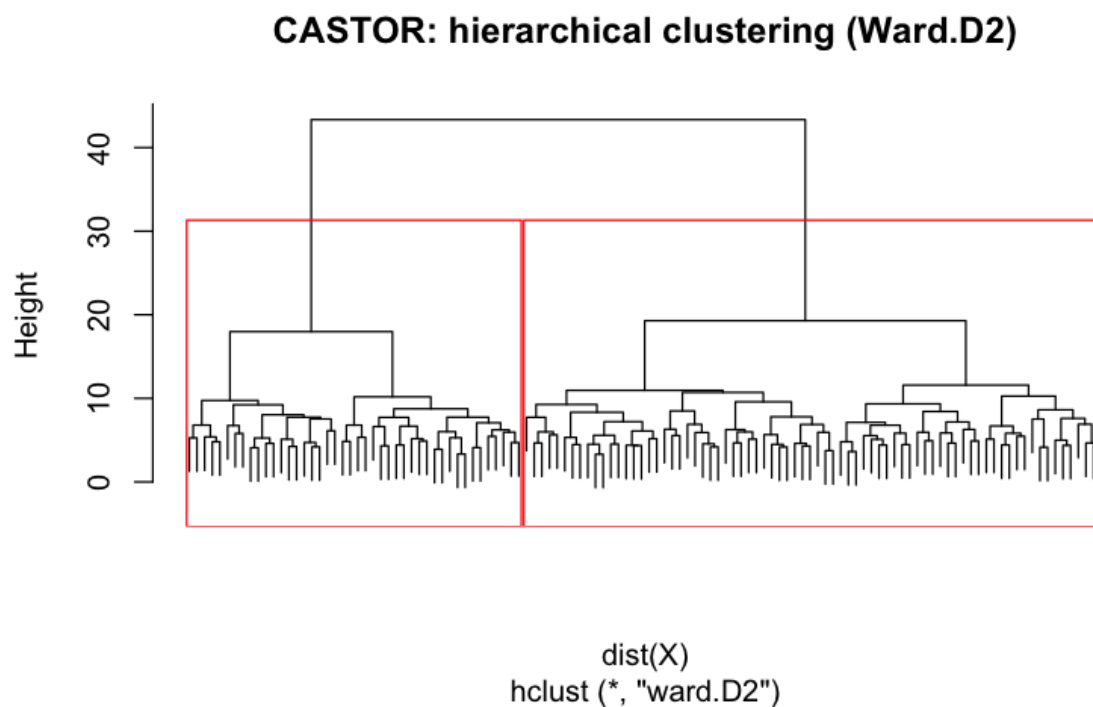


Figure 2: Ward.D2 dendrogram with $k = 2$ cut

Technique: PAM (k-medoids)

Technique card

Answers	Partition into k clusters using actual patients as centres (medoids)
Input	Scaled continuous features; distance matrix
R	<code>cluster::pam(X, k = 2)</code>
When to use	Outliers present; need a representative patient per cluster
When NOT to use	Very high p with small n ; when k-means speed is critical on huge n
Robustness	Medoids resist outliers better than k-means centroids
Does NOT prove	More “real” than k-means - still exploratory

Dual interpretation

Plain language: like k-means, but cluster centres are real patients, not averages.

Precise language: minimises sum of dissimilarities to nearest medoid (k-medoids); SWAP algorithm refines medoids.

Clinician read: medoid patients can be shown as “prototypes” in talks - but they are data points, not clinical archetypes until validated.

Caveats box

Caveat	Detail
Outliers	PAM is more robust than k-means but not immune
Speed	Slower than k-means for large n
CASTOR	Compare PAM vs k-means in method shootout (§11.10)
Interpretation	“Typical patient” narratives from medoids overstate certainty

Wrong analysis

Mistake	Present medoid case as “the eosinophilic asthma patient”
Do instead	“Representative of this exploratory cluster in this dataset”

Reporting template

Methods: PAM ($k = 2$) on z -scored markers as sensitivity to k-means.

Results: PAM agreed with k-means on XX% of patients (adjusted Rand index = ...).

Technique: Choosing k (silhouette)

Technique card

Metric	Silhouette width $[-1, 1]$; higher = better separation
Formula (patient i)	$(b_i - a_i) / \max(a_i, b_i)$ where a_i = mean distance to own cluster, b_i = mean distance to nearest other cluster
R	<code>cluster::silhouette(cl, dist(X));</code> compare mean width across k
Use	Compare candidate k values alongside domain knowledge
Does NOT prove	Correct k - geometry-dependent

Dual interpretation

Plain language: measures how well each patient fits their cluster vs the next-nearest cluster.

Precise language: average point-wise silhouette; favours compact, separated clusters.

Clinician read: a low silhouette (e.g. < 0.25) means groups overlap - do not force a crisp clinical narrative.

Caveats box

Caveat	Detail
Convex bias	Silhouette prefers spherical clusters
Best k on same data	Overfit if used to craft the publication story
Domain k	Sometimes $k = 2$ is prespecified (eosinophilic vs not) - silhouette is secondary
CASTOR	See <code>ch11_silhouette_k.png</code> for $k = 2 \dots 6$

Wrong analysis

Mistake	Pick k with highest silhouette, then test clinical outcomes on same data
Do instead	Bootstrap stability (§11.8); external cohort for k and labels

Reporting template

Results: Mean silhouette was 0.25 ($k = 2$), 0.15 ($k = 3$), ...; $k = 2$ was retained per prespecified plan [or: exploratory comparison only].

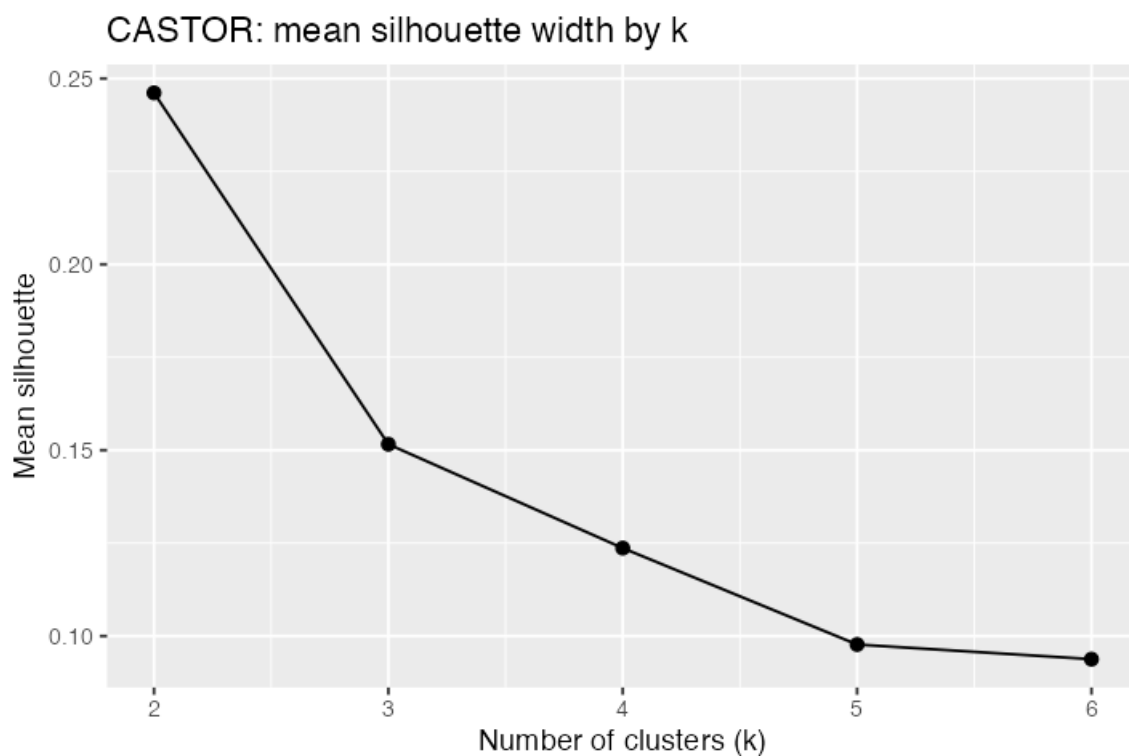


Figure 3: Mean silhouette width by k

The endotype claim ladder

This is the **governance framework** for respiratory phenotype papers. Most overclaiming happens by skipping rungs.

Rung	What you did	Language allowed	Example
1 - Exploratory	Cluster one cohort once	“Hypothesis-generating subgroups”	k-means on CASTOR markers
2 - Stable	Bootstrap / consensus / sensitivity analyses	“Internally stable clusters”	Item stability > 0.7
3 - Replicated	Same structure in independent cohort	“Reproducible subgroups”	U-BIOPRED-style replication
4 - Prognostic	Clusters predict exacerbations/LRTI not used in clustering	“Prognostic phenotypes”	Cluster at baseline → future exacerbations

5 - Pre- dictive of treat- ment	Differential treatment response in RCT	“Treatment-responsive endotype” (still cautious)	Biomarker- stratified trial
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You may not use “endotype” in a title until rung 4-5 is credibly addressed. Rung 1-2 alone → “exploratory clusters” ([Wenzel 2012](#)).

Respiratory phenotype vocabulary (clinical names vs clustering)

Clinicians use rich labels; clustering may or may not recover them:

Clinical construct	Common markers / features	Clustering caveat
Th2-high asthma	Eosinophils, FeNO, periostin	May map to one cluster - or overlap heavily
Eosinophilic vs neutrophilic	Blood/sputum cell counts	Requires validated thresholds, not just k-means
Pauci-granulocytic asthma	Low inflammation markers	Often a heterogeneous residual cluster
COPD	Low DLCO, low FEV1/FVC	Imaging/clinical variables often needed - not blood panel alone
emphysema-dominant		
ACOS overlap	Asthma + COPD features	Mixed cluster validated ACOS diagnosis

Key sentence: clinical phenotypes are **constructs**; unsupervised clustering is an **empirical partition**. Alignment must be demonstrated, not assumed.

Technique: Validation hierarchy

Level	Method	What it shows
Internal	Silhouette, bootstrap item stability, consensus clustering	Clusters not purely noise in this sample
External	Reproduce in independent cohort; transfer centroids/medoids	Structure generalises
Clinical	Prospective outcome or treatment interaction	Clusters matter for patients

Dual interpretation

Plain language: a cluster is only “real” if it shows up again and makes a difference clinically.

Precise language: unsupervised discovery requires out-of-sample replication and pre-specified validation endpoints - analogous to TRIPOD discipline for prediction models (Moons et al. 2015; Hennig 2007).

Statistician read: report stability coefficients, not only pretty heatmaps (Hennig 2007).

Wrong analysis

Mistake	Internal stability only → “validated endotype” in abstract
Why it fails	Stability generalisation; batch can be stable but technical
Do instead	External cohort + prespecified clinical endpoints

Technique: Bootstrap cluster stability

Technique card

Answers	How often is each patient assigned to a consistent cluster under resampling?
Method	Resample rows with replacement → re-cluster → item stability
R	See <code>bootstrap_item_stability()</code> in <code>ch11_clustering.R</code>
Report	Mean (and min) item stability across patients
When to use	Before naming subgroups in a paper
Does NOT prove	External validity

Reporting template

Methods: Bootstrap resampling ($B = 200$) with k-means ($k = 2$) at each draw; item stability = proportion of draws with same cluster assignment after **label alignment** (cluster numbers are arbitrary across runs) (Hennig 2007).

Results: Mean item stability = 0.78 (range 0.52-0.94). Low-stability patients flagged for sensitivity analysis.

Technique: Heatmaps and batch effects

Technique card

Answers	Visualise patient $\times$ marker matrix with joint clustering
Input	Scaled expression/abundance matrix
R	<code>pheatmap::pheatmap()</code> , <code>heatmap()</code> , or <code>ComplexHeatmap</code>
When to use	Omics exploration; supplement to quantitative validation
When NOT to use	Sole evidence for biological subgroups
Does NOT prove	Causal biology - batch can dominate colour patterns

Dual interpretation

Plain language: a coloured grid showing which markers are high/low in which patients.

Precise language: reorder rows/columns by hierarchical clustering for visual coherence; structure may reflect technical artefacts.

Clinician read: if colours align with processing site rather than diagnosis, stop - fix batch before interpreting biology.

Caveats box

Caveat	CASTOR teaching
Batch effects	<code>processing_batch</code> (SiteA/SiteB) shifts M26-M30 - check before claiming biology (McShane et al. 2011)
Double clustering	Row + column cluster simultaneously $\rightarrow$ pretty but hard to validate
Scale	Log-transform and scale omics before heatmaps

Wrong analysis

Mistake	Heatmap shows site-specific stripes $\rightarrow$ “two inflammatory endotypes”
Do instead	Batch correction or include batch in sensitivity; replicate externally

R lab

```
table(km$cluster, omics$processing_batch) # from ch11_clustering.R
```

Method shootout (CASTOR)

Run multiple algorithms on the **same scaled markers** and compare - humility is the point.

Comparison	Adjusted Rand index (illustrative)
k-means vs hierarchical	Partial agreement expected
k-means vs PAM	Often high
k-means vs <code>true_phenotype</code>	Teaching only - never available in practice
k-means vs <code>processing_batch</code>	High ARI = red flag (technical not biological)

```
source("R/examples/ch11_clustering.R") # prints shootout table
```

Interpretation: partial agreement between algorithms is normal. Perfect agreement with batch is a warning, not a discovery.

Cluster in PC space vs all markers

Chapter 10 reduced 30 markers to a few components. Clustering on **5 PCs** often:

- reduces noise,
- improves stability,
- matches full-marker clustering reasonably well.

```
# In ch11_clustering.R: ARI between k-means on 5 PCs vs all 30 markers
```

Wrong analysis : PCA on full data including outcome, then cluster on PCs, then test outcome - circular. Fit PCA on markers only; hold outcomes for external validation.

Modern alternatives (pointers, not full treatment)

When k-means is too brittle, escalate deliberately:

Method	Strength	Chapter
Latent class analysis (LCA)	Mixed discrete/continuous; probabilistic membership	Ch 11 extensions
Gaussian mixture models	Soft clustering; model-based k	Ch 11 extensions
Consensus clustering	Stability across algorithms and k	Mention in sensitivity
DBSCAN / HDBSCAN	Irregular shapes; noise points	When domain suggests

Rule: name the method in Methods; do not switch post hoc to the algorithm that “looks best” without disclosure.

CASTOR worked example (end-to-end)

Question: Are there marker-based subgroups in CASTOR?

Steps:

1. z -score M1-M30.
2. k-means $k = 2$, `nstart = 25`.
3. Silhouette for $k = 2 \dots 6$ (`ch11_silhouette_k.png`).
4. Bootstrap item stability ($B = 200$).
5. Compare k-means, hierarchical, PAM (shootout table).
6. Cluster profiles on M1-M5 (`ch11_cluster_profiles.png`).
7. Check `processing_batch` confounding.
8. Compare clustering on 5 PCs vs 30 markers.
9. Compare to `true_phenotype` (**teaching only**).

Claims allowed: “Two exploratory clusters with moderate silhouette and reasonable bootstrap stability; partial alignment with prespecified marker contrasts on M1-M5.”

Claims NOT allowed: “Validated COPD endotypes ready for stratified care.”

Results paragraph (template):

Among 120 participants with 30 blood markers, exploratory k-means ($k = 2$) identified two clusters (mean silhouette 0.25; mean bootstrap item stability 0.XX). Cluster profiles differed on M1-M5 (Figure). Agreement with hierarchical clustering was high (adjusted Rand index 1.0). Clusters were not aligned with processing site (adjusted Rand index 0.0); external validation and outcome linkage were not performed.

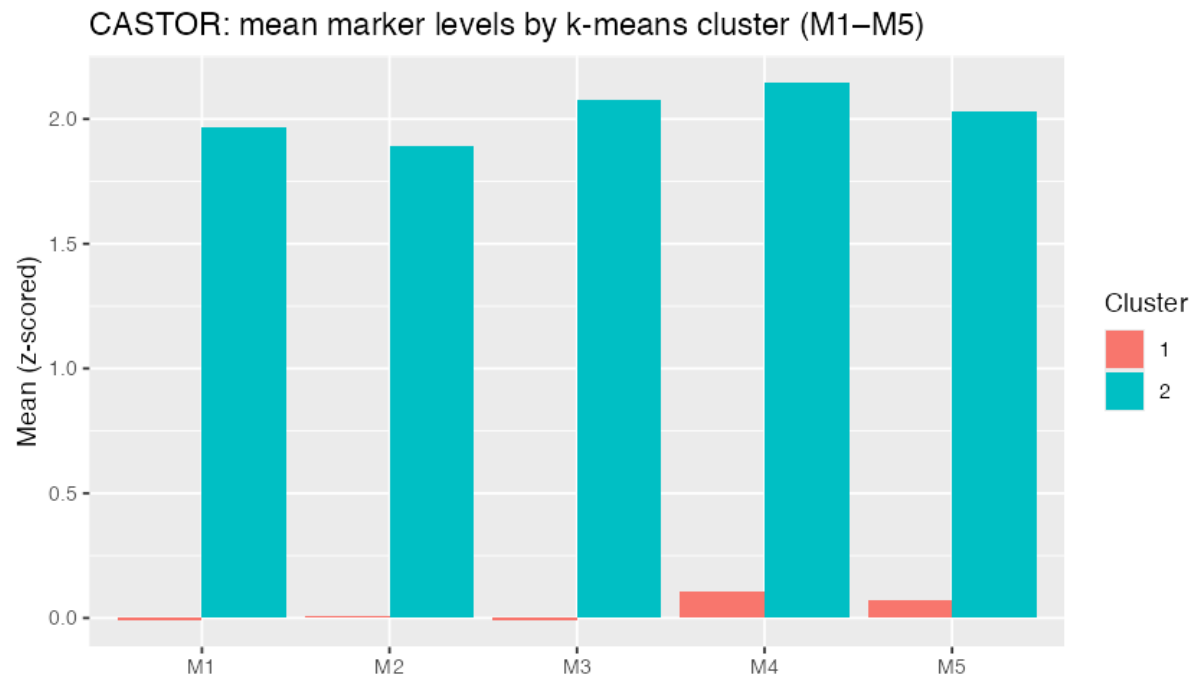


Figure 4: Mean marker levels by cluster (M1-M5)

Catalog of wrong analyses (respiratory-specific)

Wrong	Why it fails	Do instead
Cluster on markers and FEV1, then “predict” lung function	FEV1 defined groups	Cluster on markers only; test FEV1 externally
Enriched trial from exploratory clusters only	No prospective validation	Rung 5 ladder; prespecify in protocol
“Precision medicine” from $n = 80$	Underpowered for care pathways	Exploratory language; plan replication
Latent class on symptoms, then test same biomarkers	Same circularity	Hold out validation markers
Ignore <code>processing_batch</code>	Technical clusters	Batch correction / sensitivity
Endotype in title, rung 1 only	Overclaiming	“Exploratory subgroups”
k chosen post hoc	Overfit	Document all k tried; bootstrap stability

Explaining to patients and general readers

“We used computer methods to look for groups of patients with similar test results. This helps researchers form new hypotheses. It does **not** mean we have new diagnoses or that your treatment should change today. Further studies must check whether these groups behave differently over time or respond differently to medicines.”

Chapter summary

- Clustering is **exploratory unsupervised** learning - not classification, not causal inference.
- k-means, hierarchical, and PAM answer different robustness/structure needs; **compare them**.
- The **endotype claim ladder** governs language: exploratory → stable → replicated → prognostic → treatment-responsive ([Wenzel 2012](#)).
- **Bootstrap stability**, batch checks, and PC-space clustering are modern minimum standards before naming subgroups ([Hennig 2007](#)).
- Respiratory phenotype names (Th2-high, eosinophilic, etc.) are clinical constructs - demonstrate alignment, do not assume it.

Further reading

- Hennig, cluster stability assessment ([Hennig 2007](#))
- Wenzel, asthma phenotypes and endotypes ([Wenzel 2012](#))
- McShane et al., biomarker study reporting ([McShane et al. 2011](#))
- Jolliffe & Cadima, PCA (preprocessing for clustering) ([Jolliffe and Cadima 2016](#))

Exercises · [Solutions](#)

Next: [Chapter 12 - Integrated case studies](#)

Chapter 12: Integrated Case Studies

Part V: Discovery (core CASTOR capstone)

At a glance

Recurring cohort	CASTOR - full workflow
Format	End-to-end narratives with caveats, wrong analyses, reporting (template)
Cases	A: RCT FEV1 · B: Exacerbation logistic · C: PCA + clustering · D: CASTOR-HD (Ch 13–17) · E: Longitudinal + survival (Ch 18–19)
R	R/examples/ch12_case_*.R
Navigation	HANDBOOK_GUIDE · QUICK_REFERENCE · METHOD_MAP · REFERENCES

Learning objectives

1. Execute the full analysis pipeline from question to report.
2. Apply the [METHOD_MAP](#) to real CASTOR scenarios.
3. Write integrated Methods/Results paragraphs with appropriate citations.
4. Identify wrong analyses in published-style scenarios.

Prerequisites

Chapters 1–11.

Opening question

How do the CASTOR analyses from Chapters 3–11 fit together in three publishable-style case reports?

This chapter is the **capstone**: same workflow as [HANDBOOK_GUIDE](#), three complete narratives, and explicit limits on what each analysis proves.

Master workflow (every CASTOR analysis)

Step	Action	Chapter
1	Clinical question (one sentence)	1
2	Estimand + population	1-2
3	Describe sample (Table 1, plots)	3
4	Choose method via QUICK_REFERENCE	4-11
5	Fit model / test with R script	R/examples
6	Diagnostics + sensitivity	7-8
7	Report estimate + CI + limitations	8
8	State what was not proven	All

Reporting frameworks by design: CONSORT (RCT) ([Schulz et al. 2010](#)); STROBE (cohort) ([Elm et al. 2007](#)); TRIPOD (prediction) ([Moons et al. 2015](#)); biomarker discovery ([McShane et al. 2011](#)).

Case study A: Randomised trial: FEV1 comparison

Clinical question

Does intervention improve mean FEV1 at 12 weeks compared with standard care in CASTOR?

Estimand

Difference in mean FEV1 (litres) at 12 weeks: intervention — standard care, intention-to-treat population.

Design

Parallel RCT; independent groups; continuous outcome ([Schulz et al. 2010](#)).

Analysis path

Step	Done
Table 1 by group	Ch 3
Histogram / QQ FEV1	Ch 3
Welch t-test (prespecified primary)	Ch 4 (Welch 1947)
ANCOVA sensitivity (<code>spirometry_trial.csv</code>)	Ch 4-5
Bootstrap CI	Ch 8 (Efron and Tibshirani 1993)
Power note (if negative)	Ch 4, 8

```
source("R/examples/ch12_case_a_trial.R")
```

Results template

Among 400 participants, mean FEV1 was 3.85 L (SD 0.64) in intervention and 3.76 L (SD 0.64) in standard care. The mean difference was 0.09 L (95% CI -0.04 to 0.21 ; Welch t, $p = 0.20$). Results are inconclusive with respect to a prespecified MCID of 0.10 L ([Cazzola et al. 2008](#)).

Three-reader interpretation

Reader	Takeaway
Statistician	CI includes null; non-significant p does not prove equivalence (Harrell 2015)
Clinician	Cannot claim benefit or futility without MCID/power context
General	Trial did not show clear average improvement; more data or larger effect needed

Caveats

Single time point; CASTOR synthetic; no site clustering modelled. Spirometry per ATS/ERS standards ([Graham et al. 2019](#)).

Wrong analysis (Case A)

Mistake	Correct
“No significant difference → treatments equal”	CI + equivalence margin if that is goal
Compare post-BD FEV1 one arm, pre-BD other	Same spirometry protocol (Graham et al. 2019)
Skip Table 1	Always describe first (Schulz et al. 2010)

What Case A does NOT prove

Causal effect in broader population; long-term FEV1 decline; symptom benefit.

Case study B: Observational cohort: exacerbation risk

Clinical question

Is smoking associated with 1 exacerbation in 12 months after adjusting for age, FEV1 % predicted, and prior exacerbations?

Estimand

Adjusted odds ratio for smoking (binary exposure).

Design

Observational cohort; binary outcome; logistic regression ([Elm et al. 2007](#); [Hosmer et al. 2013](#)).

Analysis path

Step	Done
Event count / EPV check (~18/350)	Ch 6 (Harrell 2015)
Logistic regression (prespecified covariates)	Ch 6
Marginal risks (<code>emmeans</code>)	Ch 6
Firth sensitivity if sparse	Ch 6 (Firth 1993)
No stepwise selection	Ch 7

```
source("R/examples/ch12_case_b_exacerbation.R")
```

Results template

In 350 patients (18 exacerbation events), prior exacerbation count was associated with higher odds of a new event (OR 1.70, 95% CI 1.12 to 2.59). FEV1 % predicted OR 0.95 per 1% (95% CI 0.91 to 0.99). Smoking OR was imprecise (95% CI included 1.0). Low event count limits precision; Firth sensitivity similar (supplementary).

Three-reader interpretation

Reader	Takeaway
Statistician	EPV low; wide CIs; associative not causal (Elm et al. 2007)
Clinician	Prior history matters most; smoking signal uncertain here
General	Past flare-ups predict future ones; smoking link not confirmed in this sample

Caveats

Unmeasured confounding (adherence, SES); exacerbation definition ([Hurst et al. 2010](#)); low events.

Wrong analysis (Case B)

Mistake	Correct
lm on 0/1 outcome	Logistic (Hosmer et al. 2013)
OR reported as “30% higher risk”	Marginal RD or RR model
Stepwise 20 predictors, 18 events	Prespecified model (Harrell 2015)
“Smoking causes exacerbation”	“Associated with”

What Case B does NOT prove

Causal effect of smoking; prediction model performance (see Case C / Ch 9).

Case study C: Multi-marker panel: PCA + clustering

Clinical question

Is there exploratory evidence of patient subgroups in a 30-marker panel?

Estimand

None confirmatory - descriptive structure only.

Design

Cross-sectional marker panel; unsupervised methods ([Jolliffe and Cadima 2016](#); [Hennig 2007](#)).

Analysis path

Step	Done
Scale markers	Ch 10
PCA scree + PC1/PC2 plot	Ch 10
k-means $k = 2$, silhouette	Ch 11
Bootstrap stability + batch check	Ch 11
Compare to <code>true_phenotype</code> (teaching only)	Ch 11
Plan external validation	Ch 11

```
source("R/examples/ch12_case_c_phenotypes.R")
```

Results template

PCA: PC1 explained 27% of variance; loadings highest on M1-M5 ([Jolliffe and Cadima 2016](#)). k-means ($k = 2$) yielded clusters with silhouette 0.25. Cluster profiles differed on marker weights (Figure). Analysis was exploratory; clusters were not validated in an independent cohort or against clinical outcomes ([McShane et al. 2011](#); [Wenzel 2012](#)).

Three-reader interpretation

Reader	Takeaway
Statistician	Hypothesis-generating; multiplicity uncontrolled
Clinician	Do not change care based on these clusters
General	Possible groups in data - needs replication

Caveats

p » n risk in real omics; batch effects ([McShane et al. 2011](#)); CASTOR has simulated structure.

Wrong analysis (Case C)

Mistake	Correct
“Two validated endotypes”	Exploratory clusters (Wenzel 2012)
Test 30 markers individually then cluster	Pre-specify discovery framework
Use outcome to pick k	Unsupervised k selection + stability (Hennig 2007)
PCA on unscaled markers	Scale

What Case C does NOT prove

Biological subtypes; treatment response groups; diagnostic categories.

Case study D: CASTOR-HD discovery bridge (Ch 13–17)

Clinical question

In the CASTOR-HD extension, which molecular and immune readouts support a coherent discovery story from proteomics through confirmation assays?

Estimand

Not a single causal estimand — this is a **staged discovery pipeline**: (1) controlled false discovery among proteins; (2) batch-robust shortlist; (3) participant-level immune phenotypes; (4) confirmed antibody binding among screen hits.

Design

Observational case-control with multi-plate proteomics, RNA-seq, flow summaries, and replicate antibody screens ([McShane et al. 2011](#)).

Analysis path

Step	Chapter	Done
Per-protein DE + BH FDR	13	Top table, volcano
Batch PCA + sensitivity	14	Overlap check, discovery count with/without batch
Flow proportions (participant n)	15	Batch-adjusted cell-type effects
Screen hits → PPV + tiers	16	Prespecified threshold, Tier 1 clones
Integrated narrative + optional elastic net	17	ch17_integrated_castor_hd.R

```
source("R/00_setup.R")
source("R/examples/ch17_integrated_castor_hd.R")
```

Results template

We tested ~1000 proteins (linear models with batch/plate covariates; BH FDR). Batch overlap was assessed by PCA and group $\times$ batch tables (Figure). After batch adjustment, N proteins had $q < 0.05$; rankings were compared with and without batch covariates. Participant-level flow cytometry ($n = \dots$) showed ... Monocyte proportions differed ... (batch-adjusted). Antibody screen hits at prespecified threshold ... had PPV ... among confirmation assays; Tier 1 clones (3/3 replicate rankings) were ... Analysis was discovery-stage; external validation is required ([McShane et al. 2011](#)).

Three-reader interpretation

Reader	Takeaway
Statistician	Multiplicity controlled per modality; batch and stability audited
Clinician	No diagnostic or treatment claims from this pipeline alone
General	Hypothesis list for targeted follow-up — not a validated signature

Wrong analysis (Case D)

Mistake	Correct
“Proteomics signature validates endotype”	Separate discovery from confirmation; external cohort
Ignore batch because FDR “fixes” it	FDR does not fix confounding (Ch 14)
Report cell-level p-values as $n = \text{cells}$	Participant-level inference (Ch 15)
Screen threshold chosen after seeing PPV	Prespecify threshold; report sensitivity curve

What Case D does NOT prove

Causal mechanisms; clinical utility; transportability; antibody therapeutic potential.

Continue: [Ch 17 integrated pipeline](#) · [HIGH_DIM_REPORTING_TEMPLATES](#)

Case study E: Longitudinal FEV1 + time to exacerbation

Clinical question

In the CASTOR extension cohort, (1) does intervention improve the **FEV1 trajectory** over 52 weeks, and (2) is smoking associated with **shorter time to first exacerbation** under one-year follow-up?

Estimands

1. **Longitudinal:** difference in mean FEV1 trajectory (intervention vs standard), weeks 0–52, randomised extension population — slope and level from mixed model.
2. **Survival:** hazard of first exacerbation comparing smokers vs non-smokers, adjusted for FEV1 % predicted, therapy, and age (associational cohort estimand).

Design

- **Part 1:** parallel-group trial extension with four scheduled spirometry visits per participant (`longitudinal_spirometry.csv`).
- **Part 2:** observational time-to-event cohort with administrative censoring at 365 days (`time_to_exacerbation.csv`) ([Elm et al. 2007](#)).

Analysis path

Step	Chapter	Done
Spaghetti plot + visit counts	18	QC trajectory data
Mixed model <code>fev1 ~ weeks * group + (1 patient_id)</code>	18	Coefficient table + fitted means
Sensitivity: week-52 cross-section vs mixed	18	<code>ch18_sensitivity_mixed_vs_fixed.cs</code>
Kaplan–Meier by smoking + log-rank	19	<code>ch19_km_by_smoking.png</code>
Cox PH + Schoenfeld check	19	HR table + PH test
Integrated Case E summary	12	<code>ch12_case_e_summary.csv</code>

```
source("R/00_setup.R")
source("R/examples/ch12_case_e_longitudinal_survival.R")
```

Results template

Longitudinal (RCT extension): Among $n = \dots$ participants ($\dots$ visits), FEV1 trajectories differed by treatment (Figure). A linear mixed model with random intercepts estimated a week $\times$ intervention interaction of $\dots$ L per week (95% CI $\dots$). A sensitivity analysis using only week-52 FEV1 yielded a different standard error, illustrating pseudo-replication risk. **Survival (cohort):** During 365 days of follow-up, $\dots$ exacerbations occurred. Kaplan–Meier curves separated by smoking (log-rank $p = \dots$). The adjusted Cox hazard ratio for smoking was $\dots$ (95% CI $\dots$). Proportional hazards diagnostics: $\dots$ ([Harrell 2015](#); [Elm et al. 2007](#)).

Three-reader interpretation

Reader	Takeaway
Statistician	Correct units: patients for trajectories; events + censoring for survival
Clinician	Trajectory answers “lung function over time”; survival answers “how soon exacerbation”
General	Two related but distinct questions — do not merge into one headline

Wrong analysis (Case E)

Mistake	Correct
Pool all FEV1 visits as independent n	Mixed model with patient random effect
Use binary 12-month exacerbation only	Time-to-event with censoring
Claim smoking “causes” faster exacerbation without design	Associational Cox + confounding discussion (Ch 21)
Report HR without event counts	Table of events, censored, person-time

What Case E does NOT prove

That improving FEV1 trajectory prevents exacerbations (different endpoints); causal effect of smoking; transportability to other healthcare systems.

Continue: [Ch 20 missing data](#) · [Ch 21 causal inference](#)

Cross-case comparison

Case	Question type	Method	Inference?	Key reference
A	Treatment effect	Welch t / ANCOVA	Yes (RCT)	(Schulz et al. 2010)
B	Risk factor	Logistic	Associational	(Elm et al. 2007)
C	Structure	PCA + k-means	Exploratory only	(McShane et al. 2011)
D	Multi-omics discovery	DE + batch + flow + screen	Discovery only	(McShane et al. 2011)
E	Trajectory + time-to-event	Mixed model + Cox	RCT part + associational survival	(Harrell 2015 ; Elm et al. 2007)

Handbook synthesis checklist

Before submitting any respiratory paper, verify:

- ☐ Clinical question and estimand stated
- ☐ Table 1 and missingness ([Schulz et al. 2010](#); [Elm et al. 2007](#))
- ☐ Method matches outcome type ([QUICK_REFERENCE](#))
- ☐ Effect size + 95% CI ([Harrell 2015](#))
- ☐ Event count / n
- ☐ Sensitivity analysis or limitation noted
- ☐ No causal overclaim from observational/exploratory work
- ☐ Reproducible R script

Extended methods (now in this handbook)

- Longitudinal FEV1 → [Ch 18](#)
- Time-to-exacerbation → [Ch 19](#)
- Missing data / MICE → [Ch 20](#)
- Causal inference → [Ch 21](#)

Closing

The **core path (Ch 1–12)** is complete when you can run the CASTOR pipeline yourself — including **Case E** (longitudinal + survival). The **advanced discovery path (Ch 13–17)** extends CASTOR-HD; **Part VIII (Ch 18–21)** completes the single-volume handbook for repeated measures, time-to-event, missing data, and causal framing ([Harrell 2015](#); [Shmueli 2010](#)).

Replace author details, swap CASTOR for your cohort, and obtain statistical and clinical review before publication.

Alternatives & extensions (how these cases would change)

If your real study has...	Case change	Where it's covered
Multiple follow-up visits	Case A → Case E longitudinal	Ch 18 , Case E
Time to first exacerbation	Case B → Case E survival	Ch 19 , Case E
Many more predictors (omics)	Case C/D: penalization + validation	Ch 7, 9–11, 13–17
Clustered multi-centre design	SEs must account for centre	Ch 18
Clinical decision thresholds	Add decision-curve / net benefit	Ch 9 extensions
Proteomics + flow + screens	Case D → Ch 17 pipeline	Ch 13–17

Further reading

- Full bibliography: [REFERENCES.md](#) and `references.bib`
- Harrell, *Regression Modeling Strategies* ([Harrell 2015](#))
- CONSORT / STROBE / TRIPOD reporting ([Schulz et al. 2010](#); [Elm et al. 2007](#); [Moons et al. 2015](#))

Exercises · [Solutions](#)

End of core path (Ch 1–12) — continue with [Ch 13](#) (CASTOR-HD) or [Ch 18](#) (longitudinal/survival).

Part VI: High-dimensional biology and discovery

This part extends the handbook workflow to datasets where (p > n) and technical variation can dominate biology:

- **Proteomics** (e.g. Olink-style panels): missingness near limits of detection, plate effects, multiplicity (FDR).
- **RNA** (gene counts): count models, library size, normalization, differential expression, FDR.
- **Flow cytometry**: summaries vs per-cell data, drift, gating vs automated phenotyping.
- **Discovery screens** (antibodies): replicate agreement, hit calling, confirmation, ranking stability.

The goal is not to memorise methods. It is to make defensible claims:

1. **What changed (effect size)?**
2. **How uncertain is it (CI / stability)?**
3. **How many chances did we give ourselves to be wrong (FDR / multiplicity)?**
4. **Could this be technical (batch / plate / run)?**
5. **Does it generalise (honest validation / confirmation)?**

Recurring datasets: the CASTOR-HD synthetic files in `data/` (see [Ch 2](#) and [RECURRING_COHORT](#)).

Chapter 13: Differential analysis and false discovery rate (omics)

Part VI: High-dimensional biology and discovery

At a glance

Recurring datasets	data/proteomics_olink_like.csv, data/rnaseq_counts.csv
Format	Technique cards + Caveats + Wrong analysis + Reporting (template)
Primary goal	Estimate per-feature effects and control false discoveries
Core methods	per-feature models, Benjamini–Hochberg FDR, volcano plot (descriptive), sensitivity checks
R	R/examples/ch13_differential_fdr.R
Templates	HIGH_DIM_REPORTING_TEMPLATES

Learning objectives

1. Understand why “($p < 0.05$)” becomes meaningless when you test 1000+ features.
2. Report **effect sizes and uncertainty**, not a list of “significant proteins”.
3. Use FDR control (BH) and interpret what it means in plain language.
4. Recognise when differential results are likely batch/plate artefacts.

Prerequisites

Chapter 8 (reporting and multiplicity) and Chapters 10–11 (high-dimensional intuition).

Opening question (CASTOR-HD)

Which proteins (or genes) differ between cases and controls - and how many “discoveries” should we expect to be false if we act on them?

In omics, the two most common failure modes are:

1. **Overclaiming**: treating 50 nominal ($p < 0.05$) hits as biology.
2. **Underreporting uncertainty**: listing q-values without effect sizes.

Technique: Per-feature differential analysis + BH FDR

Technique card

Answers	Which features differ between groups, by how much, with multiplicity control?
Outcome type	Many continuous features (proteins) or many count features (genes)
Design	Usually independent groups; may include covariates and batch variables
Data required	group label, optional covariates, optional batch/plate/run
Assumptions	Model is approximately correct per feature; independence not required for BH (but strong dependence can distort)
Effect measure	mean difference or log2 fold-change (plus CI)
Multiplicity	BH FDR (q-values) for the family of tests
R	<code>p.adjust(p, method="BH")</code> after per-feature p-values
When to use	discovery with controlled false positives; prioritisation for follow-up
When NOT to use	mechanistic proof; “endotype” claims without replication
Does NOT prove	causation; diagnostic utility; pathway truth; transportability

Dual interpretation

Plain language: we tested many markers and adjusted the results so that only a small fraction of the “discoveries” are expected to be false.

Precise language: for each feature (j) we fit a model to estimate (μ_j) for group; we then applied Benjamini–Hochberg to control the expected false discovery proportion among features called significant.

Clinician read: FDR protects you from a “shopping list of biomarkers” that will not replicate. It does not tell you which marker is clinically useful.

Caveats box

Caveat	Why it matters in respiratory research
Batch and plate effects	“Significant proteins” often track run day/site rather than disease
Missingness near LOD (proteomics)	cases can have fewer detectable proteins; naive imputation can create signal
Normalization choices	log transform, scaling, library-size normalization change rankings

Caveat	Why it matters in respiratory research
Small (n), huge (p)	effect estimates are noisy; ranks are unstable without shrinkage
Confounding	smoking/age/therapy correlate with both disease and omics; adjust or stratify
Interpretation inflation	top q-values are not “most important” without effect size and uncertainty

Wrong analysis

Mistake	Run 1000 t-tests and report every ($p < 0.05$) as biology
Why it fails	expected false positives 50 even if no true differences
Do instead	control FDR (BH), report effect sizes + uncertainty, plan validation

Mistake	Volcano plot + “top hits” without mentioning preprocessing and batch
Why it fails	rankings are highly sensitive to normalization and technical drift
Do instead	show QC (PCA by batch), include batch as covariate, run sensitivity

Catalog of wrong analyses (omics discovery)

Use this as a pre-submission audit. If any row describes your workflow, rewrite.

Wrong analysis	Why it fails	Do instead
List of hits without effect sizes (“Protein X significant, $q = \dots$ ”)	You cannot judge clinical or biological importance	Report effect + 95% CI + q-value; rank by effect + stability, not only q
Nominal p-value hunting (top 20 p-values)	Multiple testing turns this into a false-positive generator	Use BH FDR; show how many tests were run
“Volcano proof” (figure implies causality)	Volcano plots are descriptive; they do not validate biology	Present as prioritisation; specify validation/confirmation plan
Batch ignored	Many top hits are technical; FDR cannot fix confounding	Diagnose (Ch 14), include batch/plate/run covariates, run sensitivity
Batch “corrected” without checking confounding	Overcorrection can erase biology or manufacture differences	Show PCA by batch and group; if confounded, state identifiability limits
LOD missingness imputed as a constant (e.g., replace NA with 0)	Creates artificial group differences if detection differs by group/batch	Prefer sensitivity: complete-case vs simple within-feature imputation; report missingness by group

Wrong analysis	Why it fails	Do instead
CPM-style normalization treated as neutral (RNA)	If some genes truly change, others can look changed due to compositional constraints	Interpret “global shifts” cautiously; prefer methods designed for counts and composition; validate with spike-ins / external data
Imputation done before train/test split (prediction)	Information leakage inflates performance	Perform all preprocessing within resampling (see Ch 9 mindset)
Genes/proteins treated as independent evidence	Correlation means “50 hits” may be 1 pathway signal	Summarise at pathway/module level as a secondary interpretation; keep per-feature results for transparency
“No hits, therefore no biology”	Low power can hide large effects; FDR can be conservative	Report effect size distributions and uncertainty; discuss detectable effect sizes
Single-cohort “signature” claim	Signatures are unstable without external validation	Treat as hypothesis; validate on external cohort or held-out batch

Reporting template

Use Template A in [HIGH_DIM_REPORTING_TEMPLATES](#).

Differential analysis + FDR: two omics examples (synthetic)

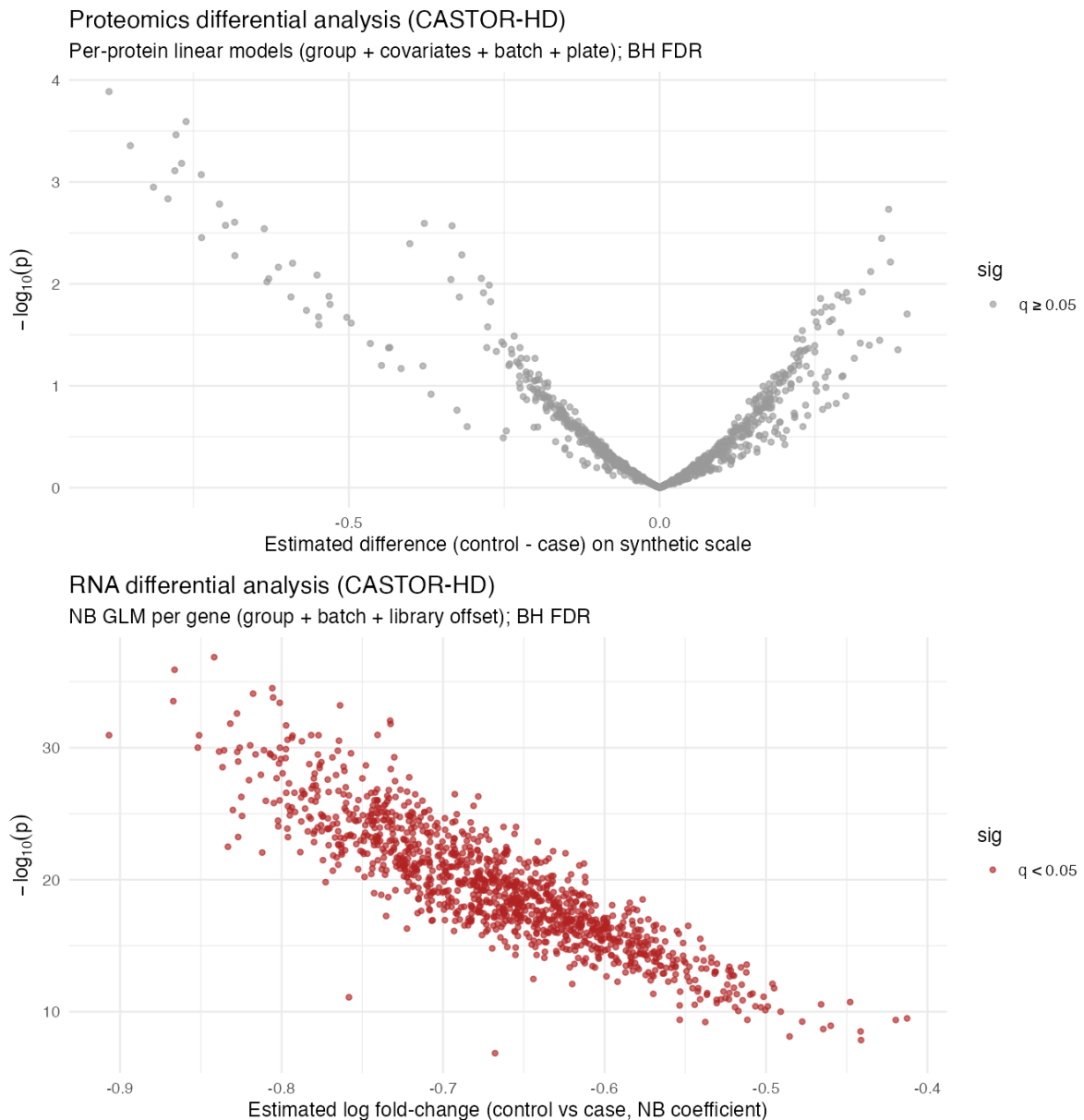


Figure 1: Volcano plots: proteomics and RNA differential analysis (BH FDR)

Alternatives & extensions (choose by goal)

Situation	Primary approach	Why
Very small (n), want stable ranking	shrinkage / empirical Bayes (conceptually)	stabilises noisy effects
Strong batch structure	handle batch explicitly (Ch 14)	prevents technical discoveries
Many missing values (LOD)	sensitivity: complete-case vs simple imputation	avoids imputation-created signal

Goal is prediction, not discovery

nested CV + calibration (Ch 9, [Ch 17](#)) prevents leakage and overfit

Mini-lab: sparse PCA pointer (exploratory)

When (p > n), dense PCA loadings are noisy. For exploratory views only, try sparse PCA (`elasticnet` / `PMA` packages) with a prespecified sparsity penalty — never treat as confirmatory DE.

```
# Teaching pointer (not run in ch13 script):  
# library(PMA)  
# sparse.pca(scale(prot_matrix), K = 2, para = c(0.5, 0.5))
```

Mini-lab: LOD missingness check (proteomics)

```
# After source("R/examples/ch13_differential_fdr.R") - or inline:  
prot <- readr::read_csv(file.path(paths$data, "proteomics_olink_like.csv"), show_col_types =  
prot %>% mutate(miss = rowMeans(is.na(dplyr::select(., starts_with("Prot_"))))) %>%  
  ggplot(aes(group, miss, fill = group)) + geom_boxplot() + theme_minimal()
```



Figure 2: Proteomics missingness by group (LOD-style)

R lab: Differential analysis on CASTOR-HD

Script: `R/examples/ch13_differential_fdr.R`

1) Proteomics (Olink-like): per-protein model + BH FDR + top-table

Your minimum output should be a table with:

- **feature** (protein)
- **effect** (control - case or log2FC)
- **95% CI**
- **p-value** and **q-value**
- **n used** (after missingness)

The script writes a copy-ready top-table to volume-01/tables/.

```
source("R/00_setup.R")
library(tidyverse)

prot <- readr::read_csv(file.path(paths$data, "proteomics_olink_like.csv"), show_col_types = FALSE)
prot %>% count(group)
```

2) RNA-seq counts: negative binomial differential expression

For RNA counts, use a **count model** (negative binomial), not a Gaussian model on raw or log-transformed counts. The teaching workflow fits one NB model per gene with an **offset for library size**:

```
library(MASS)
rna <- readr::read_csv(file.path(paths$data, "rnaseq_counts.csv"), show_col_types = FALSE)
# Per gene: glm.nb(count ~ group + batch + offset(log(library_size)))
# Then BH FDR across genes - see R/examples/ch13_differential_fdr.R
```

Teaching note: CASTOR-HD synthetic RNA includes a global expression shift, so many genes can pass FDR in this demo. In real studies, interpret discovery counts alongside MA plots and batch QC.

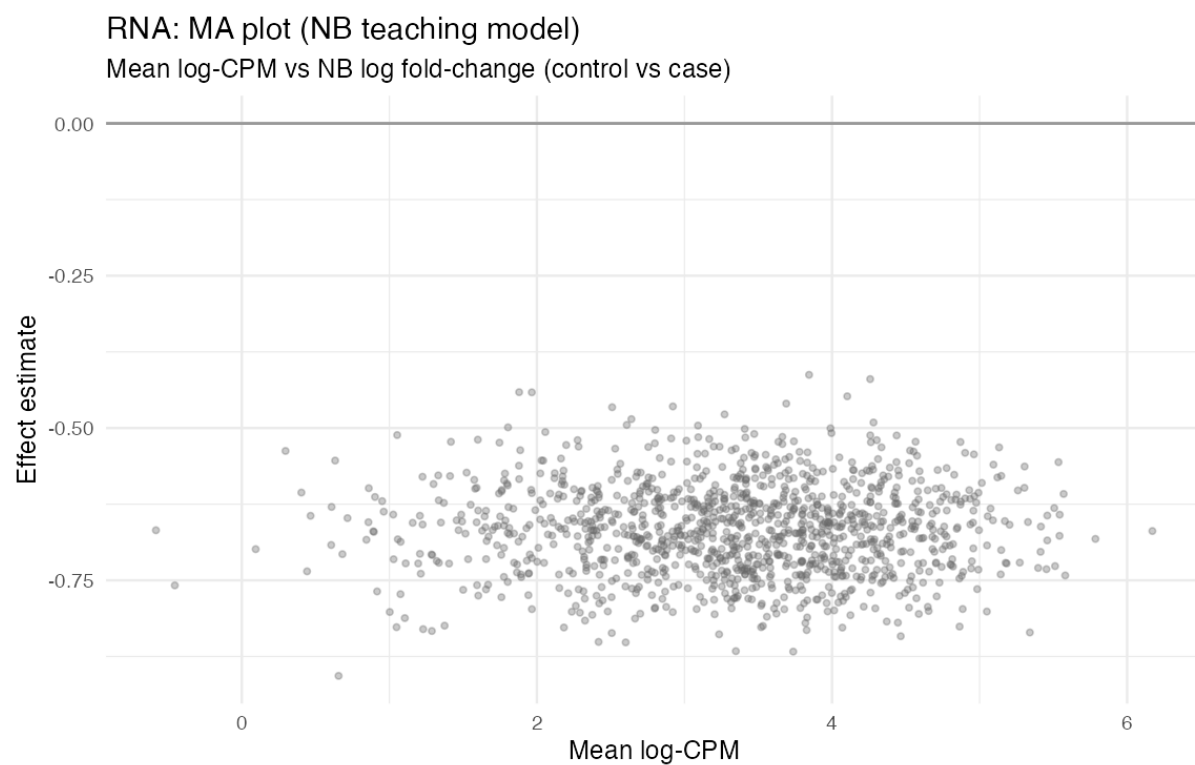


Figure 3: RNA MA plot (NB log-fold-change teaching output)

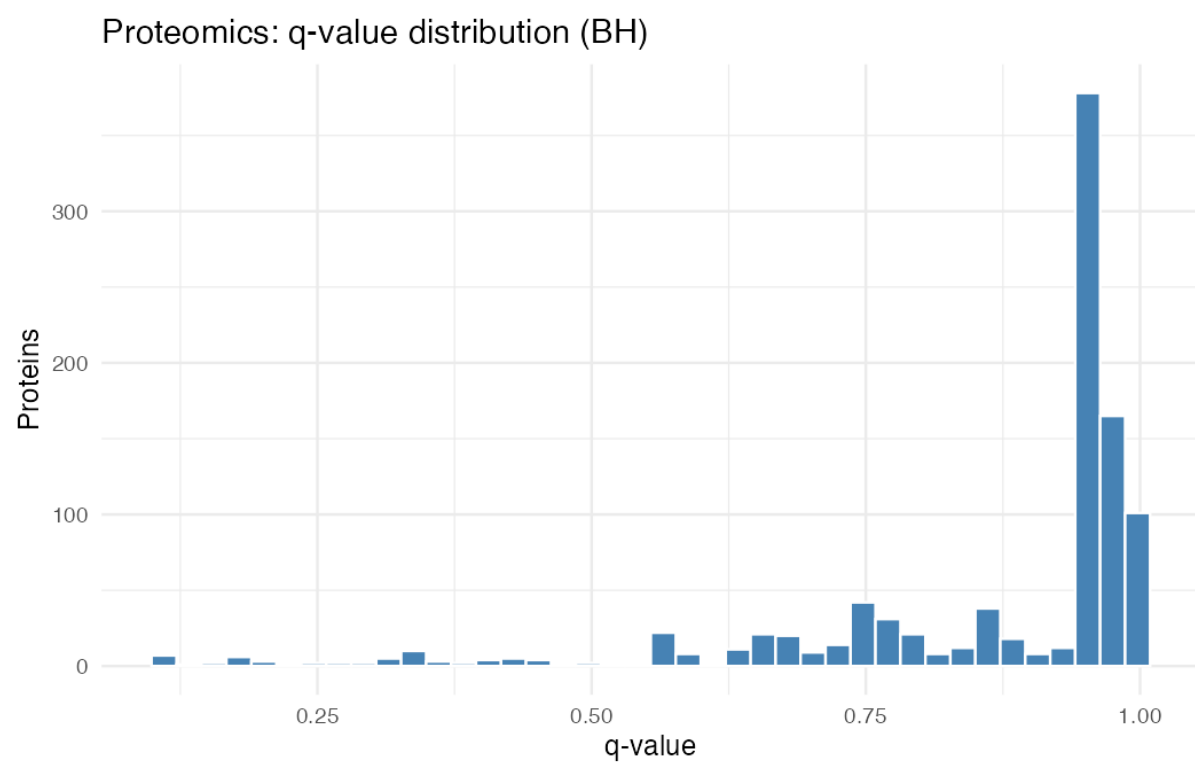


Figure 4: Proteomics q-value distribution (BH)

Sensitivity checklist (minimum)

- Run the differential analysis twice:
 - **with** batch/run as covariate
 - **without** batch/run
- Compare overlap of top 50 features and discuss stability.

Niche figures to include (recommended)

- **Missingness by group** (proteomics): if cases have more missing, LOD is part of the story.
- **MA plot** (RNA): average abundance vs effect to spot mean–variance artefacts.
- **q-value distribution**: a flat distribution suggests “mostly null”; a spike near 0 suggests signal.

These are generated by the chapter script and saved to `volume-01/figures/`:

- `ch13_proteomics_missingness_by_group.png`
- `ch13_proteomics_qvalue_hist.png`
- `ch13_rnaseq_ma_plot.png`

Exercises · Solutions

E13.1 Why is testing 1000 proteins at $\alpha = 0.05$ a problem even if only 50 are “significant”?

E13.2 What three columns must appear in a defensible DE/DA top table?

E13.3 When would you distrust a volcano plot as “proof” of biology?

Applied

1. Run `source("R/examples/ch13_differential_fdr.R")`.
2. Open `volume-01/tables/ch13_proteomics_top_table.csv` and interpret the top 5 rows (effect + CI + q).
3. Compare discovery counts with vs without batch adjustment (Ch 14 link).
4. Write a Results paragraph using [HIGH_DIM_REPORTING_TEMPLATES](#) Template A.

Chapter 14: Batch effects, plate effects, and unwanted variation

Part VI: High-dimensional biology and discovery

At a glance

Recurring datasets	data/proteomics_olink_like.csv, data/rnaseq_counts.csv, data/flowcytometry_summary.csv
Format	Technique cards + Caveats + Wrong analysis + Reporting (template)
Core idea	diagnose technical structure first, then decide how (or whether) to adjust
Primary tools	QC plots, PCA by batch, batch as covariate, sensitivity analysis
R	R/examples/ch14_batch_effects.R
Figures	FIGURE_INDEX - ch14_*.png

Learning objectives

1. Recognise when “discoveries” are likely technical.
2. Use **PCA/QC plots** to detect batch/plate/run effects.
3. Distinguish **adjustable batch** from **confounded batch** (not identifiable).
4. Apply three defensible strategies: covariate adjustment, adjustment inside resampling (prediction), or redesign.
5. Avoid overcorrection when batch removal would erase real site differences.

Prerequisites

Ch 10 (PCA intuition), Ch 13 (differential analysis logic), Ch 8 (reporting).

Opening question

Are my “signals” biological - or did I just rediscover the lab calendar?

Batch effects are not a niche issue; they are the default in real omics and cytometry:

- site differences (sample handling)

- run day / instrument drift
- plate effects
- processing batches

If batch aligns with group (e.g., all cases measured later), adjustment cannot create truth from confounding. At that point the honest conclusion is: **the study is not identifiable without stronger design.**

Worked mini-case: when adjustment is valid vs not identifiable

The chapter script compares two designs side by side.

Case A (CASTOR-HD proteomics): batch overlaps both groups

In `proteomics_olink_like.csv`, cases and controls appear in **both** batches. Batch is measurable technical variation, not a perfect proxy for disease status.

Design feature	CASTOR-HD proteomics (teaching)
Group $\times$ batch overlap	Both groups in Batch1 and Batch2
PCA pattern	Batch shifts scores, but groups are not separated by batch alone
Covariate adjustment	Defensible as sensitivity: $y \sim \text{group} + \text{batch} + \text{plate} + \text{covariates}$
Interpretation	Group effects are conditional on measured technical variables ; still need replication

Clinician read: we can attempt to separate biology from lab process because both patient types were measured under multiple conditions.

Case B (synthetic confounding): batch == group

Imagine a study where **all controls** were run on Batch1 and **all cases** on Batch2.

Design feature	Confounded mini-case
Group $\times$ batch overlap	None (perfect confounding)
PCA pattern	Batch and group occupy the same direction in PC space
Covariate adjustment	Not identifiable: group and batch are redundant
Interpretation	You cannot claim a group effect “after adjusting for batch”

Clinician read: if disease status and lab day are inseparable, the analysis cannot tell you whether the signal is biology or processing.

Decision rule (use before any “ComBat” or covariate adjustment)

1. **Tabulate** group × batch (and plate/run if available).
2. If any cell has **zero** samples for a group-batch combination you need, treat confounding as likely.
3. **Plot** PCA (or another QC embedding) colored by batch **and** group.
4. Fit a **sensitivity model** with and without batch; if the main conclusion flips, report instability.
5. If batch and group are perfectly confounded, stop claiming adjusted group effects; redesign or collect new data.

Technique: Diagnose batch with PCA + simple QC

Technique card

Answers	Is technical structure large enough to distort inference?
Outcome type	many features (proteins/genes/cytometry markers)
Design	any, but batch variables must be recorded
Data required	feature matrix + batch labels (plate/run/day/site)
Assumptions	PCA is descriptive; large PCs tracking batch suggests trouble
Effect measure	none (diagnostic)
R	<code>prcomp(X, scale.=TRUE)</code> + plot colored by batch
When to use	always before DE/ML on high-dimensional data
When NOT to use	never “skip” this step
Does NOT prove	that correction fixes the problem

Dual interpretation

Plain language: we checked whether samples cluster by lab process rather than disease.

Precise language: we examined whether leading principal components correlate with batch variables; if so, batch explains a meaningful portion of variance.

Clinician read: if batch drives the main variation, a “biomarker panel” is probably not real.

Caveats box

Caveat	Why it matters
PCA is descriptive	it does not tell you what to adjust, only what to worry about
Confounding	if batch == group, adjustment is guesswork
Overcorrection	removing batch can remove real biology if biology differs by site

Caveat	Why it matters
Leakage	correction must be inside resampling when doing prediction
Missingness	LOD missingness can mimic batch clustering (proteomics)
Reporting	you must say what batch variables existed and what you did

Wrong analysis

Mistake	“ComBat everything” without checking confounding or sensitivity
Why it fails	correction can manufacture group differences or erase biology
Do instead	show PCA by batch and group, run sensitivity, and state limits

Catalog of wrong analyses (batch effects)

Wrong analysis	Why it fails	Do instead
Skip batch QC because “the statistician will fix it”	Technical structure becomes “biology” in DE/ML outputs	Always tabulate group $\times$ batch and plot PCA by batch
Adjust for batch when batch == group	Model is not identifiable; coefficients are arbitrary	Report confounding; redesign (balance batches) or external validation
Report only the batch-adjusted model	Hides instability and post hoc storytelling	Show with vs without batch; report overlap of top features
Use ComBat on the full matrix before splitting data (prediction)	Leakage: test set informs correction	Fit correction inside training folds only
Treat plate as “random noise”	Plates are structured batch effects with known labels	Include plate/run as covariate or blocking factor when measured
Interpret PCA separation as proof	Embeddings are descriptive; separation can be driven by one outlier feature	Check feature loadings; validate with targeted QC metrics
Remove “batch genes” then test group	Feature selection using batch information contaminates group tests	Prespecify feature sets or use models that include batch explicitly
Claim transportability across sites after single-site ComBat	Correction does not create external validity	Validate on held-out site/batch or new cohort
Ignore missingness patterns by batch (proteomics)	LOD missingness can track batch and mimic disease signal	Plot missingness by group and batch (see Ch 13 figure)
Single sentence: “batch corrected” in Methods	Not auditable; reviewers cannot assess identifiability	State variables, model structure, and sensitivity outcome

Reporting template

Batch and plate variables were recorded. We tabulated group $\times$ batch overlap and examined PCA colored by batch and group. Per-feature models included batch/plate as covariates where identifiable. Sensitivity analyses with and without batch adjustment [changed/did not change] the number of discoveries at FDR $q < 0.05$. [If confounded: group effects were not identifiable after adjusting for batch.]

Technique: Adjust batch by including it as a covariate

This is the most defensible “first-line” strategy for **inference** when batch is not perfectly confounded with group.

Technique card

Answers	Group effect after accounting for measured technical variables
R (concept)	<code>lm(feature ~ group + batch + plate + covariates)</code>
When to use	DE/DA when batch is measured and partially overlaps both groups
When NOT to use	batch fully confounded with group; batch unmeasured
Does NOT prove	transportability; that unmeasured artefacts are absent

Dual interpretation

Plain language: we estimated the group difference while allowing for known lab differences.

Precise language: group effect is conditional on batch/plate indicators in a per-feature linear model; interpretability requires overlap in the group $\times$ batch table.

Clinician read: this is a reasonable attempt to separate signal from process - but only if both groups were measured across batches.

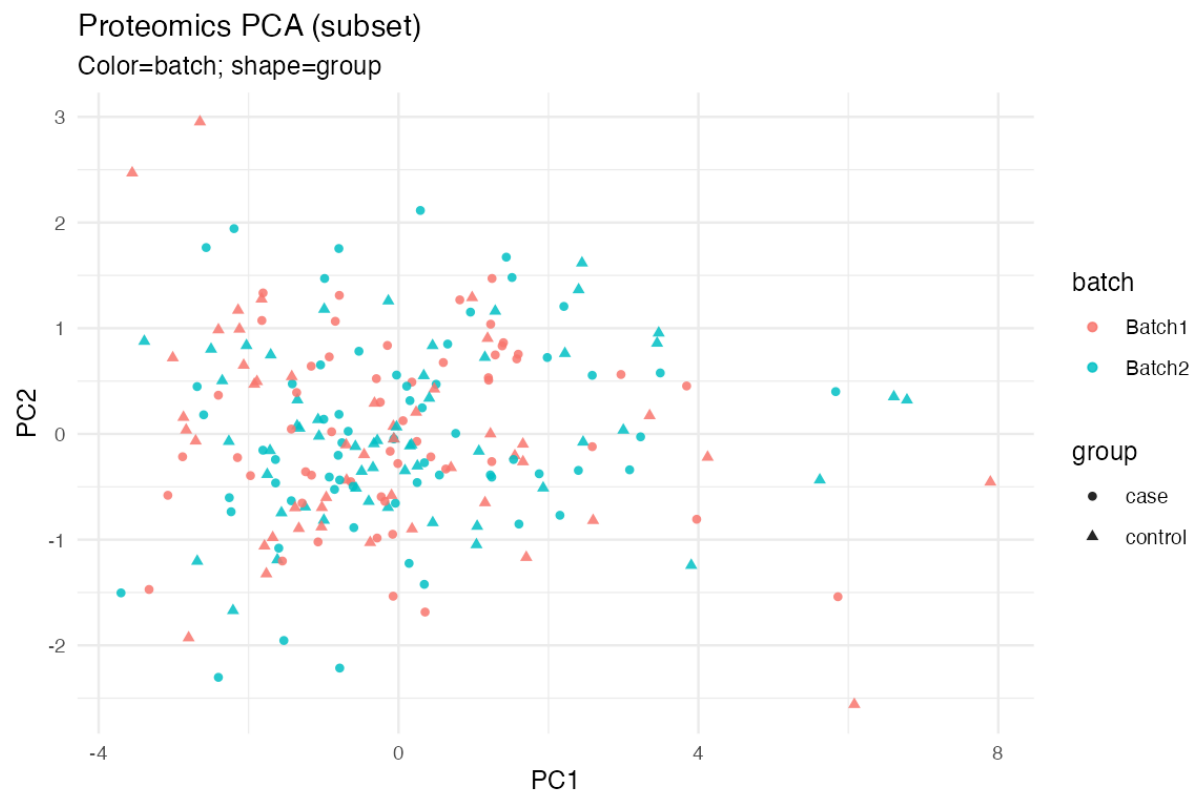


Figure 1: PCA colored by batch (proteomics subset)

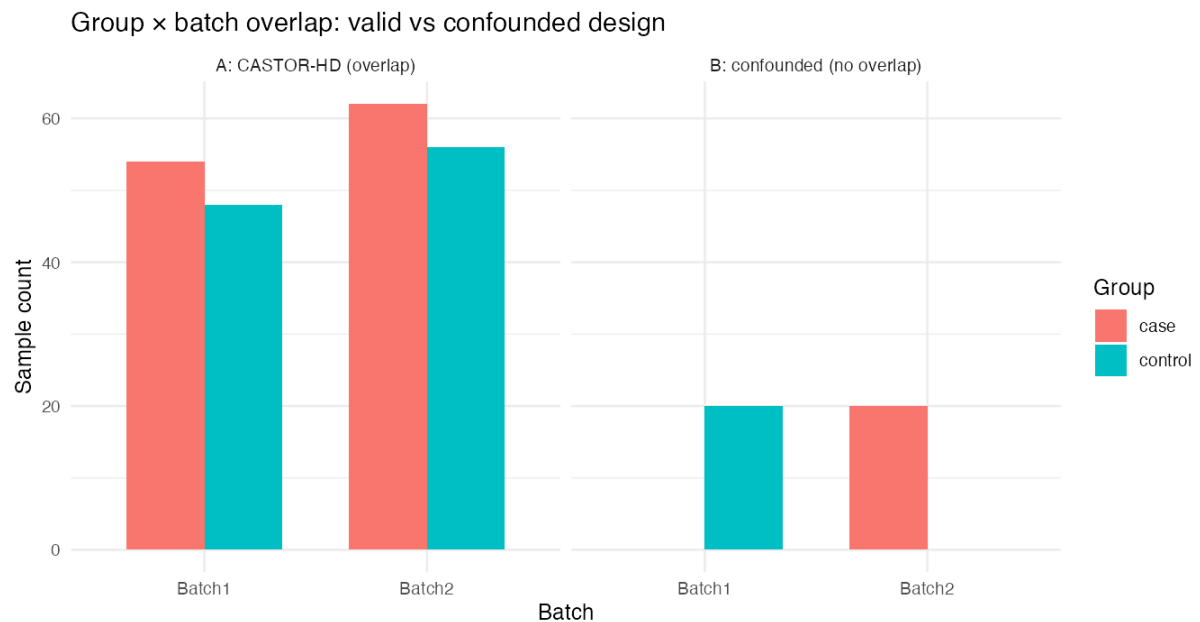


Figure 2: Valid overlap vs confounded group $\times$ batch design

Alternatives & extensions (when covariates are not enough)

Situation	Primary approach	Notes
Strong unknown technical variation	RUV / surrogate variable analysis (conceptual)	requires negative controls or careful assumptions
Prediction goal	correction inside CV folds	prevents leakage (Ch 9 mindset)
Multi-site biology is real	do not “remove site”	site may be effect modifier, not nuisance
Batch confounded	blocked redesign / prospective balancing	statistical fix cannot replace design
Flow cytometry drift	include run day + control beads in model	see Ch 15

Mini-lab: ComBat is not automatic

ComBat and similar tools can help when batch is measured and **not fully confounded** with group. Always run the overlap + PCA checks in this chapter first. If `table(group, batch)` has empty cells, report non-identifiability instead of forcing correction.

R lab: Batch effects on CASTOR-HD

Script: `R/examples/ch14_batch_effects.R`

The script produces:

- PCA diagnostics (`ch14_pca_proteomics_batch.png`, `ch14_pca_proteomics_plate.png`)
- Sensitivity bar chart (`ch14_batch_sensitivity_discoveries.png`)
- **Mini-case figures:**
 - `ch14_group_batch_overlap.png` (valid overlap vs confounding)
 - `ch14_pc1_variance_explained.png` (how much PC1 tracks batch vs group)
- Summary table: `volume-01/tables/ch14_batch_mini_case_summary.csv`

```
source("R/00_setup.R")
library(tidyverse)

prot <- readr::read_csv(file.path(paths$data, "proteomics_olink_like.csv"), show_col_types =
table(prot$group, prot$batch)
```

Sensitivity rule (non-negotiable)

If the *existence* of your main result depends on whether batch is included, say so explicitly and treat the result as **unstable** until replication / better design.

Niche figures (recommended)

- **Group $\times$ batch overlap plot:** the fastest confounding check.
- **PC1 variance explained by batch vs group:** quantifies whether technical structure dominates the leading axis.

Sensitivity to batch adjustment

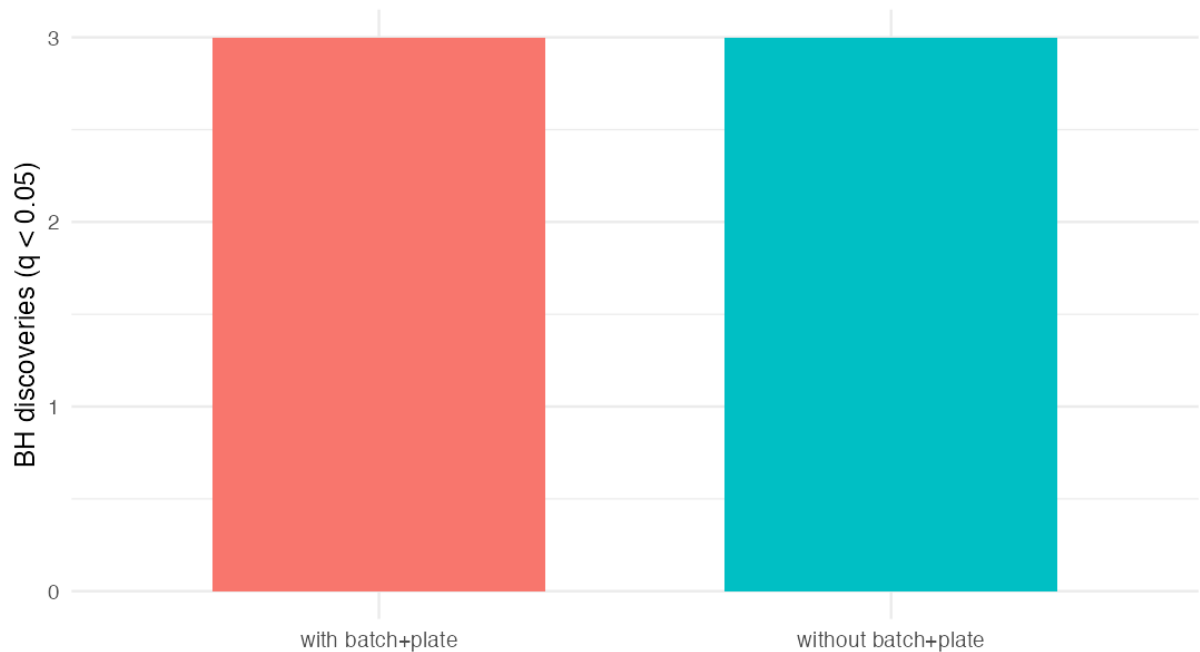


Figure 3: Discoveries with vs without batch adjustment

PC1 variance explained (proteomics subset)

If batch dominates PC1, technical structure is large

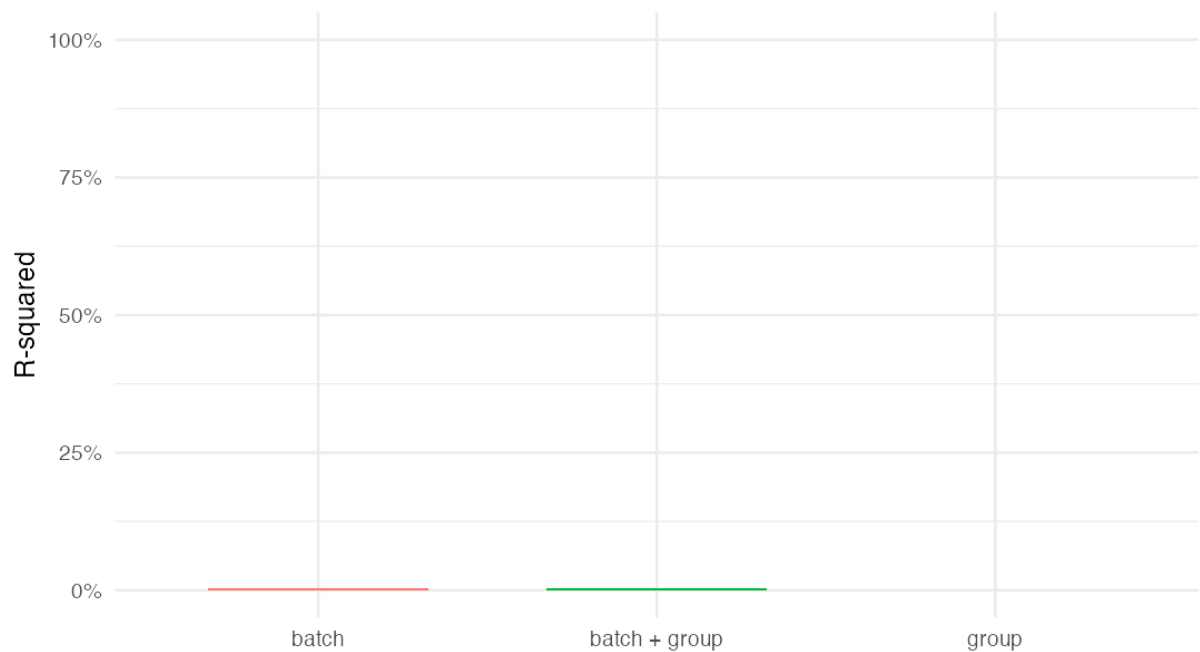


Figure 4: PC1 variance explained by batch vs group

Exercises · Solutions

E14.1 When is including batch as a covariate defensible?

E14.2 What does perfect confounding of batch and group imply for identifiability?

E14.3 Why is ComBat before train/test split a leakage problem?

Applied

1. Run `source("R/examples/ch14_batch_effects.R")`.
2. Interpret `ch14_group_batch_overlap.png` for Cases A and B.
3. Read `volume-01/tables/ch14_batch_mini_case_summary.csv`.
4. Draft a Methods sentence on batch handling for a proteomics paper.

Chapter summary

- Batch effects are expected; **diagnosis comes before correction**.
- Covariate adjustment is valid when **group and batch overlap**.
- Perfect confounding makes group effects **not identifiable** after adjusting for batch.
- Always report **sensitivity** (with vs without batch) and avoid “batch corrected” as a black box.

Chapter 15: Flow cytometry - from summaries to phenotyping

Part VI: High-dimensional biology and discovery

At a glance

Recurring datasets	data/flowcytometry_summary.csv, data/flowcytometry_cells_toy.csv
Format	Technique cards + Caveats + Wrong analysis + Reporting (template)
Main decisions	(1) unit of analysis, (2) compositional constraints, (3) drift/batch, (4) inference vs visualization
R	R/examples/ch15_flow_cytometry.R
Figures	FIGURE INDEX - ch15_*.png
Templates	HIGH_DIM_REPORTING_TEMPLATES

Learning objectives

1. Choose the correct **unit of analysis**: participant-level summaries vs cells.
2. Recognise **compositional constraints** when proportions sum to 1.
3. Compare groups on cell-type proportions with interpretable effect measures and drift adjustment.
4. Use embeddings/clustering as **descriptive** tools unless validated.
5. Avoid **pseudo-replication** and “UMAP as evidence” claims.

Prerequisites

Ch 4 (comparisons), Ch 5 (linear models), Ch 13 (multiplicity), Ch 14 (batch/drift logic).

Opening question

Do cases differ in immune composition - and can we say it without turning UMAP into “evidence”?

Flow cytometry invites a common error: using an appealing picture (embedding) as a statistical result. This chapter separates:

- **Inference** (estimates + uncertainty on **participant-level** summaries) from
- **Visualization** (embeddings, clustering, gating diagnostics).

Worked mini-case: cells vs participants (pseudo-replication)

The chapter script compares two analysis units on the same CASTOR-HD toy per-cell file.

Case A (correct): participant-level summaries

Step	What you do
1	Gate or classify cells within each participant
2	Compute one proportion per participant per cell type (e.g., % monocytes)
3	Compare groups using n = number of participants
4	Adjust for drift (run day/batch) when measured
n in CASTOR-HD	120 participants in <code>flowcytometry_summary.csv</code>

Clinician read: “Among 120 people, cases had a higher median monocyte fraction” - this is the claim you can defend.

Case B (wrong): per-cell analysis as if independent

Step	What people do (wrong)
1	Pool all cells across participants
2	Run a test comparing case vs control cells
3	Report n = thousands of cells and tiny p-values
Why it fails	Cells from the same person are correlated; SEs are far too small

Clinician read: “We studied 6,000 cells” sounds impressive but does **not** mean 6,000 independent patients.

Teaching demonstration (script output)

The script fits the same group comparison twice:

1. **Participant model:** `lm(logit(prop_Mono) ~ group + batch)` on `flowcytometry_summary.csv` (n 120)
2. **Pseudo-replication model:** `lm(CD14 ~ group)` on pooled cells (n thousands)

The pseudo-replication model will show a much smaller p-value and misleading precision. **Do not report Case B as confirmatory inference.**

Decision rule

Question	If yes →
Is each row a cell ?	Summarise to participant first (unless using a proper mixed model with random intercepts for patient)
Do proportions sum to 1 ?	Treat compositional structure explicitly or interpret one cell type at a time with caution
Is drift (run day) recorded?	Plot by batch; include in model (Ch 14 mindset)
Are you showing UMAP/t-SNE?	Label as descriptive ; support with participant summaries

Technique: Participant-level summary analysis (default)

Technique card

Answers	Are cell-type proportions / marker medians different between groups?
Outcome type	proportions (0-1), continuous marker summaries
Design	independent groups; can adjust for covariates and drift
Unit of analysis	one row per participant
Effect measure	difference in mean proportion; logit-scale difference; median difference
R	<code>lm(logit(p) ~ group + batch)</code> (simple) or beta regression (advanced)
When to use	primary reporting; transparent; supports CI and covariates
When NOT to use	claiming new cell types from one embedding
Does NOT prove	mechanistic cell identity; causal immune pathways

Dual interpretation

Plain language: we compared the fraction of each major immune population between groups.

Precise language: we modelled participant-level proportions, accounting for drift, and reported effect sizes with uncertainty.

Clinician read: these summaries are interpretable (e.g., “higher monocyte fraction”), but they are not a replacement for validated immunophenotyping.

Caveats box

Caveat	Why it matters
Compositional constraints	proportions sum to 1; changes are not independent
Drift and instrument settings	day effects can mimic disease differences
Gating subjectivity	manual gating is a measurement process; report it
Rare populations	instability and zero inflation; avoid overinterpretation
Multiple comparisons	many cell types/markers -> multiplicity (Ch 13)
Unit of analysis	do not treat cells as independent patients

Wrong analysis

Mistake	Treat 6000 cells as $n = 6000$ independent observations
Why it fails	pseudo-replication; SEs become meaningless
Do instead	summarise per participant (or use mixed models with care)

Mistake	Name clusters “new cell types” from one run of UMAP + k-means
Why it fails	embeddings distort distances; clustering is unstable
Do instead	call them “patterns”, report stability, validate with markers/replication

Catalog of wrong analyses (flow cytometry)

Wrong analysis	Why it fails	Do instead
Pooled-cell t-test / logistic on cells	Pseudo-replication inflates n and shrinks p-values	One summary per participant; $n = \text{participants}$
Report cell count as sample size	“ $n = 50,000$ events” is not $n = 50,000$ people	Report participants and events per participant separately
Ignore compositional constraint	Increasing one population mechanically affects others	Interpret one type at a time; consider compositional methods if core question
Compare proportions without drift check	Run-day drift mimics group differences	Plot by batch; adjust when identifiable (Ch 14)
UMAP separation = proof of disease subgroup	Embedding is descriptive and unstable	Show participant summaries + marker validation
Cherry-pick one cell type after scanning many	Multiplicity without FDR	Prespecify populations or control FDR (Ch 13)
Manual gating undisclosed	Not reproducible; not auditable	Document gating strategy or algorithm + QC

Wrong analysis	Why it fails	Do instead
Rare population overclaim	0.1% subsets are noisy with small n	Report uncertainty; avoid mechanistic language
Mixed model without random intercept for patient	Still treats cells as exchangeable within patient incorrectly if misspecified	Patient random effect when modelling cells directly
“Immune age” from one cohort	Signature may track batch/site	External validation; drift diagnostics

Reporting template

Use Template C in [HIGH_DIM_REPORTING_TEMPLATES](#).

Flow cytometry was performed on [panel]. Cells were gated [manual/algorithmic strategy]. For each participant we computed the proportion of [cell types] and median marker intensities. Group comparisons used participant-level models adjusting for run day/batch. Per-cell embeddings (PCA/UMAP) were used for **visual QC only**.

Technique: Compositional structure (proportions sum to 1)

When you measure five cell-type proportions, they are **not independent**: if monocytes go up, something else must go down.

Technique card

Answers	How should we interpret changes in one population given the whole?
When to use simple approach	Prespecified primary population (e.g., % regulatory T cells)
When to extend	Core question is about overall immune rebalancing
Alternatives	compositional data analysis (log-ratio transforms), Dirichlet models (advanced)
Does NOT prove	which population “caused” the shift

Practical handbook rule: for most respiratory papers, prespecify 1-3 populations for primary inference and treat the rest as exploratory.

Technique: Per-cell visualization (secondary)

Embeddings (UMAP/t-SNE) and clustering can help you **see** structure and check gating, but they are not, by themselves, confirmatory inference.

Rule: if you show an embedding, also show participant-level summaries that support the claim.

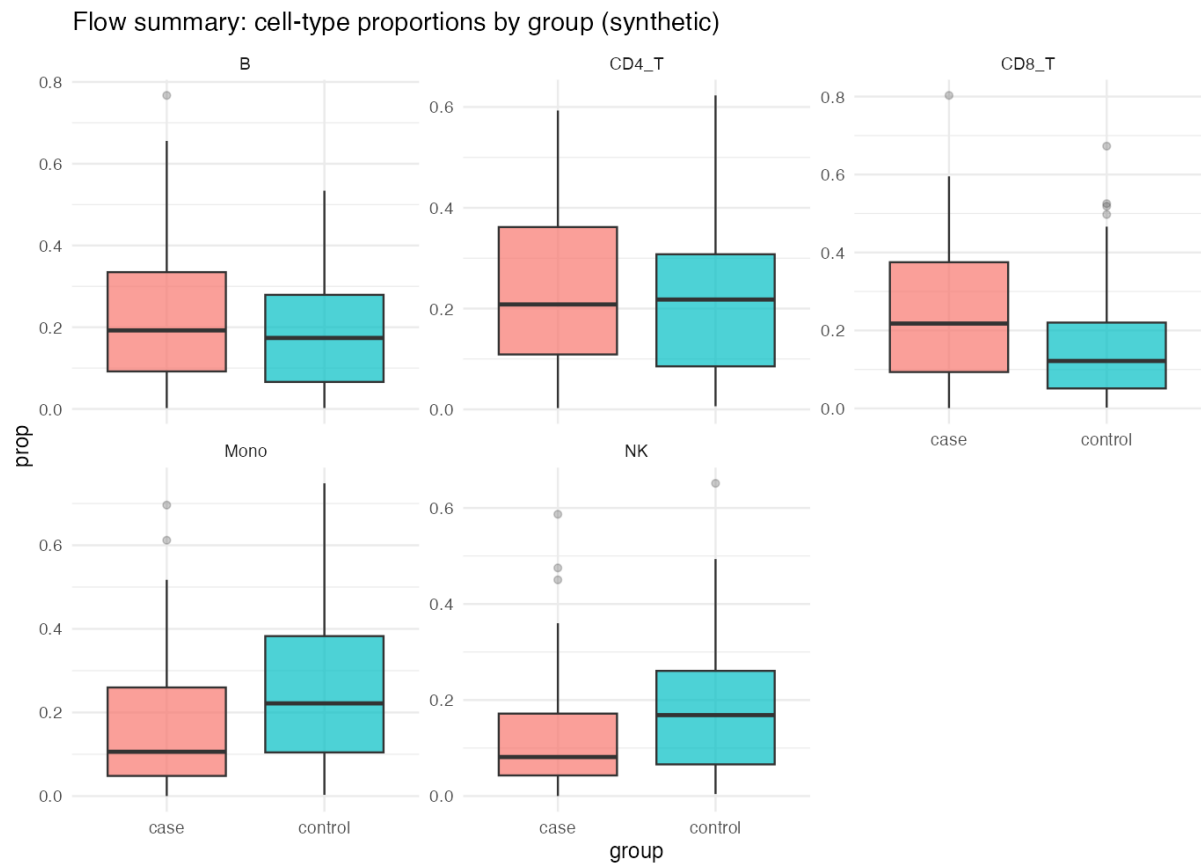


Figure 1: Cell-type proportions by group (participant-level)

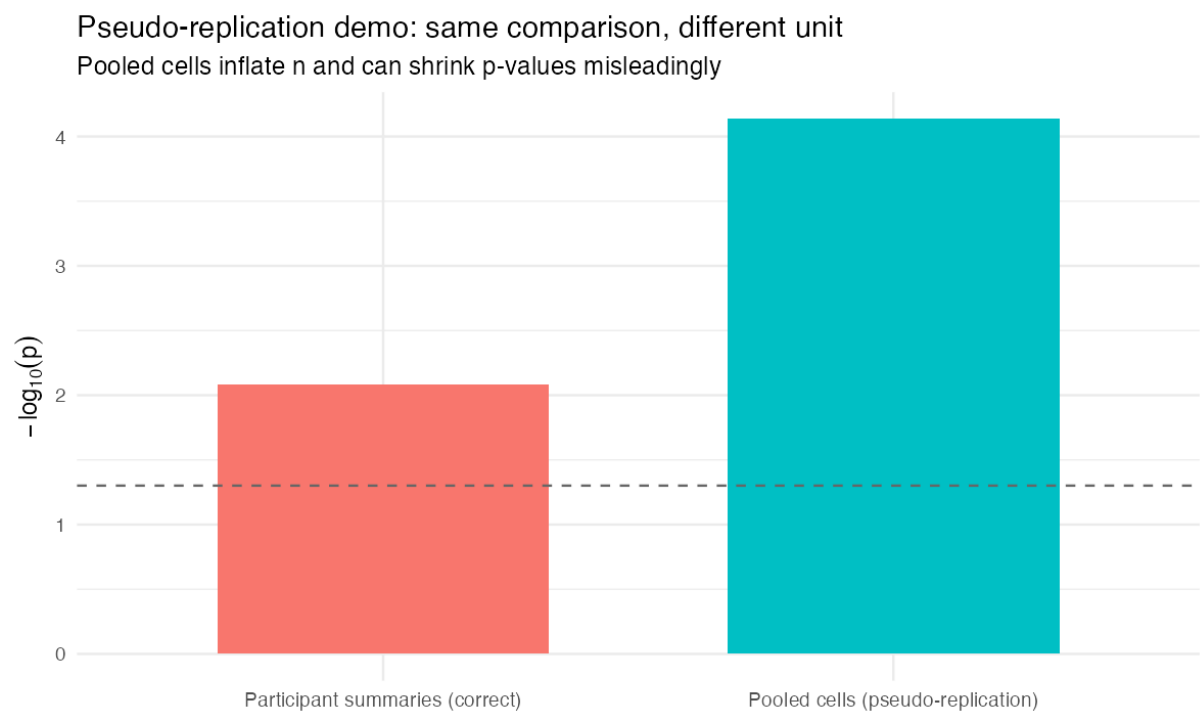


Figure 2: Pseudo-replication: participant vs pooled-cell p-values

Alternatives & extensions

Situation	Primary approach	Notes
Many cell types tested	FDR across populations (Ch 13)	prespecify primary endpoint
Strong drift	include batch + control beads QC	see Ch 14 overlap logic
Need single-cell discovery	clustering + stability (Ch 11 mindset)	do not call “endotypes”
Modelling cells directly	mixed model: <code>marker ~ group + (1 patient_id)</code>	still harder to interpret than summaries
Compositional core question	log-ratio between prespecified types	advanced; document clearly

Mini-lab: beta regression pointer (proportions)

When proportions are bounded and skewed, `lm(logit(p) ~ ...)` is a teaching default. For publication, consider beta regression (`betareg` package) on (0,1) with jitter away from 0/1.

```
# Teaching default (Ch 15 script):  
fit <- lm(logit_mono ~ group + batch, data = flow_m)
```

R lab: Flow cytometry on CASTOR-HD

Script: `R/examples/ch15_flow_cytometry.R`

Outputs:

- Participant proportions by group and batch (`ch15_flow_props_by_group.png`, `ch15_flow_props_by_batch.png`)
- **Compositional check:** stacked proportions (`ch15_compositional_stacked.png`)
- **Pseudo-replication demo:** participant vs pooled-cell p-values (`ch15_pseudoreplication_demo.png`)
- Per-cell PCA (descriptive): `ch15_flow_cells_pca.png`
- Summary table: `volume-01/tables/ch15_flow_mini_case_summary.csv`
- Effect table: `volume-01/tables/ch15_flow_effects_by_celltype.csv`

```
source("R/00_setup.R")  
library(tidyverse)  
  
flow <- readr::read_csv(file.path(paths$data, "flowcytometry_summary.csv"), show_col_types =  
table(flow$group, flow$batch))
```

Sensitivity (minimum)

- Repeat the comparison for 2-3 cell types.
- Re-fit with and without `batch` and compare effect direction and magnitude.

Niche figures (recommended)

- **Stacked compositional bars** by group (are shifts global rebalancing?)
- **Pseudo-replication contrast** (participant n vs cell n for the same comparison)

Compositional structure: proportions sum to 1 per participant

Each bar = one participant; changing one segment affects others

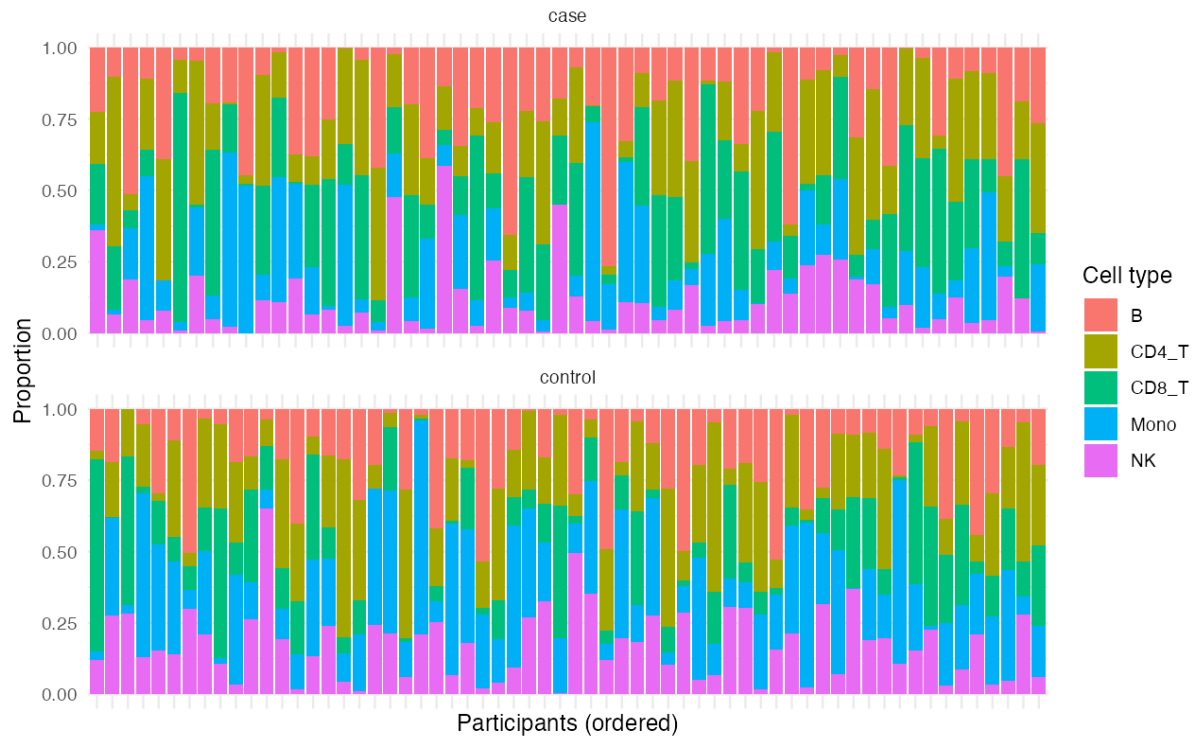


Figure 3: Compositional structure: proportions sum to 1

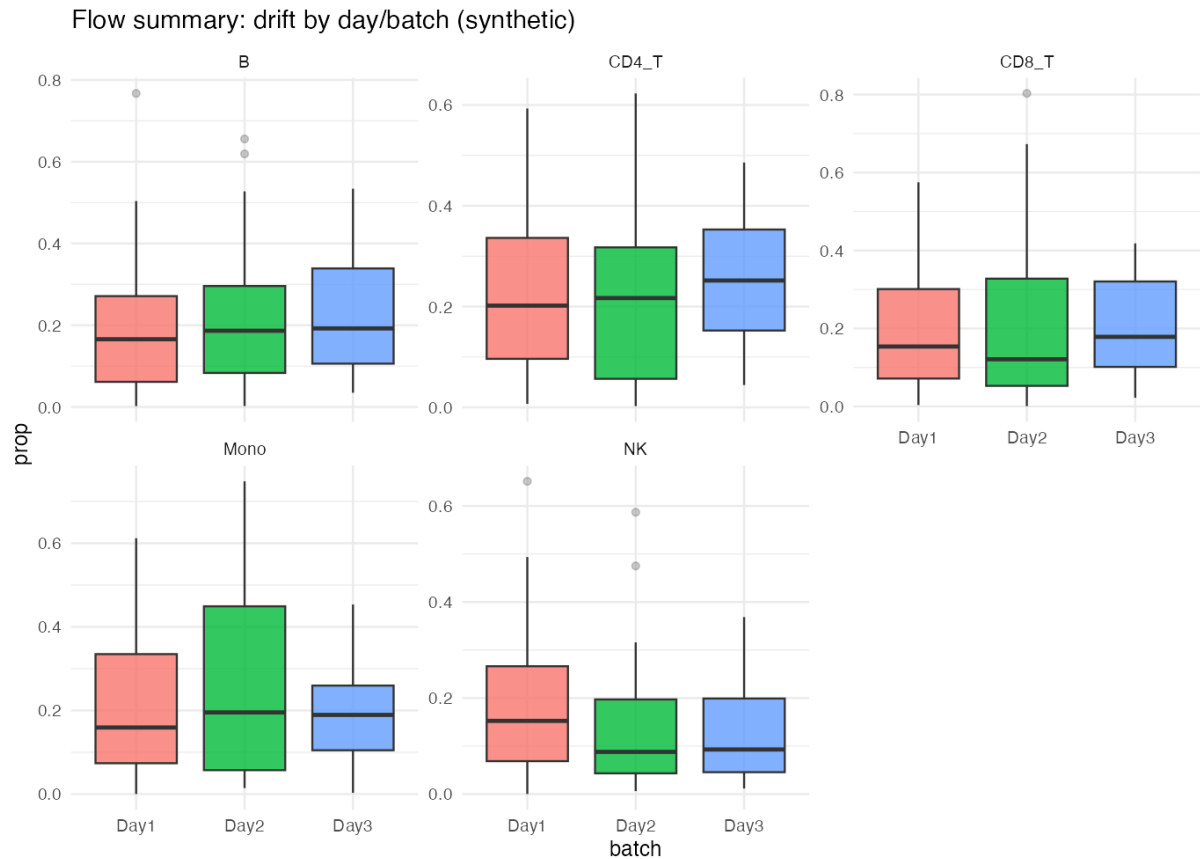


Figure 4: Drift check: proportions by batch/day

Exercises · Solutions

E15.1 Why is pooling 6,000 cells as $n = 6,000$ wrong?

E15.2 Name one compositional constraint when interpreting flow proportions.

E15.3 When is a UMAP figure acceptable in a paper?

Applied

1. Run `source("R/examples/ch15_flow_cytometry.R")`.
2. Compare participant vs pooled-cell p-values in `ch15_pseudoreplication_demo.png`.
3. Interpret monocyte results in `volume-01/tables/ch15_flow_effects_by_celltype.csv`.
4. Write a Results sentence for monocyte proportion difference with batch adjustment.

Chapter summary

- Default unit of analysis: **participant-level summaries**.
- **Pseudo-replication** (cells as n) produces false precision.
- **Compositional** structure matters when interpreting proportions.
- **Drift** diagnostics are mandatory (Ch 14 logic applies).
- Embeddings are for **QC and exploration**, not standalone proof.

Chapter 16: Antibody discovery screens - hit calling, confirmation, and stability

Part VI: High-dimensional biology and discovery

At a glance

Recurring datasets	data/antibody_screen.csv, data/antibody_confirmation.csv
Format	Technique cards + Caveats + Wrong analysis + Reporting (template)
Core idea	treat screening as triage; confirmation is where claims begin
Primary tools	replicate agreement, prespecified thresholds, PPV, stability tiers
R	R/examples/ch16_antibody_screening.R
Figures	FIGURE_INDEX - ch16_*.png
Templates	HIGH_DIM_REPORTING_TEMPLATES

Learning objectives

1. Combine replicates and quantify replicate agreement before hit calling.
2. Define hits using **prespecified thresholds** (or control-based rules), not post hoc cutoffs.
3. Evaluate screen performance with **confirmation PPV**, not screen p-values alone.
4. Assign **stability tiers** when ranks change under replicate resampling.
5. Report shortlists as tiers, not fragile 1..K rankings.

Prerequisites

Ch 8 (reporting), Ch 13 (multiplicity mindset), Ch 14 (batch), Ch 11 (do not overclaim discovery).

Opening question

Which clones should we spend money validating - and how confident are we that the top of the list stays the top when noise is resampled?

Discovery screens are not hypothesis tests. They are **prioritisation pipelines**. The “result” is not a p-value; it is a defensible shortlist and a confirmation plan.

Worked mini-case: threshold, PPV, and stability tiers

The chapter script walks through three linked decisions on CASTOR-HD.

Step 1: Replicate agreement (QC gate)

Before calling hits, check that replicates agree. Poor agreement means ranks are noise.

QC signal	Interpretation
Strong rep1 vs rep2 correlation	Ranking is reasonably stable
Batch-separated clouds	Drift may inflate false hits (Ch 14)
Wide scatter at low signal	threshold choice matters a lot

Step 2: Hit calling with a prespecified threshold

Teaching rule in the script: call a hit when mean screen signal > **1.4** (prespecified for illustration).

Quantity	What it means
Hits	clones above threshold (screen positives)
Confirmed in hits	hits with <code>confirm_positive == TRUE</code> in confirmation assay
PPV	confirmed in hits / hits

PPV answers: *of the clones we would advance, what fraction actually confirm?*

This is more actionable than a screen p-value.

Step 3: Ranking stability tiers (when ranks are noisy)

Recompute top-20 lists using replicate 1, 2, and 3 separately. Assign each clone to a **stability tier**:

Tier	Rule (AgA example)	How to report
Tier 1	Top 20 in all 3 replicates	Highest-priority confirmation
Tier 2	Top 20 in 2 of 3 replicates	Secondary shortlist
Tier 3	Top 20 in 1 of 3 replicates	Exploratory only
Below tier	Never top 20 in any replicate	Do not emphasise in main text

Clinician read: tiers are like “high / medium / low confidence shortlist” - more honest than pretending rank #7 is meaningfully better than rank #9.

Decision rule (screen to confirmation)

1. Check **replicate agreement** and batch QC.
2. Apply a **prespecified** hit rule (threshold or control-based).
3. Report **PPV** among hits (and confirmation rate by antigen).
4. Compute **stability tiers**; prefer Tier 1 for claims.
5. Confirm Tier 1/2 candidates; report KD/functional readout with prespecified positivity rule.

Technique: Hit calling + confirmation as the estimand

Technique card

Answers	Which candidates exceed a prespecified screen criterion, and how many confirm?
Outcome type	continuous screen signal; binary confirmed positive (after follow-up assay)
Design	replicates + batches; often multiple antigens
Data required	replicates, controls, batch IDs, confirmation readout
Effect measure	PPV of screen; tiered shortlist; confirmation rate
R	replicate summaries + thresholding; confusion table vs confirmation
When to use	any screening/triage workflow (antibodies, CRISPR, drug screens)
When NOT to use	claiming “best antibody” from screen alone
Does NOT prove	clinical utility; in vivo efficacy; specificity without orthogonal tests

Dual interpretation

Plain language: we used the screen to shortlist candidates, then checked which of those actually worked in a stronger assay.

Precise language: the screen defines a ranking/threshold; confirmation defines the target endpoint. Screen utility is summarised by PPV and stability of the shortlist under replicate noise.

Clinician read: a screen is like a triage test: useful if it reliably enriches true positives, but it is not the final diagnosis.

Caveats box

Caveat	Why it matters
Replicate noise	ranks can change with small measurement noise
Batch effects	plates/days shift signals; false positives cluster by batch
Threshold fishing	changing the cutoff after seeing results inflates claims

Caveat	Why it matters
Multiple antigens	multiplicity across targets; avoid cherry-picked antigen only confirming “pretty” candidates hides failure rates “confirmed positive” must be prespecified (KD cutoff etc.)
Confirmation bias	
Definition drift	

Wrong analysis

Mistake	Report the top 10 clones as “best” without confirmation
Why it fails	screens have false positives; ranks are noisy
Do instead	report tiers + confirmation rate and PPV
Mistake	Pick a threshold that maximises “story” after seeing the data
Why it fails	post hoc thresholding is hidden multiple testing
Do instead	prespecify threshold or use controls; report sensitivity analysis

Catalog of wrong analyses (antibody / discovery screens)

Wrong analysis	Why it fails	Do instead
Screen p-value as final result	Screen is triage, not confirmatory endpoint	Report PPV and confirmation outcomes
Post hoc threshold chosen to maximise hits	Inflates false discovery and PPV looks better than it is	Prespecify threshold; show sensitivity curve
Rank #1..#10 reported as precise ordering	Ranks are unstable under replicate noise	Use stability tiers
Confirm only “good-looking” clones	PPV becomes meaningless (selection bias)	Confirm prespecified shortlist rule
Ignore batch in screen QC	Hits cluster by plate/day	Plot by batch; adjust or block in design
Single replicate	No assessment of ranking stability	Require replicates or repeat measurements
Mix antigens in one threshold	Different targets have different background	Threshold/PPV per antigen
KD reported without positivity rule	“Strong binder” becomes subjective	Prespecify KD cutoff for confirm_positive
Cherry-pick one antigen for the paper	Multiplicity and hype	Report all targets or prespecify primary antigen
Claim therapeutic antibody from screen	No specificity, affinity, or function in context	Confirmation + orthogonal assays + replication

Reporting template

Use Template D in [HIGH_DIM_REPORTING_TEMPLATES](#).

Screening used [R] technical replicates per clone against [antigens]. Replicate agreement was assessed [correlation/plot]. Hits were defined prespecification as mean signal > [threshold] (or control-based rule). Confirmation used [assay] with positivity defined as KD < [X] nM. We report PPV among hits and stability tiers based on overlap of top-[K] lists across replicates.

Technique: Threshold sensitivity (prespecified vs exploratory)

A single threshold is not enough for transparency. Report a **sensitivity curve**:

- x-axis: threshold
- y-axis: number of hits and PPV (where confirmation data exist)

If PPV collapses when you move the threshold slightly, your shortlist is fragile.

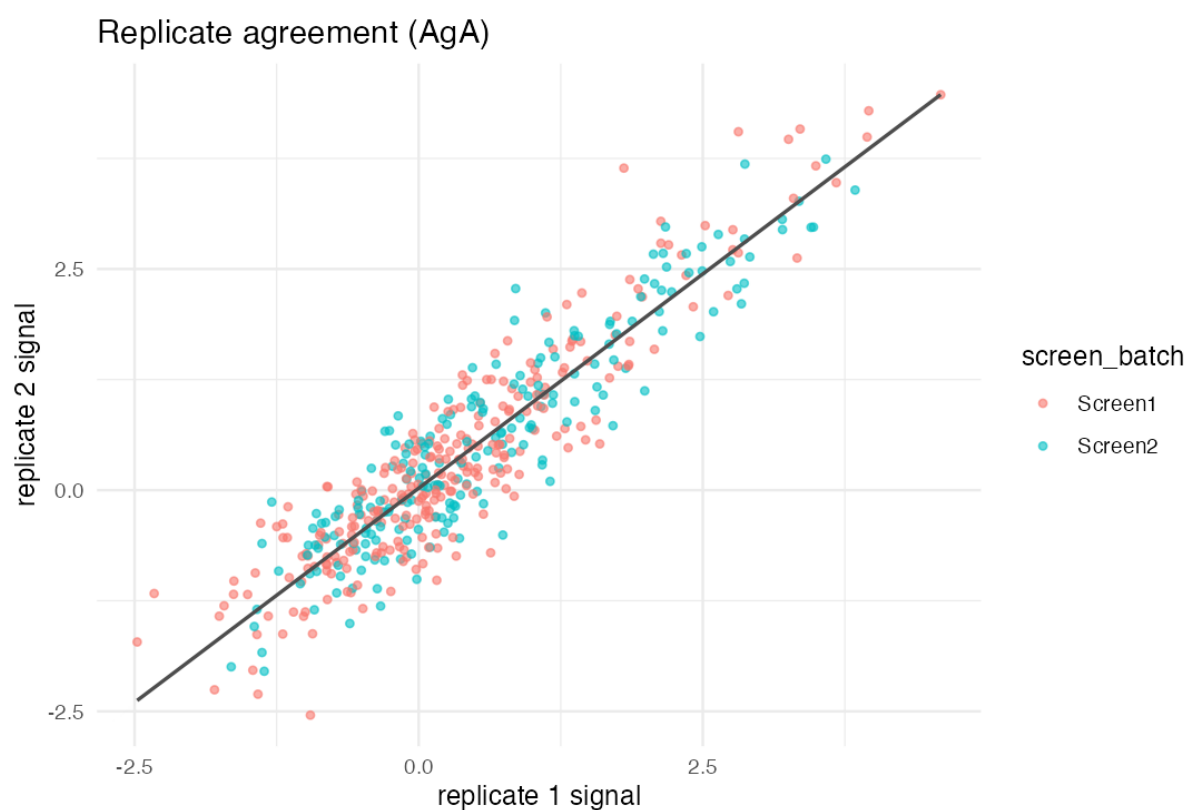


Figure 1: Replicate agreement (AgA screen)

Hit calling sensitivity: do not tune threshold post hoc

Dashed line = prespecified teaching threshold

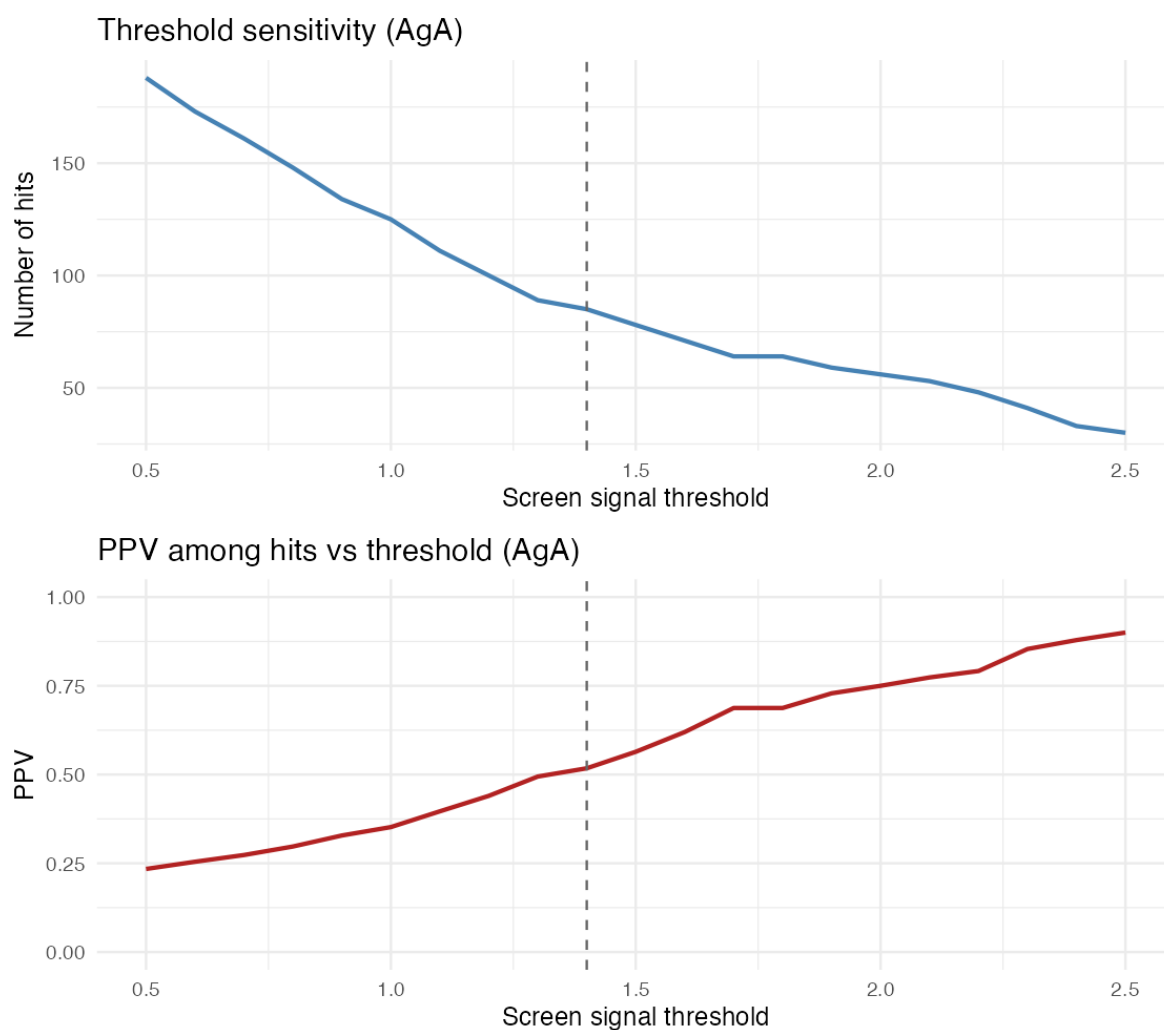


Figure 2: Threshold sensitivity: hits and PPV vs cutoff

Technique: Stability tiers (ranking under replicate resampling)

Technique card

Answers	Which clones are consistently top-ranked under noise?
Input	replicate-level signals per clone
Output	Tier 1/2/3 shortlist
When to use	whenever confirmation budget is limited
Does NOT prove	clone is “best”; only that it is stable in the screen

Alternatives & extensions

Situation	Approach	Notes
High background variability	control-normalised signal (e.g., z-score vs negative controls)	more defensible than arbitrary cutoff
Many weak binders	FDR across clones (Ch 13 mindset)	still requires confirmation
Functional screen endpoint	hit = response in primary assay; confirm in secondary	define two-stage estimands
Multi-batch screen	include batch in hit model or stratify	Ch 14 logic

Mini-lab: control-normalised screen signal

```
screen <- readr::read_csv(file.path(paths$data, "antibody_screen.csv"), show_col_types = FALSE)
# Teaching: z-score within antigen across clones (exploratory)
screen %>% group_by(antigen) %>%
  mutate(z = as.numeric(scale(signal_mean))) %>%
  filter(z > 1.5) %>% count(antigen)
```

R lab: Antibody screening on CASTOR-HD

Script: R/examples/ch16_antibody_screening.R

Outputs:

- ch16_screen_replicate_agreement.png
- ch16_screen_ppv.png (PPV by antigen at prespecified threshold)
- ch16_threshold_sensitivity.png (hits and PPV vs threshold)
- ch16_stability_tiers.png (tier counts + PPV by tier for AgA)
- ch16_ranking_stability.png (top-20 overlap across replicate pairs)
- Tables in volume-01/tables/:
 - ch16_screen_ppv_by_antigen.csv
 - ch16_threshold_sensitivity.csv
 - ch16_ranking_tiers_aga.csv
 - ch16_mini_case_summary.csv

```
source("R/00_setup.R")
library(tidyverse)

screen <- readr::read_csv(file.path(paths$data, "antibody_screen.csv"), show_col_types = FALSE)
conf <- readr::read_csv(file.path(paths$data, "antibody_confirmation.csv"), show_col_types = FALSE)
```

Sensitivity (minimum)

- Vary threshold over a prespecified grid; show PPV and hit count.
- Report stability tiers and restrict main claims to Tier 1 (and optionally Tier 2).

Prefer Tier 1/2 clones for confirmation claims

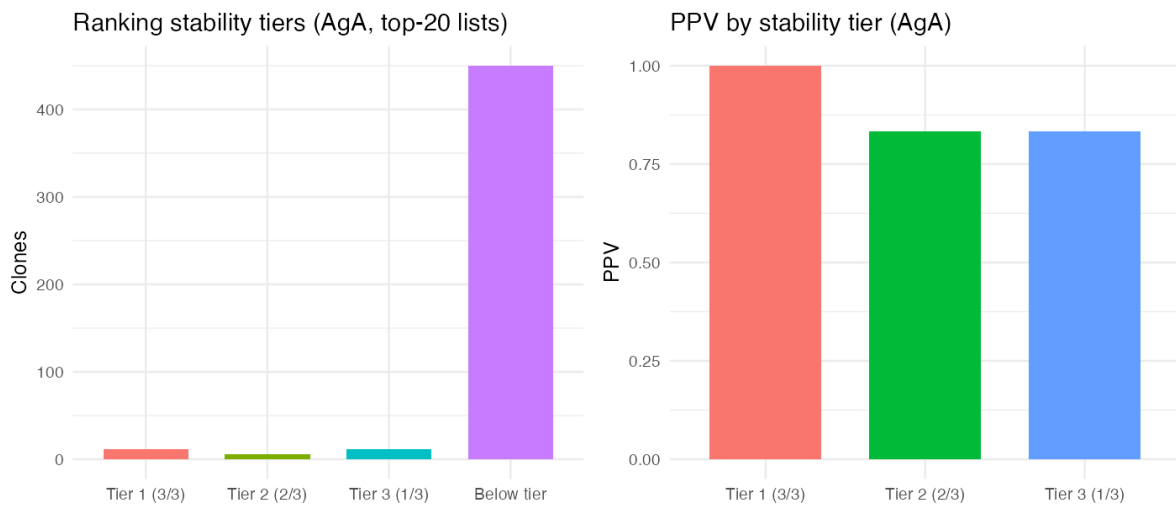


Figure 3: Stability tiers and PPV by tier (AgA)

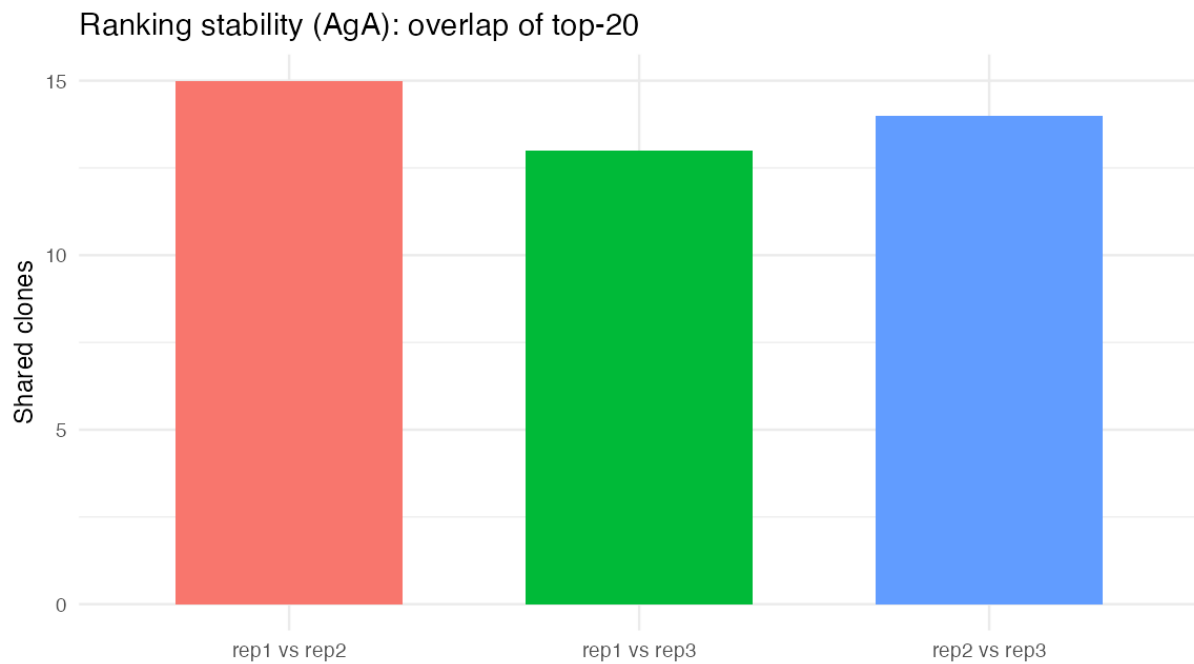


Figure 4: Top-20 overlap across replicate pairs

Exercises · Solutions

E16.1 What is PPV among screen hits, and why is it more useful than a screen p-value?

E16.2 Define Tier 1 stability in this chapter's rule.

E16.3 Why is post hoc threshold tuning dangerous?

Applied

1. Run `source("R/examples/ch16_antibody_screening.R")`.

2. Report PPV by antigen from `ch16_screen_ppv_by_antigen.csv`.
3. Interpret `ch16_threshold_sensitivity.png` at the prespecified threshold.
4. List Tier 1 clones from `ch16_ranking_tiers_aga.csv` and their confirmation status.

Chapter summary

- Screens are for **triage**, not final proof.
- Prespecify **hit rules** and **confirmation positivity**.
- Report **PPV among hits**, not just rankings.
- Use **stability tiers** when replicate resampling changes the top list.
- Batch QC and replicate agreement are part of the estimand, not optional extras.

Part VII: Integrated CASTOR-HD capstone

This part ties the high-dimensional biology chapters (13–16) into one end-to-end discovery workflow and introduces supervised (p–n) prediction with proper resampling.

Chapter 17: Integrated CASTOR-HD discovery pipeline

Part VII: Integrated CASTOR-HD capstone

At a glance

Recurring datasets	CASTOR-HD: proteomics, RNA-seq, flow, antibody screen/confirmation
Format	End-to-end narrative + supervised (p n) extension
Pipeline	DE → batch QC → shortlist → flow summary → screen confirmation → reporting
R	R/examples/ch17_integrated_castor_hd.R, R/examples/ch17_elastic_net_proteomics.R
Templates	HIGH_DIM_REPORTING_TEMPLATES

Learning objectives

1. Execute a coherent discovery workflow across omics, flow, and antibody data.
2. State what each step proves vs suggests at each stage.
3. Fit elastic net with nested cross-validation for (p n) proteomics prediction.
4. Draft an integrated Methods/Results section using the reporting templates.

Prerequisites

Chapters 13–16 and Case D in [Ch 12](#).

Opening question (CASTOR-HD)

If we start with a proteomics differential analysis, how do batch checks, immune phenotyping, and antibody confirmation change what we are willing to claim?

This chapter is the **advanced capstone**: same CASTOR-HD cohort, one narrative thread, explicit stopping rules when batch or stability fail.

Master workflow (CASTOR-HD discovery)

Step	Action	Chapter	Output
1	Per-feature DE + BH FDR	13	Top table, volcano
2	Batch/plate PCA + sensitivity	14	Overlap check, discovery count with/without batch
3	Shortlist top features for follow-up	13	Ranked list (effect + q)
4	Flow: participant-level proportions	15	Adjusted effects by cell type
5	Antibody screen → confirmation + tiers	16	PPV, Tier 1 clones
6	Integrated report	Templates	Methods + Results draft
7	(Optional) Supervised elastic net	17 R	Nested CV AUC

```
source("R/00_setup.R")
source("R/examples/ch17_integrated_castor_hd.R")
```

Step 1–2: Proteomics DE with batch awareness

Run differential analysis (Ch 13), then immediately run batch diagnostics (Ch 14). **Do not** interpret top hits until overlap is acceptable.

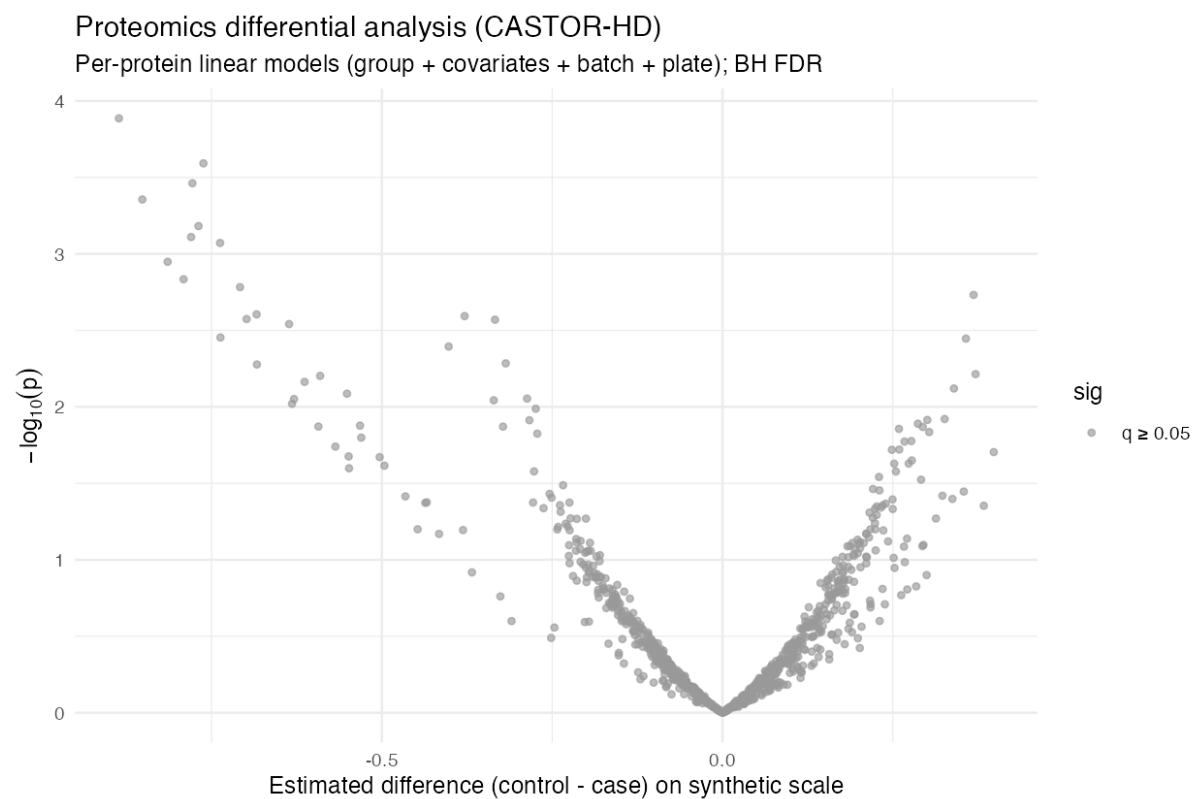


Figure 1: Proteomics volcano (BH FDR)

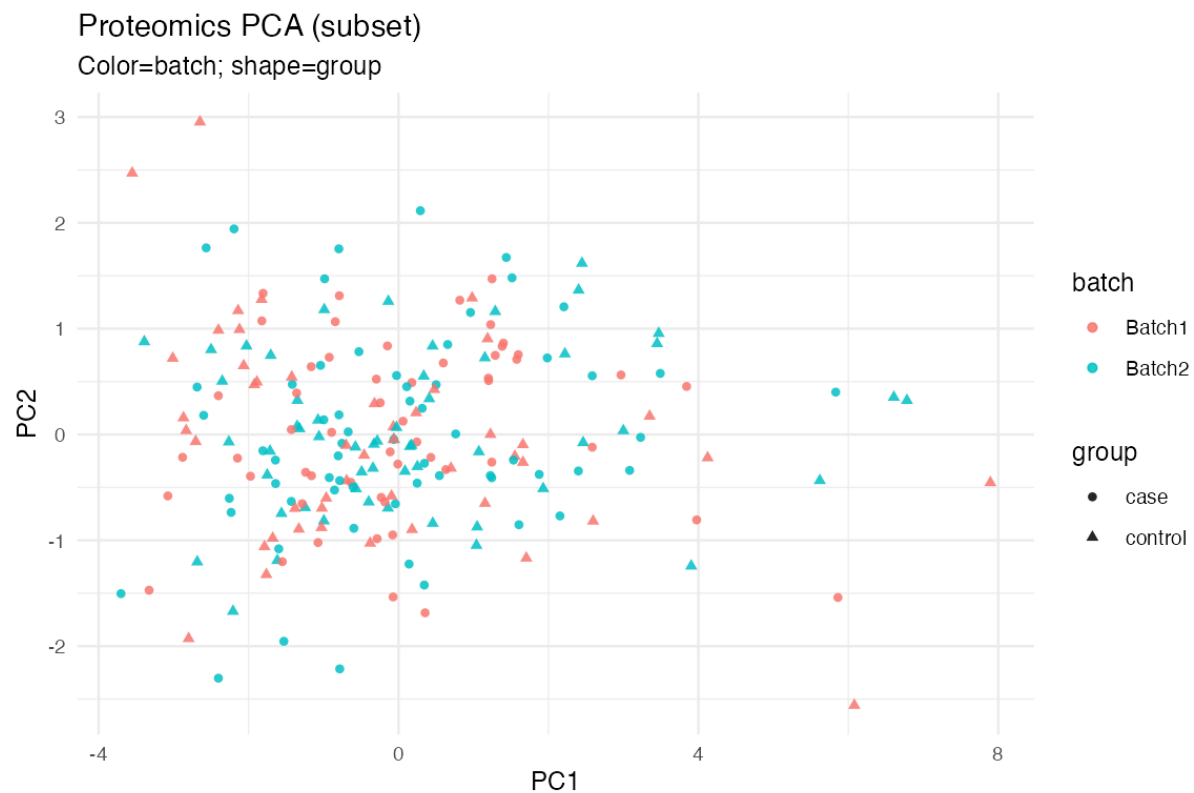


Figure 2: PCA by batch — proteomics subset

Stopping rule: if group and batch are perfectly confounded, report non-identifiability and do not claim group-specific protein differences.

Step 3: Shortlist for follow-up

Export the top 20–50 features by q-value **and** absolute effect. Prioritise features stable with vs without batch adjustment.

See `volume-01/tables/ch17_integrated_shortlist.csv` from the integrated script.

Step 4: Flow cytometry summary

Link immune phenotyping to the discovery story at the **participant** level (Ch 15). Cell embeddings are QC only.

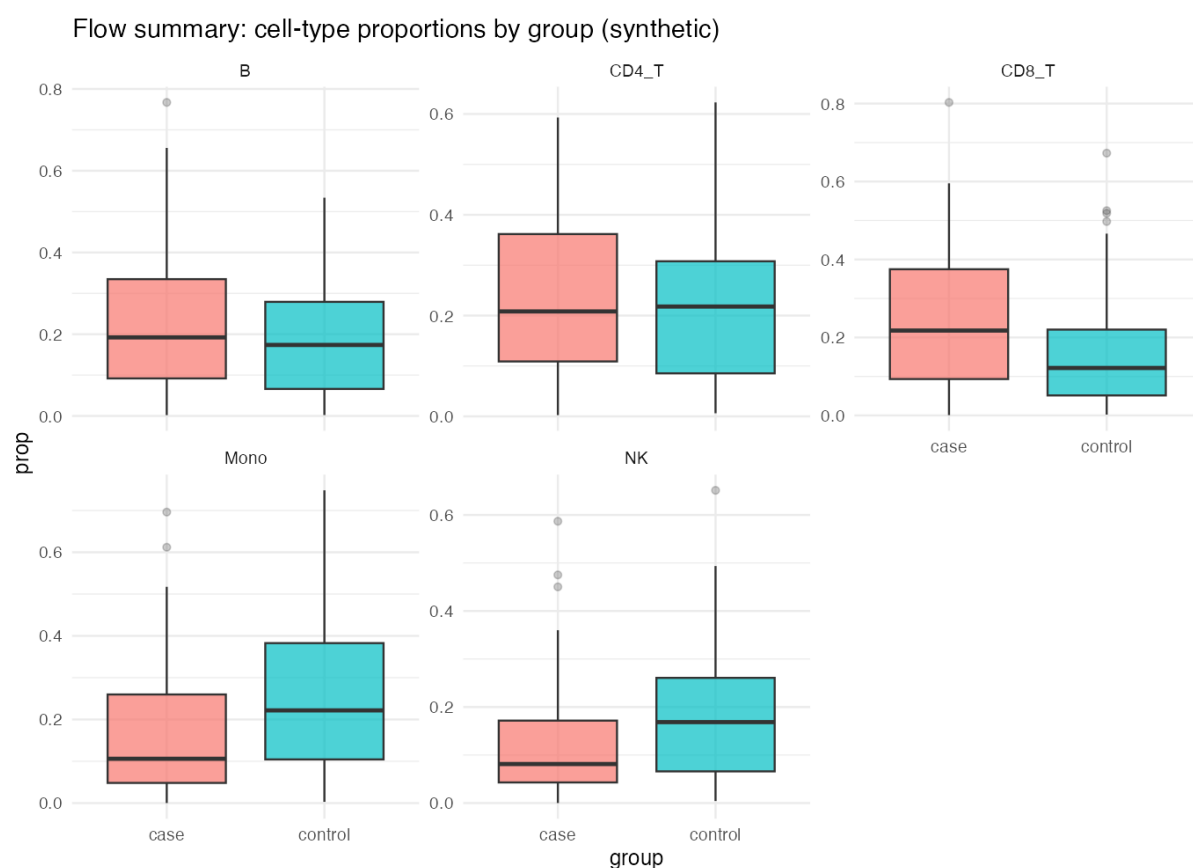


Figure 3: Monocyte proportions by group

Step 5: Antibody screen confirmation

Translate screen hits into confirmation PPV and stability tiers (Ch 16).

Hit calling sensitivity: do not tune threshold post hoc

Dashed line = prespecified teaching threshold

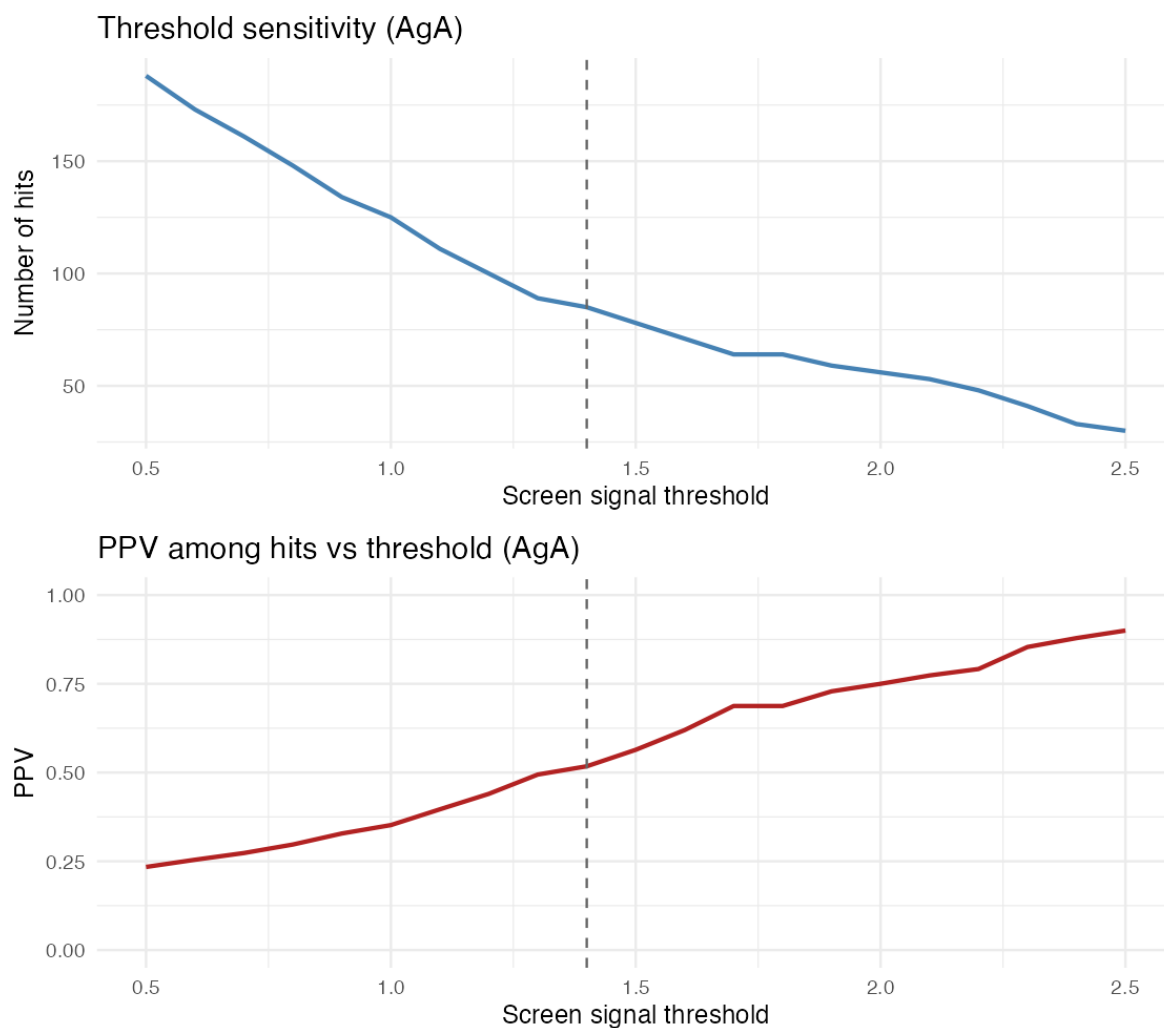


Figure 4: Threshold sensitivity: hits and PPV

Step 6: Reporting (integrated)

Use [HIGH_DIM_REPORTING_TEMPLATES](#):

- **Template A** for omics DE (n, model, FDR, batch handling)
- **Template B** for batch sensitivity
- **Template C** for flow proportions
- **Template D** for antibody discovery

Clinician read: one paragraph per modality; separate “discovery” from “confirmed binding.”

Technique: Elastic net + nested CV ((p n) prediction)

Technique card

Answers	Can proteomics predict case/control with honest internal performance?
Outcome	Binary group label
Predictors	Many proteins ((p n))
Method	glmnet elastic net inside nested CV
When to use	Exploratory risk stratification; hypothesis for external validation
When NOT to use	Causal inference; diagnostic approval without external cohort
Does NOT prove	Mechanism; transportability; clinical utility

Wrong analysis

Mistake	Do instead
Tune on the same data you evaluate	Nested CV (outer fold = performance, inner =)
Impute/batch-correct before splitting	All preprocessing inside training folds
Report training AUC as validation	Report outer-fold mean AUC $\pm$ SD

R lab

```
source("R/examples/ch17_elastic_net_proteomics.R")
```

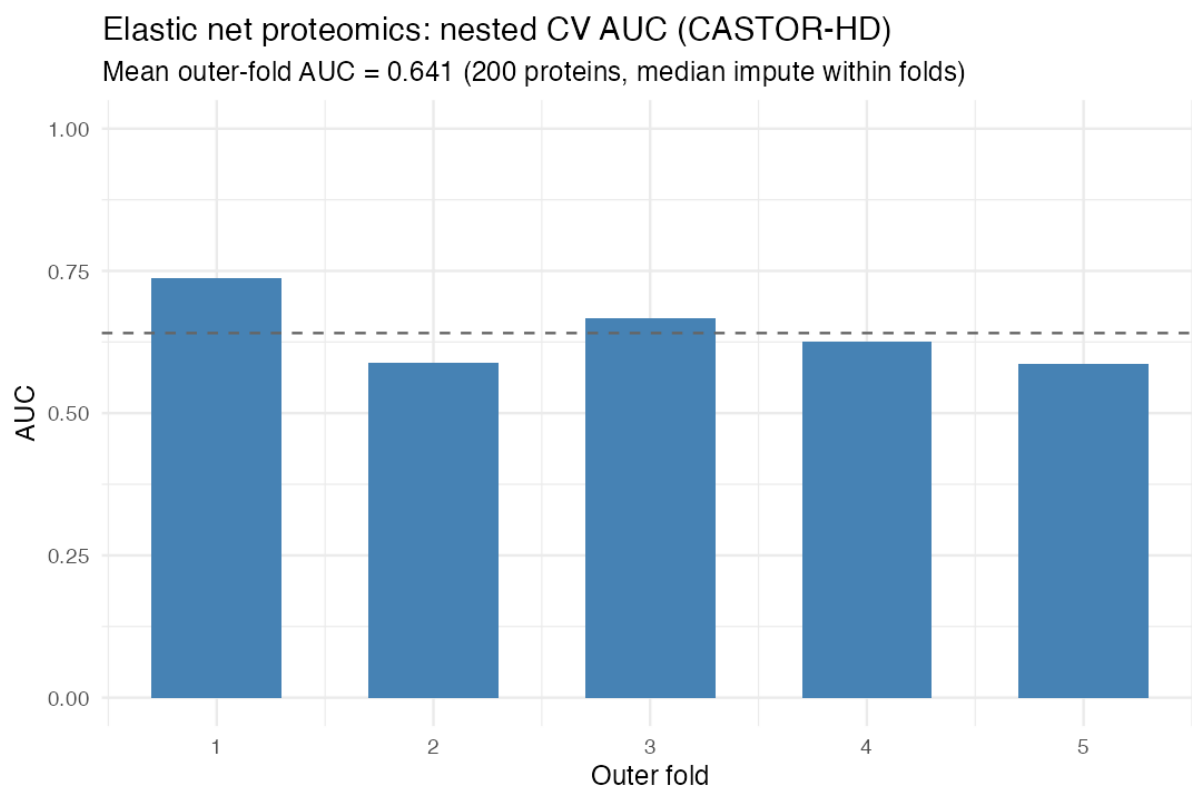


Figure 5: Nested CV AUC — elastic net on proteomics

Cross-modality synthesis

Modality	Inference level	Key metric
Proteomics DE	Discovery (FDR)	Effect + q; batch sensitivity
RNA-seq DE	Discovery (NB + FDR)	Same; library offset stated
Flow	Associational	Participant n; batch-adjusted proportion
Antibody screen	Prioritisation	PPV + Tier 1 stability
Elastic net	Prediction (internal)	Nested CV AUC

Exercises · [Solutions](#)

E17.1 At which pipeline step would you stop if batch and group are confounded?

E17.2 Why is nested CV required for elastic net on 1000 proteins?

E17.3 What is the difference between a Tier 1 antibody clone and a proteomics $q < 0.05$ hit?

Applied

1. Run `source("R/examples/ch17_integrated_castor_hd.R")`.
2. Run `source("R/examples/ch17_elastic_net_proteomics.R")`.
3. Draft a 300-word integrated Results section using the templates.
4. List three claims you would **not** make from this pipeline alone.

Further reading

- McShane et al., biomarker reporting ([McShane et al. 2011](#))
- Harrell, *Regression Modeling Strategies* ([Harrell 2015](#))
- Ch 12 Case D for the bridge from core CASTOR to CASTOR-HD

Part VIII: Longitudinal, survival, and causal inference

These chapters complete the single-volume handbook: data structures that break independence assumptions in Ch 4–7, plus principled handling of missing data and confounding.

Chapter 18: Longitudinal data and mixed models

Part VIII: Longitudinal, survival, and causal inference

At a glance

Recurring dataset	data/longitudinal_spirometry.csv (CASTOR extension)
Format	Technique cards + Caveats + Wrong analysis + Reporting (template)
Question	How does FEV1 change over time, and does treatment modify that trajectory?
Core methods	spaghetti plots, linear mixed models (<code>lmer</code>), random intercepts
R	R/examples/ch18_longitudinal_mixed_models.R
Capstone link	Ch 12 Case E

Learning objectives

1. Recognise when repeated measures violate independence (Ch 4–5).
2. Fit a random-intercept mixed model for FEV1 trajectories.
3. Interpret `weeks × group` interaction as differential change over time.
4. Report participant-level n , visit structure, and sensitivity to cross-sectional shortcuts.

Prerequisites

Chapters 4–5 (group comparisons, linear models); Ch 8 (reporting).

Opening question (CASTOR extension)

Does intervention slow FEV1 decline compared with standard care when each patient contributes four visits at weeks 0, 12, 24, and 52?

A Welch t -test on week-52 FEV1 alone **throws away** within-person information and treats visits as independent. Mixed models keep all visits while accounting for correlation within patient ([Harrell 2015](#)).

Technique: Linear mixed model (random intercept)

Technique card

Answers	Mean trajectory over time; group differences in level or slope
Outcome	Repeated continuous FEV1 (litres)
Unit of inference	Participants (n patients); visits are correlated
Design	Randomised or observational with 2 visits per person
Data required	<code>patient_id</code> , visit time, outcome, optional covariates
Assumptions	Linear trend in time (or splines); random intercepts $\sim$ Normal; MAR for dropout (discuss)
Effect measure	Fixed-effect for <code>weeks</code> , <code>group</code> , <code>weeks:group</code> (litres or litres per week)
R	<code>lme4::lmer(fev1 ~ weeks * group + covariates + (1 patient_id), data)</code>
When to use	Repeated continuous outcomes; interest in change over time
When NOT to use	One visit per person (use Ch 4–5); survival/censoring primary (Ch 19)
Does NOT prove	Causal treatment effect without randomisation, adherence, and missing-data audit

Dual interpretation

Plain language: after accounting for each patient's baseline, the intervention arm had a different average FEV1 trajectory over one year.

Precise language: we model FEV1 as a linear function of time and treatment with patient-specific random intercepts; the `weeks:group` interaction estimates differential mean change per week between arms, conditional on covariates.

Clinician read: is the modelled gap at 52 weeks clinically meaningful (MCID ~ 0.1 L in many COPD contexts)? Trajectory matters more than a single visit snapshot.

Caveats box

Caveat	Why it matters in respiratory research
Dropout / informative missingness	Sicker patients skip visits; complete-case trajectories can bias treatment effects
Unequal visit spacing	Irregular spirometry schedules need time variables in appropriate units
Learning / training effects	First post-baseline visit can differ from steady state

Caveat	Why it matters in respiratory research
Non-linear decline	FEV1 decline may accelerate; consider splines (Ch 7)
Clustered centres	Patients nested in hospitals need centre random effects or robust SEs
Cross-sectional shortcut	Week-52 <i>t</i> -test ignores baseline and correlation

Wrong analysis

Mistake	Why it fails	Do instead
<i>t</i> -test at one visit only	Discards information; wrong SEs	Mixed model on all visits
Pooled visits as independent rows	Pseudo-replication	(1 patient_id) random intercept
Change score without baseline adjustment	Noisy; regression to mean	Mixed model or ANCOVA with baseline
Ignore dropout pattern	Biased if missingness relates to health	Compare attenders; Ch 20 sensitivity
Claim causal effect from observational trajectories	Confounding by indication	State associational estimand; Ch 21

Catalog of wrong analyses (longitudinal FEV1)

Wrong analysis	Why it fails	Do instead
Last observation carried forward for dropouts	Fabricates stability	Model available data; discuss bias direction
Per-visit ANOVA without patient	Inflated <i>n</i>	Mixed model / GEE with patient cluster
Plot spaghetti only, no model	Descriptive only	Participant <i>n</i> + modelled trajectories
Spline hunting without prespecification	Overfit	Prespecify linear vs spline in protocol

Reporting template

Longitudinal FEV1 (litres) was analysed in $n = \dots$ participants contributing $\dots$ visits (weeks 0, 12, 24, 52). We fitted a linear mixed model with random intercepts for patient: $\text{FEV1} \sim \text{time} \times \text{treatment} + \text{age} + \text{sex} + \text{smoking} + (1|\text{patient})$. The estimated difference in slope (intervention – standard) was $\dots$ L per week (95% CI $\dots$). Sensitivity analysis compared a cross-sectional model at week 52 only. Missing visits were $\dots\%$.

R lab

```
source("R/00_setup.R")  
source("R/examples/ch18_longitudinal_mixed_models.R")
```

Longitudinal FEV1 (CASTOR extension)

Spaghetti plot: each line is one participant

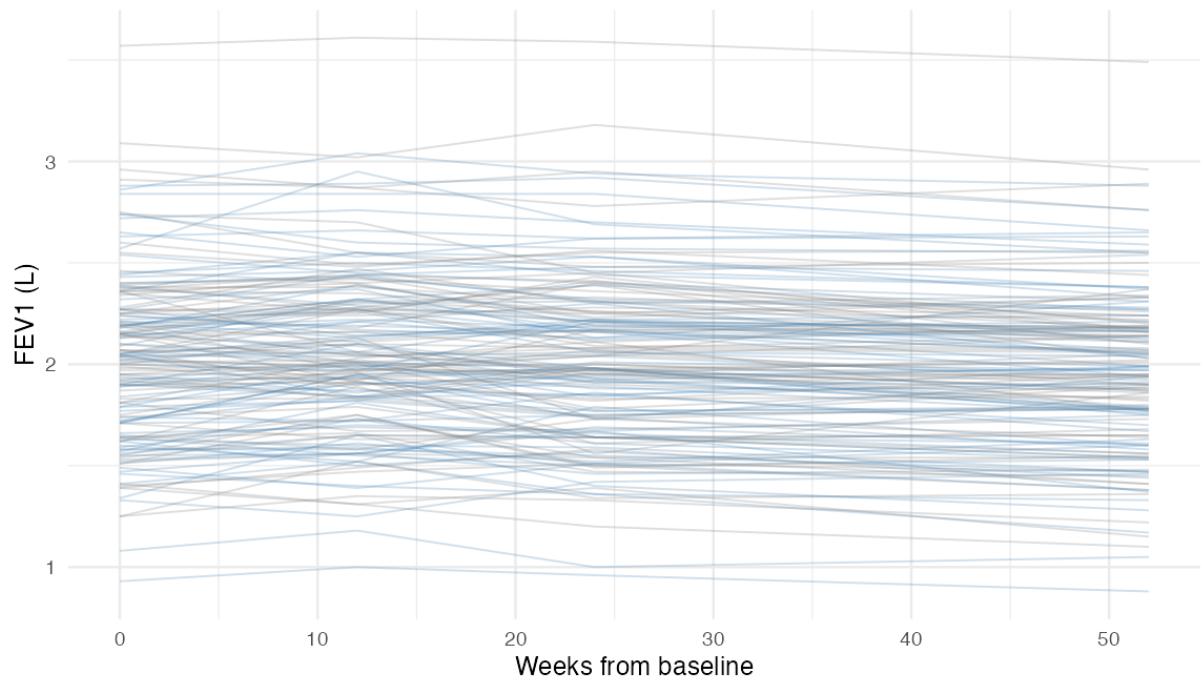


Figure 1: Spaghetti plot: FEV1 trajectories by participant

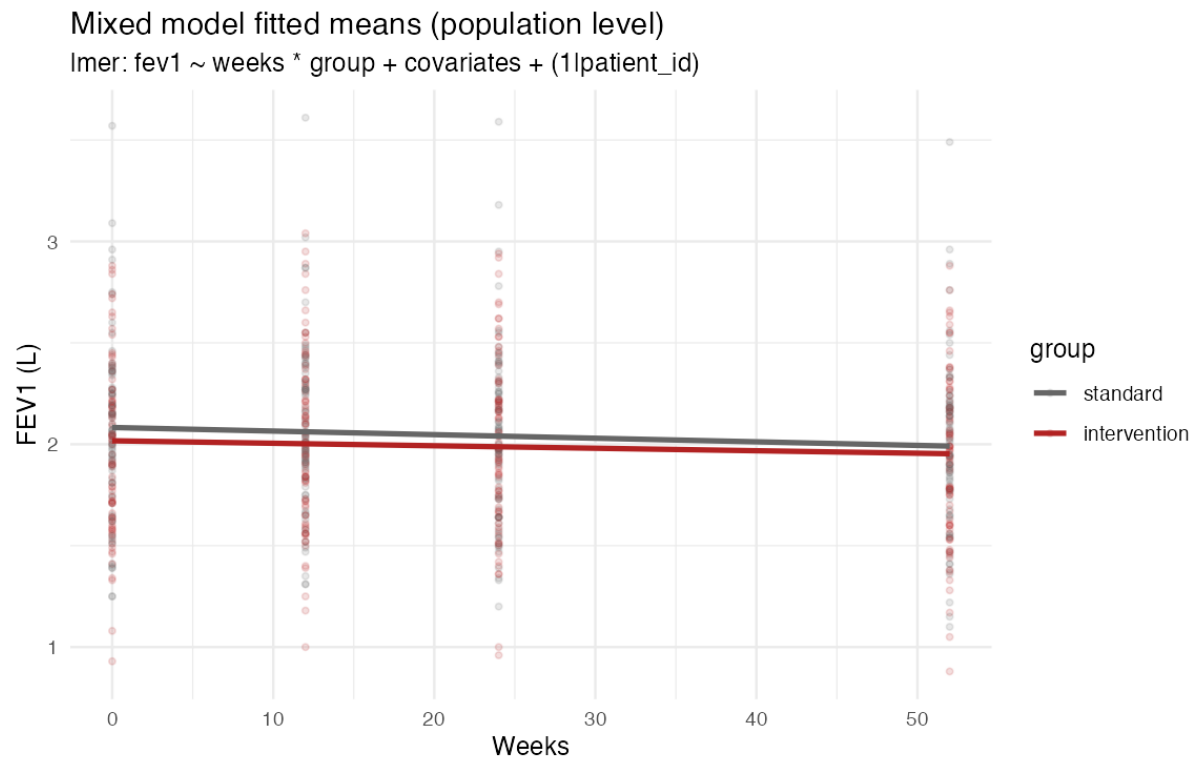


Figure 2: Mixed model fitted population trajectories

Tables: `ch18_mixed_model_coefficients.csv`, `ch18_visit_counts_by_group.csv`,
`ch18_sensitivity_mixed_vs_fixed.csv`

Mini-lab: GEE pointer

When the scientific focus is a population-averaged contrast rather than subject-specific trajectories, GEE (`geepack::geeglm`) is a common alternative. Mixed models estimate conditional (subject-specific) slopes; GEE estimates marginal effects — both are valid with different estimands.

Alternatives & extensions

Situation	Method	Note
Non-linear FEV1 decline	Splines in weeks	Prespecify knots
Clustered centres	(1 \ centre) or GEE	Multi-centre trials
Binary event over time	Survival model	Ch 19
Informative dropout	Joint model / sensitivity	Ch 20

Exercises · [Solutions](#)

E18.1 Why is independence violated with repeated FEV1?

E18.2 What does (1|patient_id) represent?

E18.3 When is a week-52 t -test misleading compared with a mixed model?

Applied

1. Run `source("R/examples/ch18_longitudinal_mixed_models.R")`.
2. Interpret `weeks:groupintervention` from the coefficient table.
3. Compare `ch18_sensitivity_mixed_vs_fixed.csv` — which SE is smaller, and why is that suspicious?

Capstone: Case E in Ch 12.

Further reading

- Harrell, *Regression Modeling Strategies* ([Harrell 2015](#))
- Fitzmaurice, Laird & Ware, *Applied Longitudinal Analysis* (mixed models / GEE)

Chapter 19: Survival analysis and time-to-event outcomes

Part VIII: Longitudinal, survival, and causal inference

At a glance

Recurring dataset	data/time_to_exacerbation.csv (CASTOR extension)
Format	Technique cards + Caveats + Wrong analysis + Reporting (template)
Question	Who reaches first exacerbation sooner, accounting for censoring?
Core methods	Kaplan–Meier, log-rank test, Cox proportional hazards
R	R/examples/ch19_survival_analysis.R
Capstone link	Ch 12 Case E

Learning objectives

1. Define event, time origin, and censoring for respiratory endpoints.
2. Plot and compare survival curves (Kaplan–Meier) with log-rank tests.
3. Fit and interpret Cox hazard ratios with 95% CI and check proportional hazards.
4. Avoid treating censored patients as permanently event-free.

Prerequisites

Chapters 4, 6 (rates and GLM intuition); Ch 8 (reporting).

Opening question (CASTOR extension)

*Among adults with COPD-style risk profiles, does smoking shorten time to **first** exacerbation within one year of follow-up?*

Binary “any exacerbation Y/N” at 12 months discards **when** events occur and how follow-up differs. Survival methods use all follow-up time until event or censoring ([Harrell 2015](#)).

Technique: Kaplan–Meier and Cox proportional hazards

Technique card (Kaplan–Meier)

Answers	What fraction remain event-free over time, by group?
Outcome	<code>Surv(time, event)</code> — time to first exacerbation
Effect measure	Survival probabilities at prespecified times; median survival (if estimable)
R	<code>survival::survfit(Surv(time_days, event) ~ smoking, data)</code>
Comparison	Log-rank test (<code>survdiff</code>)
Does NOT prove	Causation; adjusted confounding control

Technique card (Cox PH)

Answers	Instantaneous hazard of first exacerbation, adjusted for covariates
Outcome	<code>Surv(time_days, event)</code>
Effect measure	Hazard ratio (HR) with 95% CI
R	<code>coxph(Surv(time_days, event) ~ smoking + fev1_pct + therapy + age, data)</code>
Assumptions	Proportional hazards (test with <code>cox.zph</code>); independent censoring
When to use	Time-to-first-event with right censoring
When NOT to use	Competing events (death) without competing-risks model
Does NOT prove	Causation; absolute risk without extra reporting

Dual interpretation

Plain language: smokers tended to reach their first exacerbation sooner during follow-up; the adjusted hazard was higher for smokers.

Precise language: the Cox model estimates hazard ratios as multiplicative shifts in the instantaneous event rate, holding other covariates fixed, under proportional hazards and independent censoring.

Clinician read: $HR > 1$ for smoking means higher **rate** of first exacerbation, not necessarily a specific absolute risk difference — report events and person-time.

Caveats box

Caveat	Why it matters
Censoring assumed independent	Informative dropout biases HRs
Proportional hazards	HR interpretation fails if hazards cross
Sparse events	HRs unstable; report event counts
Competing risk of death	First exacerbation model mis-specified
Calendar time vs study time	Define time origin clearly (enrolment vs diagnosis)
HR risk difference	Common with frequent events

Wrong analysis

Mistake	Why it fails	Do instead
Logistic regression ignoring follow-up time	Discards when events occur	Cox or KM
Treat censored subjects as “no event ever”	Understates event rate	Censor at last follow-up
HR reported without events / n	Uninterpretable precision	Events per arm + person-time
Ignore PH assumption	Misleading adjusted HRs	<code>cox.zph</code> diagnostic
Use survival for recurrent events without extension	Multiple events per patient	Recurrent-event models (advanced)

Catalog of wrong analyses (time-to-exacerbation)

Wrong analysis	Why it fails	Do instead
Fixed 12-month binary endpoint only	Loses timing information	KM + Cox on time-to-event
Exclude early censoring	Selection bias	Intention-to-follow-up population
Unadjusted KM in RCT	Ignores randomisation balance	Pre-specified covariate adjustment in Cox
Present HR as “% risk reduction” without context	Misleading with frequent events	HR + absolute risks or NNT if appropriate

Reporting template

The primary endpoint was time to first moderate-to-severe exacerbation. Patients were followed from enrolment to event or administrative censoring at 365 days. Kaplan–Meier curves compared smokers and non-smokers (log-rank $p = \dots$). A Cox proportional hazards model adjusted for FEV1 % predicted, therapy class, and age. The adjusted hazard ratio for smoking was ... (95% CI ...). There were n events among N patients (...% censored). Proportional hazards was assessed with Schoenfeld residuals ($p = \dots$).

R lab

```
source("R/00_setup.R")  
source("R/examples/ch19_survival_analysis.R")
```

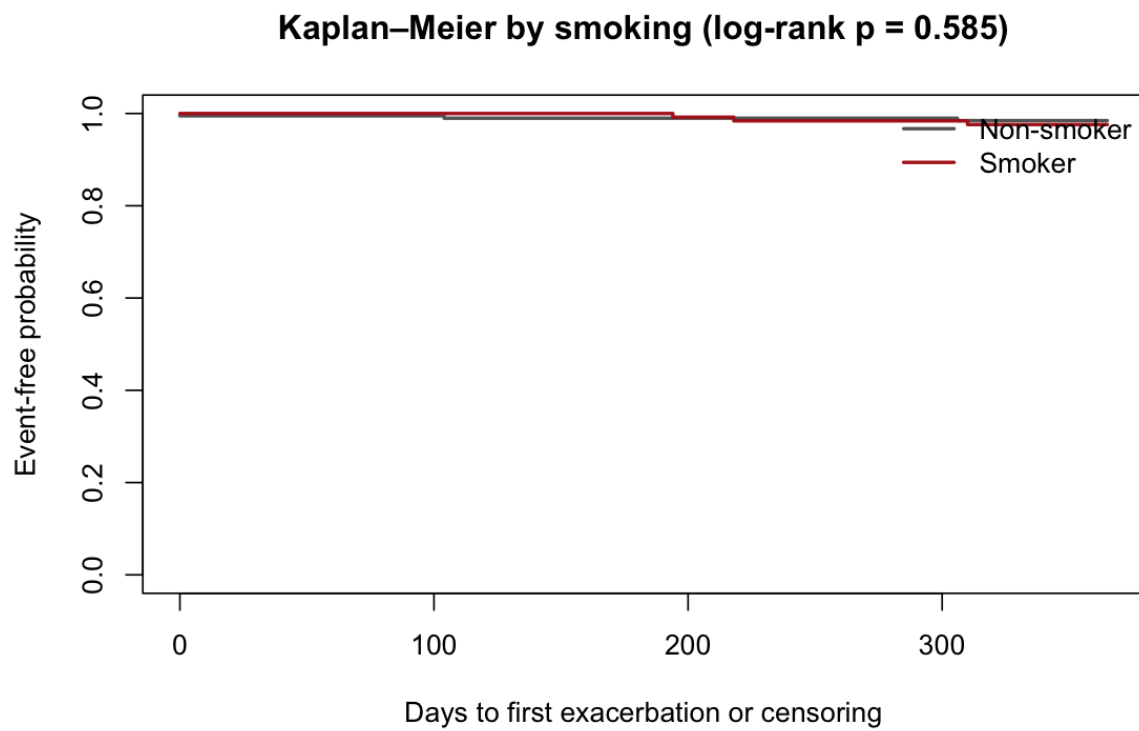


Figure 1: Kaplan–Meier curves by smoking status

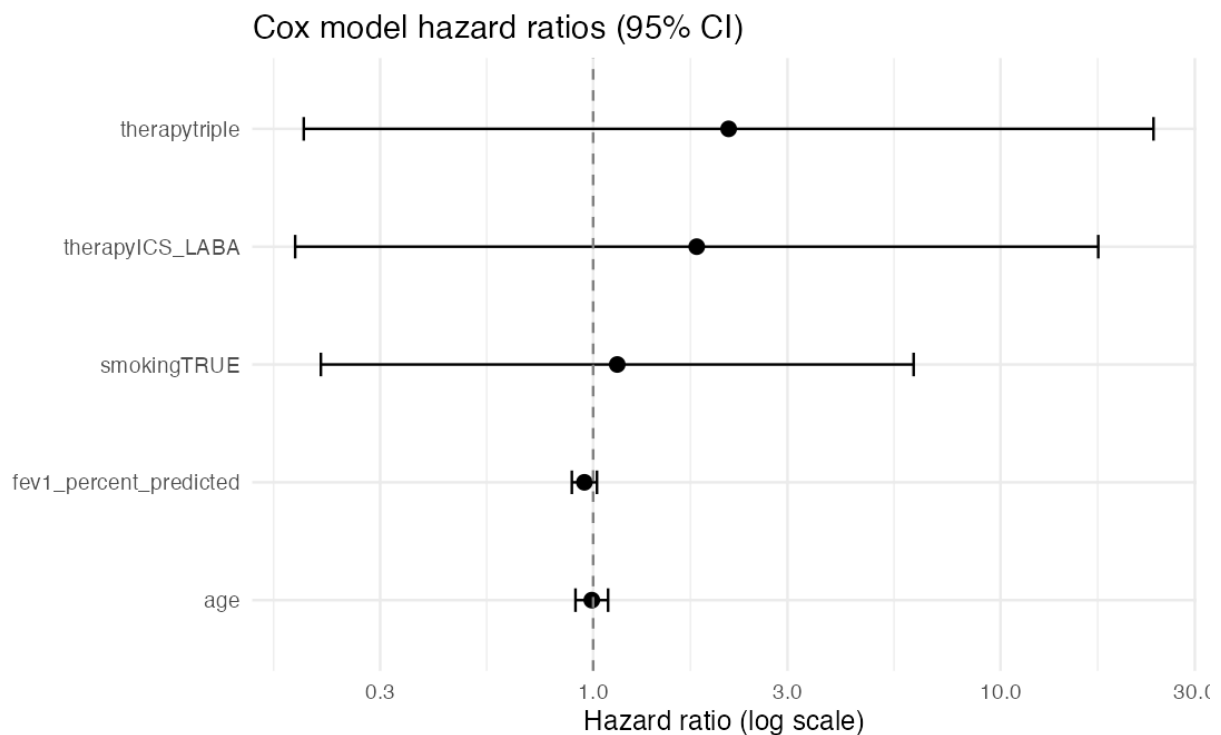


Figure 2: Cox model hazard ratios

Tables: `ch19_events_by_smoking.csv`, `ch19_cox_hazard_ratios.csv`, `ch19_cox_ph_test.csv`

Mini-lab: absolute risk at 12 months

After fitting KM, extract survival at 365 days for each smoking group and report absolute risk difference alongside HR — clinicians often need both.

Alternatives & extensions

Situation	Method	Note
Competing risk (death)	Fine–Gray subdistribution	Future expansion
Recurrent exacerbations	Andersen–Gill / PWP	Not first-event Cox
Non-proportional hazards	Stratified Cox, splines	Check <code>cox.zph</code>
Discrete time	Complementary log-log	Links to Ch 6

Exercises · Solutions

E19.1 What does censoring mean at 365 days without an event?

E19.2 When is a hazard ratio misleading for clinicians?

E19.3 What does a log-rank test compare?

Applied

1. Run `source("R/examples/ch19_survival_analysis.R")`.
2. Report smoking HR from `ch19_cox_hazard_ratios.csv`.
3. Read `ch19_cox_ph_test.csv` — is proportional hazards plausible in this teaching run?

Capstone: Case E in Ch 12.

Further reading

- Harrell, *Regression Modeling Strategies* ([Harrell 2015](#))
- Therneau & Grambsch, *Modeling Survival Data* (Cox models, diagnostics)

Chapter 20: Missing data and sensitivity analysis

Part VIII: Longitudinal, survival, and causal inference

At a glance

Recurring dataset	<code>data/spirometry.csv</code> (MAR-like missingness induced in script)
Format	Technique cards + Caveats + Wrong analysis + Reporting (template)
Question	How does missing FEV1 affect regression coefficients — and what should we report?
Core methods	MCAR/MAR/MNAR framing, complete-case analysis, single imputation contrast, MICE overview
R	<code>R/examples/ch20_missing_data.R</code>
Links	Ch 8 reporting · Ch 18 dropout

Learning objectives

1. Distinguish MCAR, MAR, and MNAR in plain language with respiratory examples.
2. Report n enrolled vs n analysed vs missingness by variable.
3. Run complete-case vs imputation sensitivity and interpret bias direction.
4. Know when multiple imputation (MICE) is the production default.

Prerequisites

Chapters 3, 5, 8 (reporting); Ch 18 (informative dropout context).

Opening question (CASTOR)

Among CASTOR participants, FEV1 is missing more often in severe obstruction. If we drop those rows, does the smoking–FEV1 association change — and is that change a feature or a bug?

Missing data is not a technical footnote. In spirometry trials and cohorts, **missing lung function often tracks disease severity**, therapy intolerance, or inability to perform the manoeuvre ([Harrell 2015](#)).

Technique: Missing data analysis and multiple imputation (overview)

Technique card

Answers	Estimates under explicit missingness assumptions; sensitivity to those assumptions
Outcome	Any (here: continuous FEV1 in <code>lm</code>)
Key quantities	% missing per variable; enrolled n ; analysed n
MCAR	Missingness unrelated to observed or unobserved values
MAR	Missingness depends on observed data only (given a rich enough model)
MNAR	Missingness depends on the unobserved value itself
Teaching script	Complete-case <code>lm</code> vs median imputation
Production default	MICE (<code>mice</code> package) + pooled estimates (Rubin's rules) when MAR is plausible
R (teaching)	<code>R/examples/ch20_missing_data.R</code>
When to use	Any non-trivial missingness in Table 1 or outcomes
When NOT to use	Single imputation without sensitivity when MNAR is plausible
Does NOT prove	That MAR holds — sensitivity and design discussion required

Dual interpretation

Plain language: we compared results using only participants with observed FEV1 versus filling missing FEV1 with a typical value; if conclusions change, missingness matters.

Precise language: under MAR, valid inference requires either a correctly specified likelihood/model integrating missingness or multiple imputation from a model predicting missing values from observed covariates; complete-case analysis is unbiased only under stronger assumptions (often MCAR or MAR with missingness independent of outcome given covariates in the analysis model).

Clinician read: if sicker patients are missing spirometry, “complete-case FEV1” may describe **healthier** subsets — not the enrolled trial population.

Caveats box

Caveat	Why it matters in respiratory research
Informative spirometry missingness	Severe dyspnoea, exacerbation, poor effort tests
MAR is untestable	You defend it with subject-matter reasoning + sensitivity
Median imputation	Shrinks variance; SEs wrong if treated as observed

Caveat	Why it matters in respiratory research
Imputing then splitting train/test	Leakage in prediction workflows (Ch 9, 17)
LOCF for FEV1 trajectories	Can create false stability (Ch 18)
MNAR for death/discontinuation	Requires dedicated models, not silent deletion

Wrong analysis

Mistake	Why it fails	Do instead
Drop missing without table	Hides selection	Report missing % by arm/severity
Listwise deletion as default	Biased under MAR/MNAR	MI or principled model
Single imputation, ignore uncertainty	SEs too small	MICE + Rubin pooling
Impute using future/outcome information	Leakage	Imputation model uses only past/ baseline covariates per protocol
“No missing data” when LOCF used	Hidden imputation	State imputation rule explicitly

Catalog of wrong analyses (missing data)

Wrong analysis	Why it fails	Do instead
Per-protocol deletion of missed visits without estimand	Changes population	Align with ITT or prespecified estimand
Replace missing FEV1 with 0	Meaningless scale abuse	Model missingness or impute within plausible range
Complete-case ML with 40% missing predictors	Biased + overfit	MI inside CV folds
“Sensitivity analysis: repeat without missing” only	One-direction sensitivity	Multiple plausible MNAR scenarios

Reporting template

Of $N = \dots$ participants enrolled, $n = \dots$ had observed FEV1 at the analysis visit ($\dots\%$ missing). Missingness differed by obstruction severity (Figure). The primary model used $\dots$ (complete-case / multiple imputation with $m = \dots$ imputations). Variables

in the imputation model were Pooled estimates for smoking were ... (95% CI ...). Results were similar / materially changed in complete-case analysis (Table).

R lab

```
source("R/00_setup.R")  
source("R/examples/ch20_missing_data.R")
```

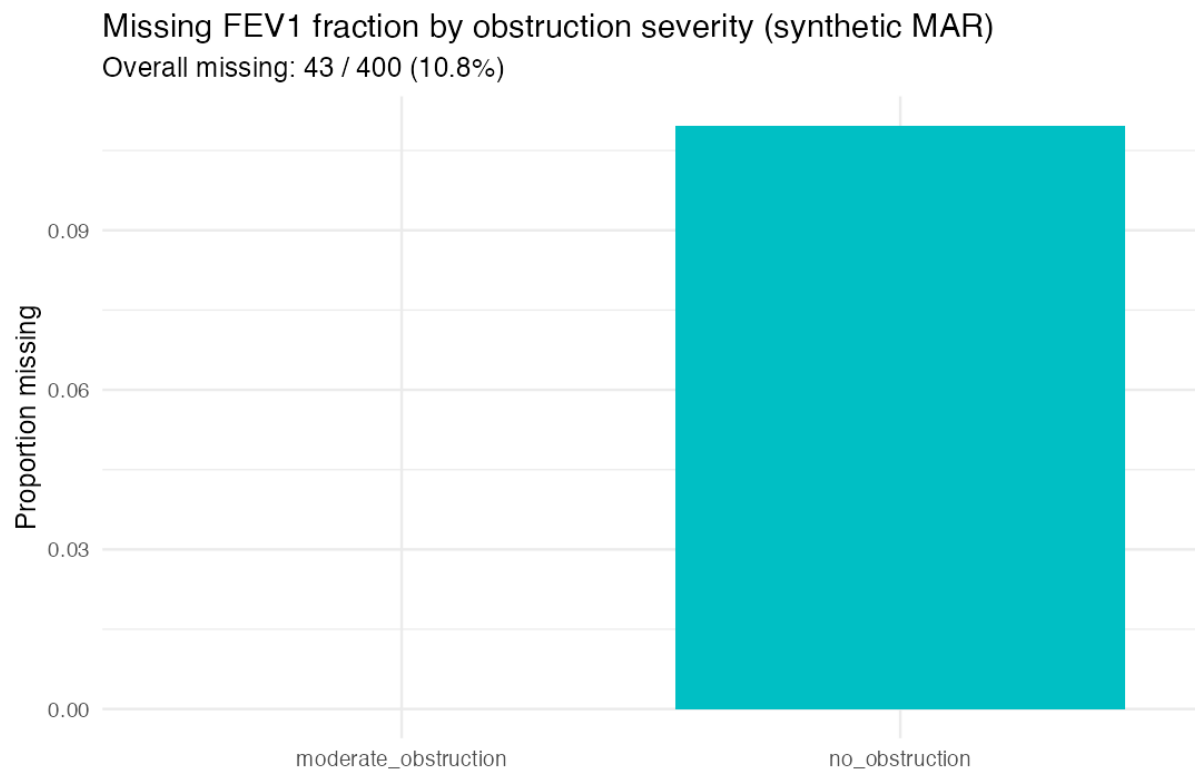


Figure 1: Missing FEV1 fraction by obstruction severity

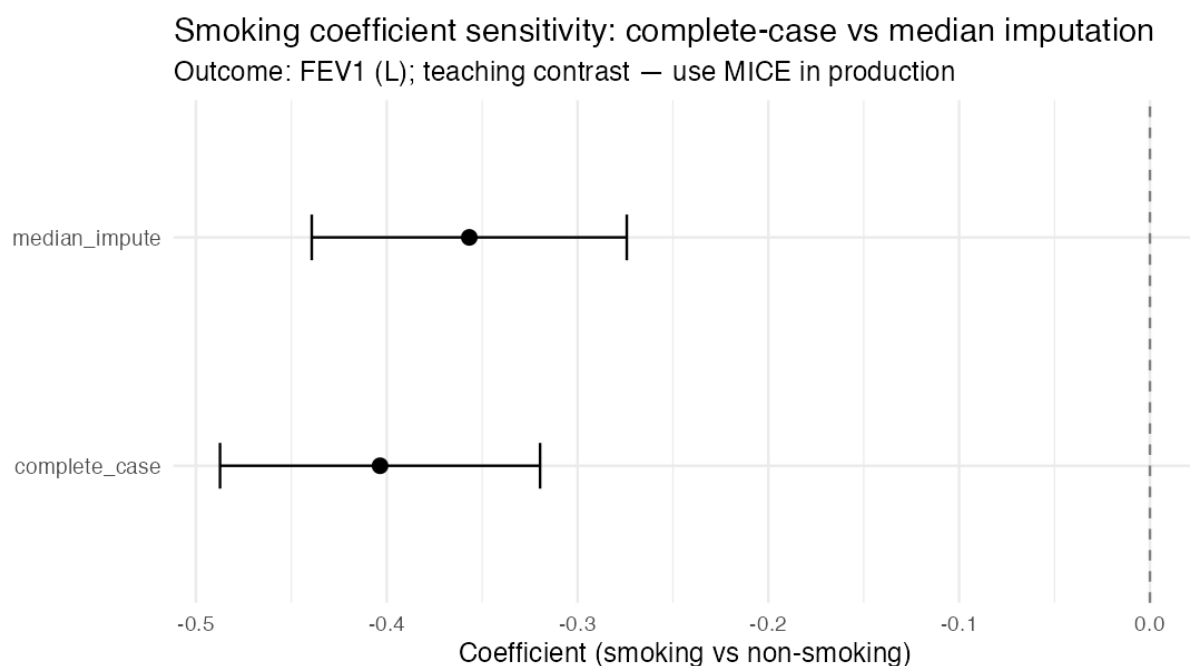


Figure 2: Smoking coefficient: complete-case vs median imputation

Tables: `ch20_enrollment_flow.csv`, `ch20_missingness_by_diagnosis.csv`, `ch20_smoking_coef_sensitivity.csv`

Mini-lab: MICE pointer (production)

```
# After install.packages("mice"):
# imp <- mice::mice(spirometry_miss, m = 20, maxit = 5, seed = 1)
# fit_mice <- with(imp, lm(fev1_obs ~ smoking + age + sex))
# pool::pool(fit_mice)
```

Use an imputation model that includes predictors of missingness (e.g. diagnosis, baseline FEV1) but avoid outcome leakage per study protocol.

Alternatives & extensions

Situation	Method	Note
Longitudinal dropout	Mixed model / joint model	Ch 18 + sensitivity
MNAR for spirometry	Pattern-mixture / selection models	Expert sensitivity
Prediction with missing predictors	MI inside resampling	Ch 9, 17
Single missing covariate, few missing	Firth / exact methods	Ch 6 sparse events

Exercises · Solutions

E20.1 Define MAR in one sentence.

E20.2 Why is complete-case analysis risky when missingness relates to obstruction severity?

E20.3 Why is median imputation inadequate as a final analysis?

Applied

1. Run `source("R/examples/ch20_missing_data.R")`.
2. Report enrolled vs analysed n from `ch20_enrollment_flow.csv`.
3. Compare smoking coefficients in `ch20_smoking_coef_sensitivity.csv`.

Further reading

- Harrell, *Regression Modeling Strategies* (missing data chapter) ([Harrell 2015](#))
- van Buuren, *Flexible Imputation of Missing Data* (MICE)

Chapter 21: Causal inference — confounding and IPW

Part VIII: Longitudinal, survival, and causal inference

At a glance

Recurring dataset	<code>data/exacerbation.csv</code>
Format	Technique cards + Caveats + Wrong analysis + Reporting (template)
Question	When can we interpret an adjusted association as closer to a causal effect?
Core methods	confounding diagram, target trial, associational vs causal estimands, introductory IPW
R	<code>R/examples/ch21_causal_inference.R</code>
Links	Ch 6 logistic · Ch 12 Case B

Learning objectives

1. State the difference between association and causation in observational respiratory studies.
2. Draw a simple confounding structure (smoking, FEV1, exacerbation).
3. Run a toy IPW sensitivity for smoking exposure and compare to naive logistic regression.
4. List requirements for causal language (design, assumptions, diagnostics, sensitivity).

Prerequisites

Chapters 1 (estimands), 6–7 (logistic regression), 8 (STROBE reporting).

Opening question (CASTOR)

*Does smoking **cause** higher 12-month exacerbation risk — or is low FEV1 a common consequence of smoking and severe disease that also drives exacerbations?*

Standard logistic regression adjusts for measured covariates, but **which** covariates belong in the model is a causal question, not a significance question ([Harrell 2015](#)).

Technique: Associational vs causal estimands; introductory IPW

Technique card (associational)

Answers	Is smoking associated with exacerbation after adjusting for measured covariates?
Estimand	Conditional odds ratio (logistic) — associational
R	<code>glm(exacerbation_12m ~ smoking + fev1_pct, family = binomial)</code>
When to use	Hypothesis generation; STROBE cohort descriptions
Does NOT prove	Smoking causes exacerbations

Technique card (IPW — introductory)

Answers	Weighted pseudo-population where exposure is balanced on measured confounders (toy)
Steps	Model exposure ~ confounders → weights = $1/P(\text{exposure} \text{confounders})$ → weighted outcome model
Teaching script	IPW for smoking using FEV1 tertile marginal weights
Assumptions	No unmeasured confounding; positivity (all exposure levels possible within confounder strata); correct exposure model
R	<code>R/examples/ch21_causal_inference.R</code>
When to use	Sensitivity to covariate adjustment; motivation for propensity methods
When NOT to use	As automatic proof of causation; with extreme weights untrimmed
Does NOT prove	Causation without design + unmeasured confounding sensitivity

Dual interpretation

Plain language: we reweighted patients so smoking groups looked more comparable on FEV1 strata, then re-estimated the smoking–exacerbation association.

Precise language: IPW estimates a marginal or appropriately weighted estimand under a structural model where conditioning on measured confounders blocks back-door paths; residual confounding remains if unmeasured common causes exist.

Clinician read: “Adjusted OR” and “IPW OR” still describe **observational** data — randomised trials (Case A) remain the gold standard for causal treatment effects.

Caveats box

Caveat	Why it matters
Unmeasured confounding	Adherence, SES, exacerbation history incompletely captured
Collider adjustment	Conditioning on post-exposure variables (e.g. antibiotics) opens bias
Mediator adjustment	Adjusting for variables on causal path blocks total effect
Positivity / extreme weights	Sparse cells → unstable weights; trim and report ESS
Overlap	No empirical overlap → estimand not supported in data
Transportability	CASTOR-like synthetic cohort your clinic population

Wrong analysis

Mistake	Why it fails	Do instead
“Adjusted OR proves smoking causes exacerbations”	Observational; model-dependent	State associational estimand; Ch 12 Case B limits
Throw all predictors into propensity model	Overfitting; bias	DAG / subject-matter covariate set
IPW without checking weights	Extreme weights dominate	Summary of weight distribution; trimming
Adjust for mediators when total effect is target	Blocks part of effect	Prespecify estimand (total vs direct)
Causal language from IPW alone	Untestable assumptions	Sensitivity + design discussion

Catalog of wrong analyses (causal claims)

Wrong analysis	Why it fails	Do instead
Significant covariate → include in PS model	Data-driven confounding control	Prespecified adjustment set
Compare IPW and OLS, pick “better” p-value	Multiplicity / fishing	Pre-register estimand and method
Ignore immortal time bias in therapy cohorts	Misaligned time zero	Target trial emulation
Use future FEV1 as confounder	Post-baseline leakage	Baseline covariates only unless clear protocol

Reporting template

We estimated the association between smoking and 12-month exacerbation in an observational cohort ($n = \dots$). The naive logistic model adjusting for FEV1 % predicted yielded OR = ... (95% CI ...). As a sensitivity analysis, we applied inverse probability weights for smoking based on ... (weight range ...; mean ...). The IPW OR

was ... (95% CI ...). These analyses assume no unmeasured confounding and sufficient overlap of smokers and non-smokers within confounder strata. Causal interpretation is not claimed; findings are associational ([Elm et al. 2007](#)).

Target trial (concept)

Ask: *What randomized experiment would answer this question?* Then emulate its eligibility, treatment strategies, assignment, follow-up, and outcomes using observational data — misalignment at any step introduces bias.

R lab

```
source("R/00_setup.R")
source("R/examples/ch21_causal_inference.R")
```

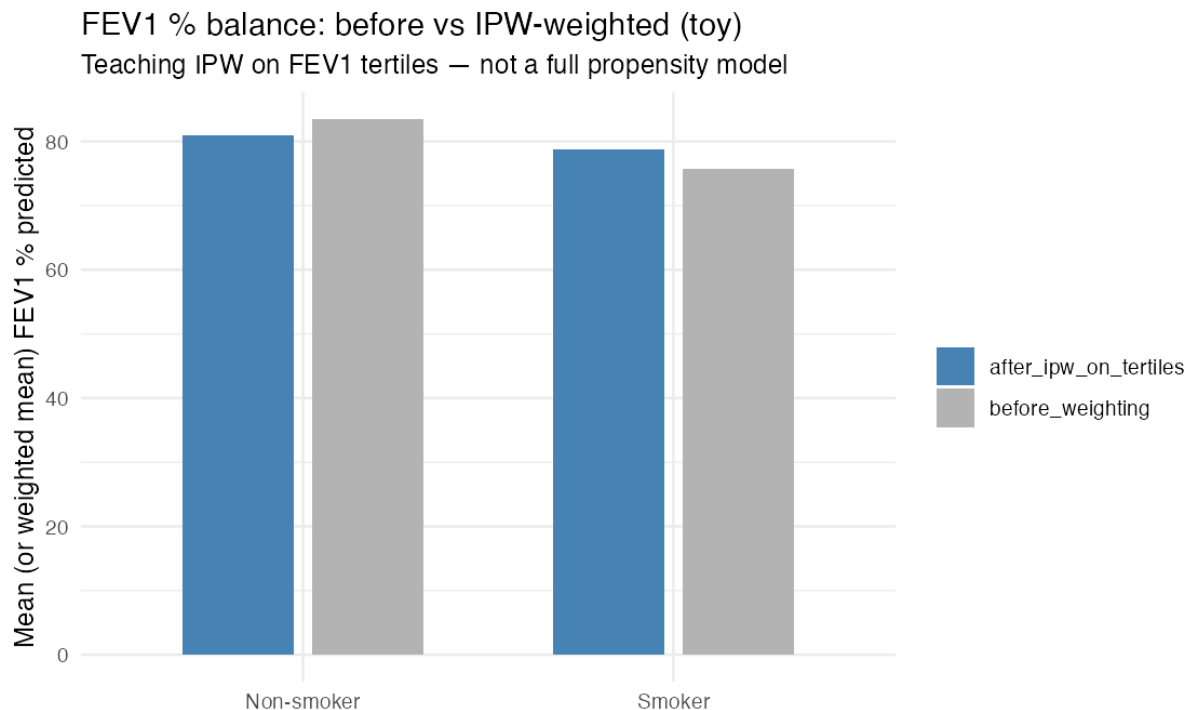


Figure 1: FEV1 balance before vs after IPW (toy)

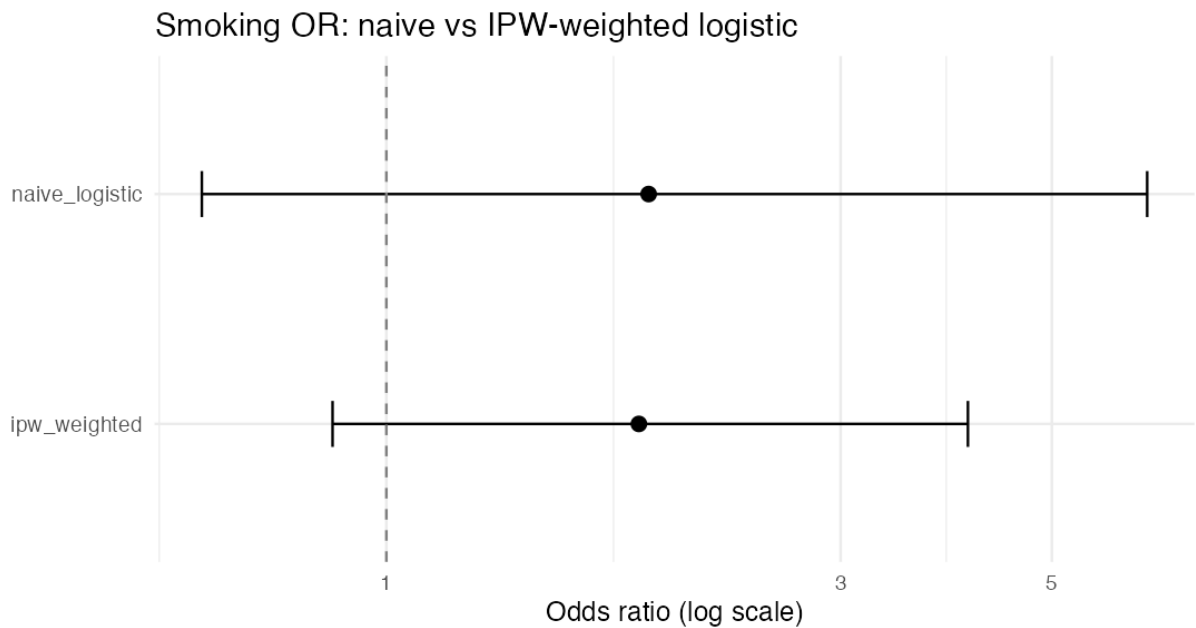


Figure 2: Smoking OR: naive vs IPW

Tables: `ch21_smoking_or_naive_vs_ipw.csv`, `ch21_balance_before_after_ipw.csv`, `ch21_ipw_weight_summary.csv`

Mini-lab: propensity score pointer

Full propensity score workflow: estimate $e(X) = P(\text{smoking}=1|X)$, check overlap, weight or match, then outcome model. Packages: `WeightIt`, `MatchIt`. Always report balance diagnostics (standardised mean differences).

Alternatives & extensions

Situation	Method	Note
Binary treatment, many covariates	Propensity matching / weighting	Overlap plots
Time-varying treatment	Marginal structural models	Advanced
Unmeasured confounding	Sensitivity analysis (E-values)	Report bounds
RCT subgroup	No causal adjustment needed for randomisation	Case A

Exercises · Solutions

E21.1 Name one confounder on the smoking → exacerbation path in CASTOR.

E21.2 What does the “no unmeasured confounding” assumption mean?

E21.3 Why check weight distributions after IPW?

Applied

1. Run `source("R/examples/ch21_causal_inference.R")`.
2. Compare naive vs IPW OR in `ch21_smoking_or_naive_vs_ipw.csv`.
3. Read `ch21_ipw_weight_summary.csv` — any extreme weights?

Capstone link: [Case B](#) (associational logistic) vs this chapter (explicit causal framing).

Further reading

- Hernán & Robins, *Causal Inference: What If* (free online)
- Harrell, *Regression Modeling Strategies* ([Harrell 2015](#))
- STROBE reporting for observational studies ([Elm et al. 2007](#))

References

This appendix lists every source **cited** in handbook chapters ([@key] citations). Entries are drawn from [references.bib](#).

For a **topic-organized reading guide** (textbooks, CONSORT/STROBE/TRIPOD, respiratory methods), see [REFERENCES.md](#).

Agresti, Alan. 2018. *An Introduction to Categorical Data Analysis*. 3rd ed. Wiley.

Benchimol, Eric I, Liam Smeeth, Astrid Guttman, et al. 2015. “The REporting of Studies Conducted Using Observational Routinely-Collected Health Data (RECORD) Statement.” *PLoS Medicine* 12 (10): e1001885. <https://doi.org/10.1371/journal.pmed.1001885>.

Benjamini, Yoav, and Yosef Hochberg. 1995. “Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing.” *Journal of the Royal Statistical Society Series B* 57 (1): 289–300.

Breiman, Leo. 2001. “Random Forests.” *Machine Learning* 45 (1): 5–32.

Cameron, A Colin, and Pravin K Trivedi. 2013. *Regression Analysis of Count Data*. 2nd ed. Cambridge University Press.

Cazzola, Mario, William MacNee, Fernando J Martinez, et al. 2008. “Minimal Clinically Important Difference in COPD Lung Function.” *European Respiratory Journal* 31 (4): 718–19.

Celli, Bartolome R, Peter J Barnes, Stephen I Rennard, et al. 2015. “An Official American Thoracic Society/European Respiratory Society Statement: Research Questions in Chronic Obstructive Pulmonary Disease.” *American Journal of Respiratory and Critical Care Medicine* 191 (3): e4–27. <https://doi.org/10.1164/rccm.201501-0045ST>.

Efron, Bradley, and Robert J Tibshirani. 1993. *An Introduction to the Bootstrap*. Chapman; Hall.

Elm, Erik von, Douglas G Altman, Matthias Egger, et al. 2007. “The Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) Statement: Guidelines for Reporting Observational Studies.” *PLoS Medicine* 4 (10): e296. <https://doi.org/10.1371/journal.pmed.0040296>.

Firth, David. 1993. “Bias Reduction of Maximum Likelihood Estimates.” *Biometrika* 80 (1): 27–38.

Graham, Brendan L, Irene Steenbruggen, Matthew R Miller, et al. 2019. “Standardization of Spirometry 2019 Update. An Official American Thoracic Society and European Respiratory Society Technical Statement.” *American Journal of Respiratory and Critical Care Medicine* 200 (8): e70–88. <https://doi.org/10.1164/rccm.201908-1590ST>.

- Harrell, Frank E. 2015. *Regression Modeling Strategies: With Applications to Linear Models, Logistic and Ordinal Regression, and Survival Analysis*. 2nd ed. Springer.
- Hennig, Christian. 2007. “Cluster-Wise Assessment of Cluster Stability.” *Computational Statistics & Data Analysis* 52 (1): 258–71.
- Hilbe, Joseph M. 2011. *Negative Binomial Regression*. 2nd ed. Cambridge University Press.
- Hilbe, Joseph M. 2014. *Modeling Count Data*. Cambridge University Press.
- Hosmer, David W, Stanley Lemeshow, and Rodney X Sturdivant. 2013. *Applied Logistic Regression*. 3rd ed. Wiley.
- Hurst, John R, Jorgen Vestbo, Antonio Anzueto, et al. 2010. “Understanding the Impact of Chronic Obstructive Pulmonary Disease Exacerbations on Patient Quality of Life.” *European Respiratory Journal* 36 (4): 818–28. <https://doi.org/10.1183/09031936.00040810>.
- James, Gareth, Daniela Witten, Trevor Hastie, and Robert Tibshirani. 2023. *An Introduction to Statistical Learning: With Applications in r*. 2nd ed. Springer.
- Jolliffe, Ian T, and Jorge Cadima. 2016. “Principal Component Analysis: A Review and Recent Developments.” *Philosophical Transactions of the Royal Society A* 374 (20150202): 20150202.
- Mann, Henry B, and Donald R Whitney. 1947. “On a Test of Whether One of Two Random Variables Is Stochastically Larger Than the Other.” *Annals of Mathematical Statistics* 18 (1): 50–60.
- McShane, Lisa M, Douglas G Altman, Willi Sauerbrei, et al. 2011. “Reporting Recommendations for Tumor Marker Prognostic Studies (REMARK): Explanation and Elaboration.” *PLoS Medicine* 8 (10): e1001006. <https://doi.org/10.1371/journal.pmed.1001006>.
- Moons, Karel G M, Douglas G Altman, Johannes B Reitsma, et al. 2015. “Transparent Reporting of a Multivariable Prediction Model for Individual Prognosis or Diagnosis (TRIPOD): Explanation and Elaboration.” *Annals of Internal Medicine* 162 (1): W1–73. <https://doi.org/10.7326/M14-0698>.
- Schulz, Kenneth F, Douglas G Altman, David Moher, et al. 2010. “CONSORT 2010 Statement: Updated Guidelines for Reporting Parallel Group Randomised Trials.” *BMJ* 340: c332. <https://doi.org/10.1136/bmj.c332>.
- Shmueli, Galit. 2010. “To Explain or to Predict?” *Statistical Science* 25 (3): 289–310.
- Steyerberg, Ewout W. 2019. *Clinical Prediction Models: A Practical Approach to Development, Validation, and Updating*. 2nd ed. Springer.
- Stoltzfus, Steve. 2019. *Biostatistics for Health and Biological Science Users of r*. Leanpub.
- Venables, W N, and B D Ripley. 2002. *Modern Applied Statistics with s*. 4th ed. Springer.
- Welch, Bernard L. 1947. “The Generalization of Student’s Problem When Several Different Population Variances Are Involved.” *Biometrika* 34 (1/2): 28–35.

Wenzel, Sally E. 2012. “Asthma Phenotypes: The Evolution from Clinical to Molecular Approaches.” *Nature Medicine* 18 (5): 716–25. <https://doi.org/10.1038/nm.2678>.

Wickham, Hadley. 2016. *Ggplot2: Elegant Graphics for Data Analysis*. 2nd ed. Springer.

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